Mentions and Discussions of “Nomenclature” or “Definitions” In Rheology Leaflets/Bulletins, or early Transactions

The Rheology Leaflet No. 1, March 1937, pg. 4 (6)

REPORT OF THE COMMITTEE ON DEFINITIONS AND NOMENCLATURE

Comments and criticisms concerning the following report will be welcomed, whether coming from members or non-members of the Society. They should be addressed to Prof. E. C. Bingham, Lafayette College, Easton, PA

A REPORT ON RHEOLOGICAL DEFINITIONS AND NOMENCLATURE SUGGESTED FOR DISCUSSION

Foreword

In the following report by the Committee on Definitions and Nomenclature, an attempt is made to develop a complete, logical and self-consistent system of classification of materials according to their rheological properties. It is intended that the system shall provide a classification for all conceivable types of behavior of materials in deformation. The system has been developed on the basis of current usage to the extent permitted by logic and self-consistency.

The classification and the accompanying definitions have been developed in terms of idealized materials to which actual materials can be compared. In other words, ideal types of behavior and deformation are postulated which are susceptible of exact definition and classification. Actual materials can then be described in terms of the ideal types to which their behavior approximates. It should be borne in mind that an actual material with rheological properties coinciding exactly with those of a single idealized material is the exception rather than the rule. In applying to actual materials the quantitative definitions for idealized materials, great caution must be used. It is recognized that the classification of a particular material may sometimes depend upon the method of measurement. This is particularly true as applied to the precision attained. The recognition of a measurable “yield stress,” for example, may depend upon the precision of the measurement and upon the duration of the test. These uncertainties in the classification of materials result from the fact that it is sometimes either impossible or unimportant to deter mine for actual materials some of the characteristics which can be theoretically distinguished

Transcribed by James J Griebler, April 2026

In this system of classification and definitions, only shearing strains and stresses are of interest, and the-relations involved are expressed with reference to simple shear. Many of the concepts defined here in terms of simple shear are closely related to those usually defined by engineers in terms of more complex types of deformation, such as those arising in tension and compression tests. These deformations, while involving simultaneous shears in three dimensions and hence analytically complex, are often easier to produce experimentally than a simple shear. They, therefore, serve as a basis for many useful and practical concepts and definitions in the field of engineering. Where an analysis of the fundamental behavior of the material in deformation is required, these engineering tests may be reduced to a combination of shears, and described in terms of the concepts used in this report. A general relation between strain and stress applies to any point within a material, and, except in extraordinary circumstances, these. values will vary at different points. The quantities involved in these | relations applying to individual points within the material cannot be observed directly. Observable quantities such as efflux-rate, pressure, etc., can be obtained by integrating the basic point relations is, and can then be compared with the values obtained experimentally. It however, always possible, at least formally, to devise experiments and methods of analysis by which the fundamental relationship can be completely determined from experimental data. Although all materials exhibit the inertia effects associated with acceleration of the mass elements within the material, these effects are ignored as being irrelevant in the classification and definitions given below. It should be noted that turbulence and related hydro dynamical properties belong in this category.

Classification

The classification of idealized types of materials is in terms of their behavior under load, or the relation between strain and stress. The stress at any point within a material is a function of the strain and its time derivative, or rate of strain. In all materials the second time derivative of strain is effective in accordance with the inertia effect. As noted above, this effect is ignored in these definitions as common to all materials.

In addition to the instantaneous values of the strain and rate of Strain, the stress may depend on the prior strain or deformation history of the material. Such dependence upon the prior strain is regarded for convenience as a secondary characteristic, materials being classified primarily by the relation between stress and instantaneous strain or rate of strain without regard to whether or not this relationship is affected by the prior strain.

When materials are classified according to their behavior under a shearing stress system, the usual classification is into fluids and solids, fluids including the two sub-classifications, gases and liquids.

Fluid:

A fluid is a substance which undergoes continuous deformation when subjected to any system of finite shearing stress.

Solid:

A solid is a substance which undergoes permanent deformation or rupture only when subjected to a system of shearing stresses which exceeds a certain minimum value.

Classes of Fluids:

As stated above, fluids include gases and liquids. The distinction between a gas and a liquid depends upon the behavior under hydrostatic pressure changes. The precise nature of this distinction and its disappearance above the critical temperature are complicated matters which are adequately discussed in the standard text books. It is unnecessary to repeat such a discussion here, especially in view of the fact that the basis for rheological classifications and definitions is the behavior under shearing stresses, as contrasted with uniform pressures.

Gases:

All gases are similar in their rheological properties, obeying the equation

$$\frac{dv}{dr} = \frac{1}{\eta} F \qquad \text{(See Summary)}$$

Classes of Liquids:

(a) A **Simple** (Newtonian) **Liquid** is one in which the rate of shear is proportional the shearing stress, i.e.:

$$\frac{dv}{dr} = \frac{1}{\eta} F \qquad \text{(See Summary)}$$

(b) A **Complex** (Non-Newtonian) **Liquid** is one in which the rate of shear is not proportional the shearing stress. In general a complex liquid obeys the relation:

$$\frac{dv}{dr} > 0, F > 0 \qquad \text{(See Summary)}$$

Classes of Solids:

(a) A **Plastic Solid** is one which does not deform under a shearing stress until the stress attains a critical value (yield stress), when the solid deforms so readily or so quickly that, except for inertia effects, the stress never exceeds the yield stress.

$$\varepsilon = 0, \qquad 0 < \mathrm{F} < \mathrm{F}_0$$

$$\varepsilon > 0, \qquad \frac{\mathrm{dv}}{\mathrm{dr}} > 0, \qquad \mathrm{F} = \mathrm{F}_0 \ \ (\text{See Summary})$$

Reduction of the shearing stress below the yield stress results in the rate of deformation vanishing. No recovery takes place.

(b) An **Elastic Solid** is one for which the shearing strain is a single valued function of the shearing stress for all values of the shearing stress below the rupture stress (shear strength). Symbolically

$$\varepsilon = \mathrm{f(F)} > 0, \qquad \frac{\mathrm{d}\varepsilon}{\mathrm{dF}} > 0, \qquad 0 < F < F_\infty \qquad (\text{See Summary})$$

(c) A special case of an elastic solid is the **Hookian Solid** for which the shearing strain is proportional to the shearing stress for all values of the shearing stress below the rupture stress. Symbolically, this may be expressed by the relation

$$\varepsilon = \frac{1}{\mathrm{G}}\mathrm{F}, \qquad 0 < F < F_\infty \qquad (\text{See Summary})$$

Reduction of the shearing stress to zero results in the strain vanishing. There is no permanent set in an elastic solid.

(d) An **Elastico-Plastic Solid** is one which obeys the law of an elastic solid for values of the shearing stress below the critical stress corresponding to the elastic limit in shear, and deforms plastically when the shearing stress exceeds that value. Symbolically, this may be expressed by the relations:

$$\varepsilon = \mathrm{f(F)} > 0, \qquad \frac{\mathrm{d}\varepsilon}{\mathrm{dF}} > 0, \qquad 0 < F < F_C$$

$$\varepsilon > 0, \qquad \frac{\mathrm{dv}}{\mathrm{dr}} > 0, \qquad \mathrm{F} = \mathrm{F_c} \qquad (\text{See Summary})$$

Reduction of the shearing stress to zero results in an immediate recovery. The permanent deformation is equal to the plastic deformation undergone.

(e) An **Elastico-Viscous Solid** is one which obeys the laws of an elastic solid for values of the shearing stress below the critical stress corresponding to the elastic limit in shear and deforms continuously at a rate of shear which is a function of the shearing stress, when the shearing stress exceeds that value. Symbolically, this may be expressed by the relations:

$$\varepsilon = \mathrm{f(F)} > 0, \qquad \frac{\mathrm{d}\varepsilon}{\mathrm{dF}} > 0, \qquad 0 < F < F_C$$

$\varepsilon > 0, \qquad \frac{dv}{dr} > 0, \qquad F > F_c \qquad$ (See Summary)

Reduction of the shearing stress to zero results in a recovery which proceeds at a continuously decreasing rate. After an infinite recovery time, no permanent deformation remains

(f) A **Plastico-Viscous Solid** is one for which no deformation occurs for values of the shearing stress below the yield stress and, for values of the shearing stress above that value, deforms continuously at a rate of shear which is a function of the shearing stress. Symbolically, this may be expressed by the relations:

$\varepsilon = 0, \qquad 0 < F < F_0$

$\varepsilon > 0, \qquad \frac{dv}{dr} > 0, \qquad F > F_0 \qquad$ (See Summary)

Reduction of the shearing stress to a value below the yield stress results in the rate of shear vanishing. No recovery takes place.

Prior Strain:

The classification given above is without regard to prior strain. When the terms defined above are used without qualification, it is implied that the stress-strain relation is independent of the strain previously experienced. Materials for which the stress-strain relation is dependent on prior strain should be described as **Thixotropic** or **Strain Hardening** in the senses in which these terms are defined below.

General Remarks:

Again it should be noted that actual materials seldom yield rheological curves which exactly coincide with the idealized diagrams which have been employed in this report. However, no general relation exists which will describe the rheological behavior of all materials and it thus seems necessary to resort to these idealized classifications. It is hoped that they will serve at least for the approximate cataloging of the great number of actual materials.

Definitions

Descriptive Definitions:

Consistency is the resistance to flow of a material. (See Quantitative Definitions).

Plasticity is that property of a body by virtue of which it retains a fraction of its deformation after reduction of the deforming stress to zero

Elasticity is that property of a body by virtue of which it recovers its original size and shape after deformation.

A material, solid or liquid, may exhibit **Thixotropy**, that is, the form of its rheological curve (rate of shear vs. shearing stress) may depend on the previous extent and duration of shearing deformation. Such thixotropic materials are usually of the complex liquid or the plastico-viscous solid types. The action of shearing a thixotropic material results in reducing the shearing stress necessary for a given rate of shear.

Some materials, notably those of the elastico-plastic type, exhibit an effect known as **Strain Hardening**, which is, in a sense, the inverse of thixotropy, i.e., yield stress increases with increasingly shearing strain.

Materials for which the stress-strain curve obtained with increasing stress does not coincide with the curve obtained with decreasing strain are said to exhibit **Hysteresis** and are known as **Hysteretics**. Elastic solids show no hysteresis whereas all other types of solids show it in varying amounts. The classification "hysteretic" is most useful with regard to materials which are predominantly elastic; i.e., elastico-plastic and elastico-viscous solids

Quantitative Definitions:

From an analytical point of view, the simplest type of deformation without volume change is the simple shear. It is convenient and logical, therefore, to take the simple shear as the basic type of deformation and to base upon it our definitions of the primary quantitative rheological properties. This policy seems justified from a practical standpoint also since all the analytically more complex but experimentally simpler types of 'deformations can be resolved into simple shears. A simple shear is a deformation in which the material at any point has moved parallel to a fixed line in a reference plane by an amount which is proportional to the distance of the point from the plane. Such a deformation is shown in Figure 1, the full lines showing the initial shape and the dotted lines the final shape. $\overline{OO}$ is the fixed line in the reference plane, P is the original position of a point in the body, and P' its position after deformation. The extent of the shear or shearing strain, ε is determined by the ratio of the displacement, ds, to the distance, dr of the point, P, from $\overline{OO}$. Thus $\varepsilon = \mathrm{ds}/\mathrm{dr}$. The angle α is known as the angle of shear ($\alpha = \tan^{-1} \mathrm{ds}/\mathrm{dr}$). If the shear is continuous the rate of shear $(\mathrm{dv}/\mathrm{dr})$ is simply the time rate of change of the quotient ds/dr.

Complex shears are possible, involving two or more simple Shears taking place simultaneously in three orthogonal surfaces; but they need not be considered for present purposes, and in what follows, the word "shear" is to be taken as signifying a simple shear only.

The **Consistency** of a material is the ratio of the shearing stress to the rate of shear. The

Viscosity of a simple liquid is the constant ratio of shearing stress to the rate of shear. The unit of viscosity in the C.G:S. is called the **Poise**.

The term “viscosity” is applied only to simple liquids. How ever, the adjective “viscous” is not so limited in its application; and it will be noted in the definitions given above that the terms “elastico viscous” and “plastico-viscous” are not limited to materials with a straight-line rheological diagram.

The **Fluidity** of a simple liquid is the reciprocal of the viscosity. The unit of fluidity in the C.G.S. system is called the **Rhe.**

In the special case where the rate of shear is a linear function of the shearing stress with a positive intercept on the stress axis, the **Mobility** $\boldsymbol{\mu}$, may be defined by the equation

$$\frac{\mathrm{dv}}{\mathrm{dr}} = \mu\,(\mathrm{F} - \mathrm{F}_0)$$

where dv/dr is the rate of shear, F_0 is the yield stress, and F is the shearing stress

The **Elastic Limit In Shear** of a solid is the maximum deformation that is elastic and is the minimum deformation that results in rupture, or in a permanent shear deformation.

The **Shear Strength** of a solid is the minimum value of the shearing stress at which macroscopic rupture occurs.

The **Shear Modulus of Elasticity** (modulus of rigidity). of an elastic solid is the constant ratio of shearing stress to shear strain for values of the shear strain below the elastic limit in shear.

Rheological Diagram

It is recommended that, wherever possible, published data on complex liquids should include not only the directly observed quantities such as pressure, time of efflux, etc.; but, in addition, the calculated basic quantities. The preferred curve should be either consistency, fluidity, or rate of shear plotted against shearing stress; rather than efflux rate against pressure or rate of rotation against torque. **Rheological Curve** or **Rheological Diagram** are suggested as short convenient terms for curves of these preferred types

Transcribed by James J Griebler, April 2026

Summary of Rheological Nomenclature

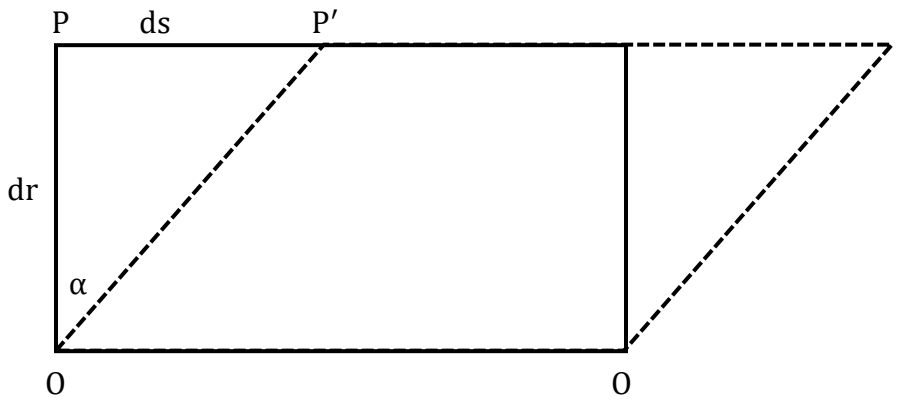
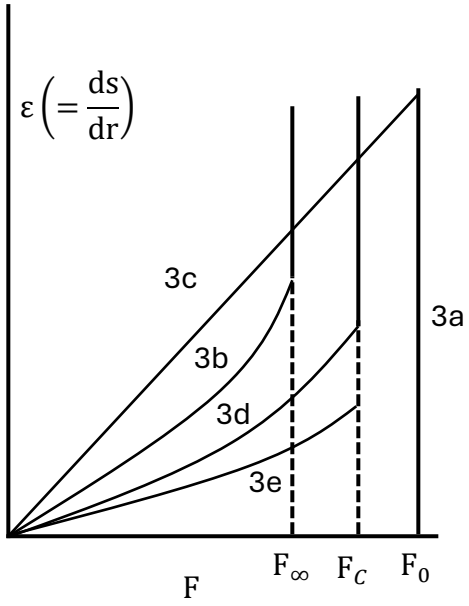


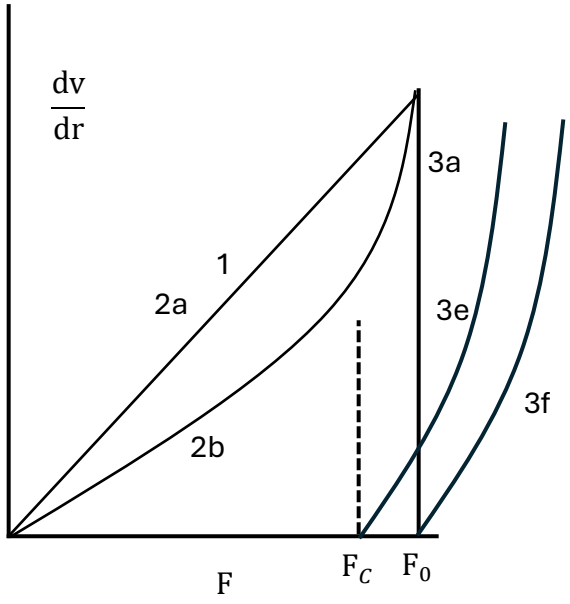
Figure 1 Simple Shear

: Linear displacement parallel to direction of deformation Linear coordinate in the plane of deformation and normal to direction to deformation ds/dt: linear velocity

- $\mathbf{s}$: Linear displacement parallel to direction of deformation.
- $\mathbf{r}$: Linear coordinate in the plate of the deformation and normal to direction to deformation.
- ds/dt: linear velocity.
- dv/dr: Rate of Shear.
- $\mathbf{F}$: Shearing stress.
- $\boldsymbol{\varepsilon} = ds/dr$: Shear strain.
- $\mathbf{F_C}$: Critical shearing stress at elastic limit in shear.
- $\mathbf{F_0}$: Yield stress.
- $\mathbf{F_\infty}$: Shearing stress at rupture (shear strength).
- $\boldsymbol{\eta}$: Coefficient of viscosity
- $\mathbf{G}$: Shear modulus of elasticity (modulus of rigidity)
- $\boldsymbol{\mu}$: Coefficient of mobility.
- $\mathbf{F}/\frac{\mathbf{dv}}{\mathbf{dr}}$: Consistency.

Transcribed by James J Griebler, April 2026





1. Gases

$$\frac{dv}{dr}=\frac{1}{\eta}F$$

(a) Plastic

$$\varepsilon = 0, 0 < F < F_0$$
$$\varepsilon > 0, \frac{dv}{dr} > 0, F = F_0$$

(b) Elastic

$$\varepsilon = f(F), 0 < F < F_\infty$$

(c) Hookian

$$\varepsilon = \frac{1}{G}F, \quad 0 < F < F_\infty$$

2. Liquids

(a) Simple (Newtonian)

$$\frac{dv}{dr}=\frac{1}{\eta}F$$

(b) Complex (Non-Newtonian)

$$\frac{dv}{dr} > 0, \qquad F > 0$$

3. Solids

(d) Elastico-Plastic

$$\varepsilon = f(F) > 0, 0 < F < F_C$$
$$\varepsilon > 0, \frac{dv}{dr} > 0, F = F_C$$

(e) Elastico-Viscous

$$\varepsilon = f(F) > 0, 0 < F < F_C$$
$$\varepsilon > 0, \frac{dv}{dr} > 0, F > F_C$$

(f) Plastico-Viscous

$$\varepsilon = 0, 0 < F < F_0$$

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$$\varepsilon > 0, \frac{dv}{dr} > 0, F = F_0$$

The Rheology Leaflet No. 2, July 1937, pg. 3 (25)

Definitions of Consistency, etc.

Another feature of the A. S. T. M. meeting of special interest to theologists was a meeting of Technical Committee II of Committee E-1. This committee, of which Prof. Bingham is chairman, is responsible for formulating definitions and testing methods in the field of Consistency Plasticity, etc. Some years ago it adopted and published a tentative series of definitions. At this meeting a new series of definitions, prepared by Mr. R. Burns, of the Bell Telephone Laboratories, were introduced as a proposed substitute for the series previously adopted. After some discussion and revision, these new definitions were adopted by the committee, which recommended that they be adopted as a new Tentative Standard.

These new definitions are largely based upon, and are wholly consistent with the definitions developed by our own Committee on Definitions, and published in the first number of this Leaflet. The adoption of these new definitions by the A. S. T. M. is further evidence of the increasing co-operation between Rheologists and Materials Testing Engineers.

The Rheology Leaflet No. 3, October 1937, pg. 3 (37)

Contributed Papers

9:50 A.M. 1. The Definitions of Rheology, by Eugene C. Bingham, Lafayette College, Easton, Penna.

The definitions have been published in the Leaflet. They were further discussed by the A.S.T.M. committee, of which several members of this Society are members. The resulting definitions have been submitted to the members for vote. It would seem to be the appropriate time for this Society also to take action.

The Rheology Leaflet No. 4, February 1938, pg. 4 (56)

DEFINITIONS AND NOMENCLATURE

After valiant efforts on the part of our Committee On Definitions and Nomenclature to attain a satisfactory degree of unanimity in the matter of definitions, a lengthy discussion

at the Akron meeting reveals the persistence of marked difference of opinion. The matter is one of wide-spread interest. Professor Bingham has recently received a letter from the laboratory of N. V. de Bataafsche Petroleum Maatschappij in which the laboratory desires, (1) a more precise definition of ""yield stress"; (2) the inclusion of "elastico-plastico viscous solids" along with elastic, plastic, and viscous solids; (3) substitution of new symbols for F and r; (4) the substitution of "solid" for "body" in the definition of plasticity, (see Rheology Leaflet, No. 1, Page 9.)

The Rheology Leaflet No. 6, August 1938, pg. 4 (95)

NEWS OF RHEOLOGY AND OF RHEOLOGISTS

American Society for Testing Materials, Tentative Standards, 1937, covers many tests involving deformation such as compressive strengths of cements and ceramic materials, viscosities of petroleum products, rubber, both vulcanized and in solution, and definitions of terms relating to rheological properties of matter.

The Rheology Leaflet No. 7, November 1938, pg. 1 (116)

TENTH ANNUAL MEETING

In addition to the program, this issue of the Leaflet contains abstracts of the papers to be presented at the meeting, and also the revised report of the Committee on Definitions and Nomenclature. The original report was published in Leaflet No. 1, May 1937, and discussed at the last annual meeting. As a result of this discussion, the Society adopted a resolution referring the report to the Committee for revision. The report as given here has been revised in the light of the discussion on which this action was based, and is now offered to the Society for adoption. Discussion of the report is scheduled as part of the program for the session to be held the morning of December 29.

The Rheology Leaflet No. 7, November 1938, pg. 11 (125)

A REPORT ON RHEOLOGICAL DEFINITIONS AND NOMENCLATURE

Foreword

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Transcribed by James J Griebler, April 2026

The classification and the accompanying definitions have been developed in terms of idealized materials to which actual materials can be compared. In other words, ideal types of behavior and deformation are postulated which are susceptible of exact definition and classification. Actual materials can then be described in terms of the ideal types to which their behavior approximates. It should be borne in mind that an actual material with rheological properties coinciding exactly with those of a single idealized material is the exception rather than the rule. In applying to actual materials the quantitative definitions for idealized materials, great caution must be used.

It is recognized that the classification of a particular material may sometimes depend upon the method of measurement. This is particularly true as applied to the precision attained. The recognition of a measurable "'yield stress," for example, may depend upon the precision of the measurement and upon the duration of the test. These uncertainties in the classification of materials result from the fact that it is sometimes either impossible or unimportant to determine for actual materials some of the characteristics which can be theoretically distinguished. In this system of classification and definitions, only shearing strains and stresses are of interest, and the relations involved are expressed with reference to simple shear. Many of the concepts defined here in terms of simple shear are closely related to those usually defined by engineers in terms of more complex types of deformation, such as those arising in tension and compression tests. These deformations, while involving simultaneous shears in three dimensions and hence analytically complex, are often easier to produce experimentally than a simple shear. They therefore serve as a basis for many useful and practical concepts and definitions in the field of engineering. Where an analysis of the fundamental behavior of the material in deformation is required, these engineering tests may be reduced to a combination of shears, and described in terms of the concepts used in this report.

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Although all materials exhibit the inertia effects associated with acceleration of the mass elements within the material, these effects are ignored as being irrelevant in the

classification and definitions given below. It should be noted that turbulence and related hydrodynamical properties belong in this category.

Descriptive Definitions:

Consistency is the resistance to flow of a material. (See Quantitative Definitions).

Plasticity is that property of a body by virtue of which it retains a fraction of its deformation after reduction of the deforming stress to zero

Elasticity is that property of a body by virtue of which it recovers its original size and shape after deformation.

Thixotropy is that property of a body by virtue of which its consistency is reduced by previous deformation.

Strain hardening is that property of a body by virtue of which the stress required for permanent deformation increases with the extent of previous permanent deformation.

Hysteresis is that property of a body by virtue of which the relation between stress and strain for decreasing stress differs from that for increasing stress. Materials exhibiting hysteresis are known as **hysteretics**.

A **fluid** is a substance which undergoes continuous deformation when subjected to shear stress.

A **solid** is a substance which undergoes permanent deformation or rupture only when subjected to shear stress in excess of a certain minimum value.

Quantitative Definitions:

A **simple shear** is a deformation in which the material at any point has moved parallel to a fixed line in a reference plane by an amount which is proportional to the distance of the point from the plane. Such a deformation is shown in Figure 1, the full lines showing the initial shape and the dotted lines the final shape. $\overline{OO}$ is the fixed line in the reference plane, P is the original position of a point in the body, and P' its position after deformation. The shear strain, ε, is given by the ratio of the displacement $\mathrm{d}s$ to the distance $\mathrm{d}x$ of the point P from $\overline{OO}$, ore $\varepsilon = \mathrm{d}s/\mathrm{d}x$.

The **angle of shear** is the angle whose tangent is the shear strain. In Figure 1 the angle α is the angle of shear ($\alpha = \tan^{-1} \mathrm{d}s/\mathrm{d}x$.)

The **rate of shear** is the time rate of change of shear strain. In Figure 1 the rate of shear is given by:

$$\frac{dv}{dx} = \frac{d}{dt}\left(\frac{ds}{dx}\right)$$

A **complex shear** is the resultant of two or more simple shears. For simplicity, the definitions given in this report are stated in terms of simple shear, it being understood that the same definitions apply in the case of complex shears resulting from simple shears of the same type.

The **consistency** of a material is the ratio of the shear stress to the rate of shear.

The **viscosity** of a simple liquid is the constant ratio of shear stress to the rate of shear.

Note: The term "viscosity" is applied only to simple liquids. How ever, the adjective "viscous" does not necessarily imply a linear relation between shear stress and rate of shear and is used in the more general sense in the term "plastico-viscous"', defined below.

The **poise** is the unit of viscosity in the C.G.S. system.

The **fluidity** of a simple liquid is the reciprocal of the viscosity.

The **rhe** is the unit of fluidity in the C.G.S. system.

The **mobility μ** is the quantity defined by the equation:

$$\frac{dv}{dx} = \mu\,(F - F_0)$$

characterizing the deformation of a body for which the rate of shear $\frac{dv}{dx}$ is a linear function of the shear stress F for stresses greater than the yield stress F_0, and is zero for stresses less than F_0

The **yield stress** of a solid is the shear stress which must be equalled or exceeded to produce permanent deformation

The **elastic limit in shear** of a solid is the maximum deformation that is elastic and the minimum deformation that results in rupture or in permanent deformation.

The **shear strength** of a solid is the minimum value of shear stress at which macroscopic rupture occurs.

The **shear modulus of elasticity** (modulus of rigidity) of an elastic solid is the constant ratio of shear stress to shear strain for values of the shear strain below the elastic limit.

Transcribed by James J Griebler, April 2026

Classification

The classification of idealized types of materials is in terms of their behavior under load, or the relation between strain and stress. The stress at any point within a material is a function of the strain and its time derivative, or rate of strain. In all materials the second time derivative of strain determines the inertia effect. As noted above, this effect is ignored in these definitions.

The stress may depend to some extent on the prior strain or deformation history of the material. Such dependence upon the prior strain is regarded for convenience as a secondary characteristic, materials being classified primarily by the actual relation between stress and instantaneous strain or rate of strain without regard to differences in this relationship that would result from a different strain history.

Classes of Fluids

(a) A **gas** is a fluid distinguished from a liquid by its behavior under hydrostatic pressure changes. This distinction is irrelevant to this classification, as it does not involve shear stresses. Under shear stress the behavior of a gas is identical with that of a simple liquid.

(b) A simple liquid (Newtonian liquid) is one for which the rate of shear is proportional to the shear stress. Symbolically:

$$\frac{dv}{dr} = \frac{1}{\eta} F \qquad \text{(See Summary)}$$

(c) A **complex liquid** (Non-Newtonian liquid) is one for which the rate of shear is a function of shear stress only, this function being other than that of direct proportionality. Symbolically:

$$\frac{dv}{dr} = f(F) > 0 \qquad \text{(See Summary)}$$

Classes of Solids

(a) An **elastic solid** is one for which the shear strain is a single valued monotonic function of the shear stress for all values of the latter below the shear strength. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_\alpha \qquad \text{(See Summary)}$$

(b) A **Hookian solid** is an elastic solid for which the shear strain is directly proportional to the shear stress for all values of the latter below the shear strength. Symbolically:

$$\varepsilon = \frac{F}{G}, 0 < F < F_\alpha \qquad \text{(See Summary)}$$

Transcribed by James J Griebler, April 2026

Reduction of the shearing stress to zero results in the immediate and complete disappearance of the strain.

(c) A **plastic solid** is one which obeys the law of an elastic solid for shear stress below the yield stress, and for shear stress in excess of this value is permanently deformed at a rate controlled only by the inertia reaction. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_0$$

$$\varepsilon > 0, \qquad \frac{dv}{dx} > 0, \qquad F_0 < F < F_\alpha \ \text{(See Summary)}$$

Reduction of the shear stress to zero results in an immediate recovery of the elastic portion of the strain. The strain in excess of that experienced up to the yield point remains as permanent deformation.

(d) A **plastico-viscous solid** is one for which the deformation for shear stress below the yield stress is that of an elastic solid, and which for shear stress above that value deforms continuously at a rate of shear which is a function of the shear stress. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_\alpha$$

$$\frac{dv}{dx} = f(F) > 0, \qquad F_0 < F < F_\alpha \ \text{(See Summary)}$$

Reduction of the shear stress to a value below the yield stress results in the immediate reduction of the rate of shear to zero. The strain in excess of that experienced up to the yield point remains as permanent deformation.

(e) A **complex solid** is one for which the shear stress required for deformation is a function of both the shear strain and rate of Strain. Symbolically:

$$\varepsilon = f\left(F, \frac{dv}{dx}\right)$$

Prior Strain

The classification given above is without regard to prior strain. When the terms defined above are used without qualification it is implied that the stress-strain relation is independent of the strain previously experienced. Materials for which the stress-strain relation is dependent on prior strain should be described as **Thixotropic** or **Strain Hardening** in the senses in which these terms are defined above.

Rheological Diagram

It recommended that wherever possible published data on is complex liquids should include not only the observed quantities such as pressure, time of efflux, etc., but in

addition the calculated basic quantities. The preferred curve should show consistency, fluidity, or rate of shear plotted against shear stress, rather, than efflux rate against pressure, or rate of rotation against torque. **Rheological Curve** or **Rheological Diagram** are suggested as short convenient terms for curves of these preferred types

Summary of Rheological Nomenclature

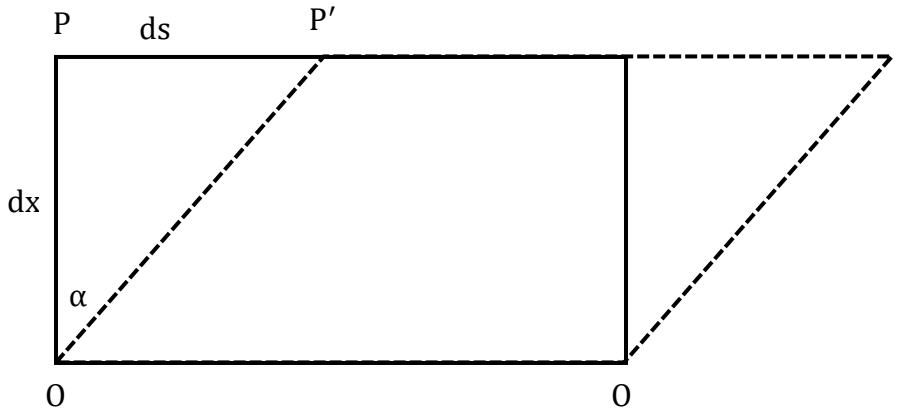


Figure 1 Simple Shear

$\mathbf{s}$: Linear displacement parallel to direction of deformation.

$\mathbf{x}$: Linear coordinate in the plate of the deformation and normal to direction to deformation.

$\mathrm{ds/dt}$: linear velocity.

$\mathrm{dv/dr}$: Rate of Shear.

$\mathbf{F}$: Shearing stress.

$\boldsymbol{\varepsilon} = \mathrm{ds/dr}$: Shear strain.

$\mathbf{F_0}$: Yield stress.

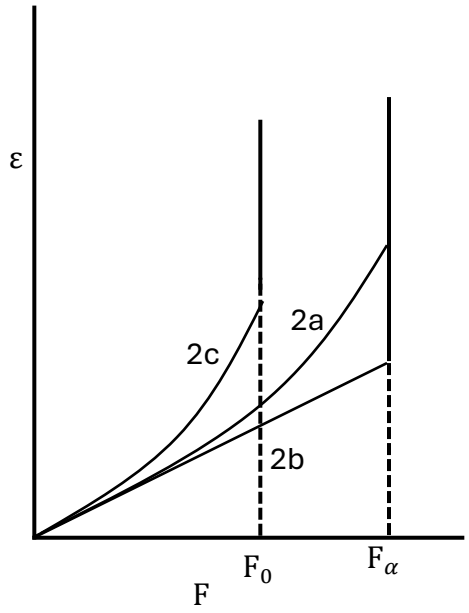
$\mathbf{F_\infty}$: Shearing stress at rupture (shear strength).

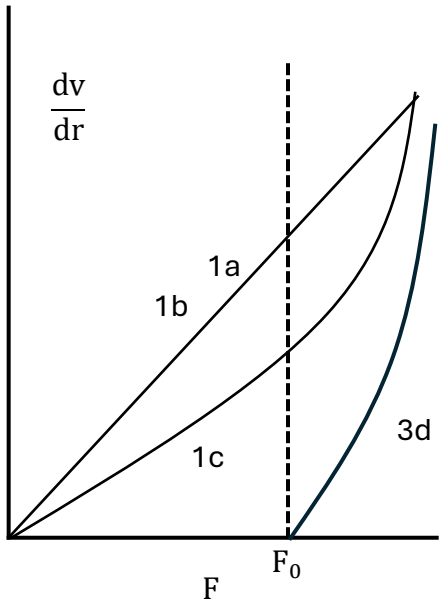
$\boldsymbol{\eta}$: Coefficient of viscosity

$\mathbf{G}$: Shear modulus of elasticity (modulus of rigidity)

$\boldsymbol{\mu}$: Coefficient of mobility.

$\mathbf{F}/\frac{\mathbf{dv}}{\mathbf{dr}}$: Consistency.





1. Fluids

(a) Gas

$$\frac{dv}{dx} = \frac{1}{\eta} F$$

(b) Simple Liquid

$$\frac{dv}{dx} = \frac{1}{\eta} F$$

(c) Complex Liquid

$$\frac{dv}{dx} = f(F) > 0$$

2. Solids

(a) Elastic Solid

$$\varepsilon = f(F) > 0, 0 < F < F_\alpha$$

(b) Hookian Solid

$$\varepsilon = \frac{F}{G}, \quad 0 < F < F_\alpha$$

(c) Plastic Solid

$$\varepsilon = f(F) > 0\ , 0 < F < F_0$$

$$\varepsilon > 0, \frac{dv}{dx} > 0, F_0 < F < F_\alpha$$

(d) Plastico-Viscous

$$\varepsilon = f(F) > 0, 0 < F < F_0$$

$$\frac{dv}{dx} = f(F) > 0, F_0 < F < F_\alpha$$

(e) Complex Solid

$$F = f\left(\varepsilon, \frac{dv}{dx}\right)$$

Transcribed by James J Griebler, April 2026

COMMITTEE ON DEFINITIONS AND NOMENCLATURE

Although three quarters of an hour had been provided for discus Sion of the report which was published in the November issue of the Rheology Leaflet, program delays reduced the available time to about fifteen minutes. Drs. Mayo D, Hersey and A. Nadai spoke in opposition to the report as printed and no final action was taken on the report as a whole. The group did express their belief that the term “viscosity” should not be limited to simple liquids. Aside from the formal objection to this generalization of earlier scientific usage, the fact is that the word is widely used in industry to describe fluids in general and a limitation of its use by members of the Society would result in a failure to be understood. Wherever possible scientific terminology should be consistent with the vocabulary of the industrialist and technologist. It is the hope of the officers of the Society that the entire question of definitions and nomenclature can be discussed in some detail in the coming numbers of the Leaflet and that at the next annual meeting, time may be allowed for a full and conclusive agreement.

THE RHEOLOGY LEAFLET

After the Annual Dinner on December 28, those present held an informal discussion of publication policy. This was done in order to obtain an expression of opinion from as many members as possible under conditions more favorable than those normally provided by the business meeting. This proved to be a very successful procedure and a thorough discussion disclosed a virtual unanimity of opinion on the following points:

1. That our present policy of submitting contributed articles, particularly those on original research, for publication in the Journal of Applied Physics should be continued.

2. That the Leaflet should be enlarged and improved by the inclusion of additional technical material.

The proposal that was most favorably received was for the inclusion of invitation review papers. It was agreed that abstracts and bibliographies should be limited to those covering specific topics and that we should not attempt any complete coverage of rheological literature. Reviews and abstracts will be given of certain books and of the more important articles. Discussions of the Definitions Report could very well be carried to such a point in the Leaflet that final decisions could be made during the annual meeting

Transcribed by James J Griebler, April 2026

The Rheology Leaflet No. 9, May 1939, pg. 1 (162)

ELEVENTH ANNUAL MEETING

"The Science of Rheology and the Plastics Industry" is the title of an interesting article written by Dr. G. W. Scott Blair and published in British Plastics for March, 1939. The principal theme is that the plastics industry may profit by a scientific approach to the phenomena of deformation. On the other hand the author suggests that the scientists adopt more practical and less confused definitions of such commonly used terms as "plasticity." He takes exception to the definition of a plastic solid as given on Page 15 of The Rheology Leaflet No. 7. It appears that the word "plastic" means an entirely different thing to a physicist than to the so-called "plastics industry"

The Rheology Leaflet No. 11, November 1939, pg. 10 (215)

ANNUAL MEETING FOR 1939

SATURDAY MORNING PROGRAM

The first hour was devoted to consideration of the report of the Committee on Definitions and Nomenclature. A Summary of Criticisms had been prepared by Dr. Mooney, and these criticisms were discussed in considerable detail but with no general agreement. The final decision of the members present was that the report should be revised in some manner so that the criticisms could be embodied in it and that it should then be presented to the entire membership either by mail or through the Leaflet so that a ballot could be taken. This will be carried out during the first part of 1940

The Rheology Leaflet No. 11, November 1939, pg. 22 (227)

MISCELLANEOUS NOTES OF INTEREST TO RHEOLOGISTS

It is interesting to note in view of the discussion now going on in regard to the Rheological Definitions that the American Society for Testing Materials has a committee on Rheological Definitions. Mr. Robert Burns, Acting Chairman, reports favorable action of that committee on the new revision of the "Tentative Definitions of Terms Relating to Rheological Properties of Matter," with A. S. T. M. designation E34-37T. 'The definitions must now be submitted to the entire Society.

Transcribed by James J Griebler, April 2026

THE FLOW PROPERTIES OF BITUMINOUS MATERIALS (III)

At the June, 1940 meeting of the American Society for Testing Materials varied information of interest to bituminous technologists was announced.

Proposed revised tentative definitions of terms relating to rheological properties of matter were presented for publication as tabulated below:

Consistency:	That property of a body by virtue of which it tends to resist deformation.
Plasticity:	That property of a body by virtue of which it tends to retain its deformation after reduction of the deforming stress to its yield stress.
Elasticity:	That property of a body by virtue of which it tends to recover its original size and shape after deformation.
Liquid:	A substance which undergoes continuous deformation when subjected to shearing stress.
Simple Liquid:	A liquid in which the rate of shear is proportional to the shearing stress. The constant ratio of shearing stress to rate of shear of a simple liquid is the viscosity of the liquid.
Complex Liquid:	A liquid in which the rate of shear is not proportional to the shearing stress.
Solid:	A substance which undergoes permanent de formation only when subjected to shearing stress in excess of some finite value characteristic of the substance (yield stress).
Plastic Solid:	A substance which does not deform under a shearing stress until the stress attains the yield stress, when the solid deforms permanently.
Elastic Solid:	A substance in which, for all values of the shearing stress below the rupture stress (shear strength), the strain is fully determined by the stress regardless of

whether the stress is increasing or decreasing.

The Rheology Bulletin, Vol 12, No. 1 February 1941, pg. 6 (108)

A REPORT ON RHEOLOGICAL DEFINITIONS AND NOMENCLATURE

Foreword

In the following report by the Committee on Definitions and Nomenclature, an attempt is made to develop a logical and self-consistent system of classification of materials according to their rheological properties. The classification and the accompanying definitions have been developed in terms of idealized materials to which actual materials can be compared. In other words, ideal types of behavior and deformation are postulated which are susceptible of exact definition and classification. Actual materials can then be described in terms of the ideal types to which their behavior approximates. It should be borne in mind that an actual material with rheological properties coinciding exactly with those of a single idealized material is the exception rather than the rule. In applying to actual materials the quantitative definitions for idealized materials, great caution must be used.

It is recognized that the classification of a particular material may sometimes depend upon the method of measurement. This is particularly true as applied to the precision attained. The recognition of a measurable “yield stress,” for example, may depend upon the precision of the measurements and upon the duration of the test. These uncertainties in the classification of materials result from the fact that it is sometimes either impossible or unimportant to determine for actual materials some of the characteristics which can be theoretically distinguished.

In this system of classification and definitions, only shearing strains and stresses are of interest, and the relations involved are expressed with reference to simple shear. Many of the concepts defined here in terms of simple shear are closely related to those usually defined by engineers in terms of more complex types of deformation, Such as those arising in tension and compression tests. These de formations, while involving simultaneous shears in three dimensions and hence analytically complex, are often easier to produce experimentally than a simple shear. They therefore serve as a basis for many useful and practical concepts and definitions in the field of engineering. Where an analysis of the fundamental behavior of the material in deformation is required, these engineering tests may be reduced to a combination of shears and described in terms of the con cepts used in this report.

Transcribed by James J Griebler, April 2026

A general relation between strain and stress applies to any point within a material, and except in extraordinary circumstances, these values will vary at different points. The quantities involved in these relations applying to individual points within the material cannot be observed directly. Observable quantities such as efflux-rate, pres sure, etc., can be obtained by integrating the basic point relations and can then be compared with the values obtained experimentally. It is, however, always possible, at least formally, to devise experiments and methods of analysis by which the fundamental relationship can be completely determined from experimental data. Although all materials exhibit the inertia effects associated with acceleration of the mass elements within the material, these effects are ignored as being irrelevant in the classification and definitions given below. {t should be noted that turbulence and related hydro dynamical properties belong in this category.

Descriptive Definitions.

Plasticity is that property of a body by virtue of which it retains a fraction of its deformation after reduction of the deforming stress to zero.

Elasticity is that property of a body by virtue of which it recovers its original size and shape after deformation.

Thixotropy is that property of a body by virtue of which the ratio of shear stress to rate of deformation is temporarily reduced by previous deformation.

Strain hardening is that property of a body by virtue of which the stress required for permanent deformation increases with the extent of previous permanent deformation.

A **fluid** is a substance which undergoes continuous deformation when subjected to shear stress.

A **solid** is a substance capable of remaining in static equilibrium under a shear stress, less than some value called the **Yield Stress**.

Quantitative Definitions

A **simple shear strain** is a deformation in which the material at any point has moved parallel to a fixed line in a reference plane by an amount which is proportional to the distance of the point from the plane. Such a deformation is shown in Figure 1, the full lines showing the initial shape and the dotted lines the final shape. $\overline{OO}$ is the fixed line in the reference plane, P is the original position of a point in the body, and P' its position after deformation. The shear strain, ε, is given by the ratio of the displacement $\mathrm{d}s$ to the distance $\mathrm{d}x$ of the point P from $\overline{OO}$, or $\varepsilon = \mathrm{d}s/\mathrm{d}x$. This simple relation does not apply if there is rotation of the reference line $\overline{OO}$

Transcribed by James J Griebler, April 2026

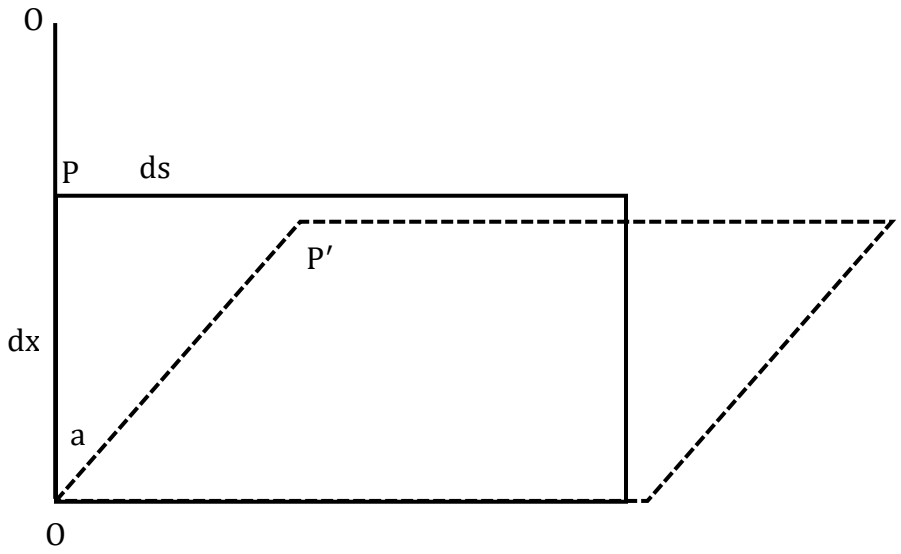


Figure 1

The **angle of shear** is the angle whose tangent is the shear strain. In Figure 1 the angle a is the angle of shear ($a = \tan^{-1} ds/dx$).

The **rate of shear** is the time rate of change of shear strain. In Figure 1 the rate of shear is given by:

$$\frac{dv}{dx} = \frac{d}{dt}\left(\frac{ds}{dx}\right)$$

A **complex shear** is the resultant of two or more simple shears. For simplicity, the definitions given in this report are stated in terms of simple shear, it being understood that the same definitions apply in the case of complex resulting from simple shears of the same type.

When the ratio of shear stress to rate of shear is constant, then the fluid may be called "simple." Under these conditions the constant may be called "coefficient of viscosity."

The **poise** is the unit of viscosity in the C.G.S. system.

The **fluidity** of a simple liquid is the reciprocal of the viscosity.

The **rhe** is the unit of fluidity in the C.G.S. system.

The **mobility μ** is the quantity defined by the equation:

$$\mu = \frac{d}{dF}\left(\frac{dv}{dx}\right)$$

The **yield stress F_0** of a solid is the shear stress which must be equalled or exceeded to produce permanent deformation

The **elastic limit in shear** in shear of a solid is the maximum shear deformation that exhibits complete elastic recovery

The **shear modulus of elasticity** (modulus of rigidity) of an elastic solid is the constant ratio of shear stress to shear strain for values of the shear strain below the elastic limit.

Classification

The classification of idealized types of materials is in terms of their behavior-under stress, or the relation between strain and stress. The stress at any point within a material is a function of the strain and its time derivative, or rate of strain.

The stress may depend to some extent on the prior strain or deformation history of the material. Such dependence upon the prior strain is regarded for convenience as a secondary characteristic, materials being classified primarily by the actual relation between stress and instantaneous strain or rate of strain without regard to differences in this relationship that would result from a different strain history.

Classes of Fluids.

(a) A simple fluid (Newtonian fluid) is one for which the rate of shear is proportional to the shear stress. Symbolically:

$$\frac{dv}{dr} = \frac{1}{\eta}F \quad (\text{See note.})$$

(b) A **complex fluid** (Non-Newtonian fluid) is one for which the rate of shear is a function of shear stress only, this function being other than that of direct proportionality. Symbolically:

$$\frac{dv}{dx} = f(F) > 0 \neq \frac{F}{\eta} \quad (\text{See Note})$$

Classes of Solids.

(a) An **elastic solid** is one for which the shear strain is a single valued monotonic function of the shear stress for all values of the latter below the shear strength. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_\infty \quad \text{(See note)}$$

Reduction of the shearing stress to zero results in the imme diate and complete disappearance of the strain.

(b) A **Hookean solid, (ideal elastic solid)** is an elastic solid for which the shear strain is directly proportional to the shear stress for all values of the latter below the shear strength. Symbolically:

$$\varepsilon = \frac{F}{G}, 0 < F < F_\infty \quad \text{(See Note)}$$

Reduction of the shearing stress to zero results in the immediate and complete disappearance of the strain.

(c) A **plastic solid** is one which obeys the law of an elastic solid for shear stress below the yield stress, and for shear stress in excess of this value is permanently deformed without further increase in the shear stress. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_0$$

$$\varepsilon > 0, \qquad \frac{dv}{dx} > 0, \qquad F_0 < F < F_\infty \ \text{(See note)}$$

Reduction of the shear stress to zero results in an immediate recovery of the elastic portion of the strain. The strain in excess of that experienced up to the yield point remains as permanent deformation.

(d) A **plastico-viscous solid** is one for which the deformation for shear stress below the yield stress is that of an elastic solid, and which for shear stress above that value deforms continuously at a rate of shear which is a function of the shear stress. Symbolically:

$$\varepsilon = f(F) > 0, \qquad \frac{d\varepsilon}{dF} > 0, \qquad 0 < F < F_0$$

$$\frac{dv}{dx} = f(F) > 0, \qquad F_0 < F < F_\infty \ \text{(See note)}$$

Reduction of the shear stress to a value below the yield stress results in the immediate reduction of the rate of shear to zero. The strain in excess of that experienced up to the yield point remains as permanent deformation

(e) A **complex solid** is one for which the shear stress required for deformation is a function of both the shear strain and rate of strain. Symbolically:

$$\varepsilon = f\left(F, \frac{dv}{dx}\right)$$

NOTE

The classification given above is without regard to prior strain. When the terms defined above are used without qualification it is implied that the stress-strain relation is independent of the strain previously experienced. Materials for which the stress-strain relation is dependent on prior strain should be described as **Thixotropic or Strain Hardening** in the senses in which these terms are defined above.

For purposes of easy comparison the following is recorded here

TENTATIVE DEFINITIONS OF TERMS RELATING TO RHEOLOGICAL PROPERTIES OF MATTER

A.S.T.M Designation. E 24-37 T

(Under the standardization of the Society, these definitions are under the jurisdiction of the A. S. T. M. Committee E-1 on Methods of Testing.)

Consistency—The resistance to deformation of material.

Quantitatively it is the ratio of the shearing stress to the rate of shear.

Plasticity—The ability of a body to retain a part or all of its deformation after reduction of the deforming stress to zero.

Elasticity—That property of a body by virtue of which it re covers its original size and shape after deformation.

Liquids—Substances which undergo continuous deformation when subjected to shearing stress.

Simple Liquids—Those substances in which the rate of shear is proportional to the shearing stress. The constant ratio of shearing Stress to rate of shear of a simple liquid is the viscosity of the liquid.

Complex Liquids—Those substances in which the rate of shear is not proportional to the shearing stress.

Solids—Substances which undergo permanent deformation only when subjected to shearing stress in excess of some finite value characteristic of the substance.

Plastic Solids—Those substances which do not deform under a shearing stress until the stress attains a critical value (yield stress) when the solids deform so readily or so quickly that, except for inertia effects, the stress never exceeds the yield stress.

Elastic Solids—Those substances in which, for all values of the shearing stress below the rupture stress (shear strength), the strain is fully determined by the stress regardless of whether the stress is increasing or decreasing. for

NOTE: Most solids exhibit combinations of properties. Metals, example, are mainly elastic-plastic, while some petroleum residues, waxes and similar materials are plastic-viscous. Some materials exhibit even more complex deformation, for example, rub ber compounds.

The Rheology Bulletin, Vol 12, No. 2 May 1941, pg. 12 (154)

CORRELATED ABSTRACTS

PLASTICS

A significant point about the article is the use of the word “plasticity” by one of the author’s standing in the plastics industry. As used in this article, “plasticity” is synomous with f low. This is the common, if not universal, usage in the industry. It is contrary to the usage recommended by the ASTM, to the definition in Webster’s International dictionary and to the classical meaning in the field of physics and engineering. Examples of this same changing use of the word can be found in most of the technical articles dealing with plastics and is most common in the advertising and sales literature. The word “‘plastic’”’ is often used to characterize the solid material remaining after application of a solution of an oil, resin or organic high polymer to solid surfaces or as an impregnation into cloth. There is nothing one can do about this trend but it may be that in the future, a new word will be required to describe the amount of retained deformation.

The Rheology Bulletin, Vol 12, No. 3 August 1941, pg. 12 (182)

BRITISH RHEOLOGIST’S CLUB

American Co-operation

We are glad to note an appreciative paragraph about the club in the November (1940) number of the “Rheology Leaflet” of the American Society of Rheology. Co-operation with our American col leagues is most welcome and we hope to keep in touch with them especially when the very difficult and controversial questions of nomenclature and definition are being discussed.

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 12, No. 3 August 1941, pg. 19 (189)

The following letter is placed in the Plastics section for it concerns the review of the article 'Plasticity of Molding Compounds" which appeared on page 12 of Rheology Bulletin No. 2.
Dear Mr. Wakefield,

Many thanks for permitting me to comment on your remarks in the RHEOLOGY BULLETIN for May.

I shall have to plead guilty to your charge that I used the word "plasticity" improperly in my discussion of the multicapillary die. The offense is particularly heinous when it is recalled that the A.S.T.M. definition of plasticity, which you use to confound me, was written by the accused himself, in collaboration with Mr. R. L. Peek. Further more, the particular A.S.T.M. committee meeting which officially adopted the definition was presided over by none other than myself. Perhaps the cruelest cut of all was your reference to Webster. Your words carried the unmistakable implication that Nathaniel was a better plastics man than I. Well, perhaps he was; I don't see how he could be much worse.

Seriously, I cannot agree with your statement that nothing can be done about misuse of words in the plastics industry. We should retain the classical definition of 'plasticity' and find another term to describe the results of empirical flow tests. Since they are designed primarily to determine "moldability" by integrating temperatures, pressures and other factors related to commercial molding practice, perhaps that term could be used notwithstanding that the correlation between actual moldability and results of our various flow tests is not too good.

You were misinformed regarding my conceding the difficulty of obtaining reproducible data with the multicapillary apparatus. Our A.S.T.M. round-robin tests demonstrated very good reproducibility within the individual cooperating laboratories. Reproducibility between laboratories was poor. Similarly unsuccessful was our attempt to obtain concordant results between several laboratories using the Rossi-Peakes flow tester. The general situation was summed up by Mr. L. M. Debing of Monsanto Chemical in a recent letter to all the committee members when he wrote "... we are getting nowhere fast on the problem of setting up a standard test method for determining flow..."'. While this is the bare truth as of the present, the future, to me at least, looks very promising. It can be stated beyond any reasonable doubt that the average plastics technician has learned in the last few years more about flow phenomena than the previous entire stock of knowledge from the beginning of the industry. Out of this will come some progress.

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 12, No. 4 November 1941, pg. 36 (206)

Paper No. 17. Fundamental Principles for the Nomenclature of Deformation

The Viscosity Committee of the Academy of Sciences has this year printed a 44-page bulletin entitled, 'Fundamental Principles for the Nomenclature of Deformation," by J. M. Burgers, R. N. J. Saal and C. B. Biezeno. It is published as Section One of Volume 18 of the Transaction of the Academy of Science. Due to the present conditions, the publication is in Dutch.

The authors distinguish five different yield values, viz., elastic deformation, creep, viscous flow, accelerating flow and rupture. They introduce several terms which are new such as aliminary flow and deciatoric stresses, firmo-viscosity, dynamic and static hysteresis. Liquids are divided into

1. hardening liquids
2. perfect liquids
3. true liquids
    a. Newtonian
    b. Non-Newtonian
4. Excessive liquids

Solids show a purely elastic deformation at sufficiently small shearing stresses. The total shear γ is the sum of the elastic shear γ_e and the permanent shear γ_p

The Rheology Bulletin, Vol 13, No. 4 November 1942, pg. 40 (342)

LETTER TO THE EDITOR

Sir: In the September issue of the Rheology Bulletin comments on the British Rheologists' Club's Classification of Rheological Properties are invited.

I suggest that we cooperate with our British colleagues in establishing an international standard system of notation.

For example, our British friends represent strain by σ, whereas we use ε. To us, σ always means stress but S is used for this in England. A unified system of symbols would prevent much confusion.

If such a project were successful, it could, perhaps, be extended to include standard definitions

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 14, No. 4 November 1943, pg. 4 (450)

Minutes of the 1943 Annual Meeting of the Society of Rheology

The need for getting theological material to the members, and ways and means of improving the Bulletin, particularly as regards abstracts, were discussed at length by President Mark, Professor Bingham and Mr. Wakefield, Professor Bingham discussed a system of numbering abstracts and the great desirability of having bibliofilms of scientific papers. It was the sense of the meeting that Dr. Bingham prepare an explanatory statement along these lines for the Bulletin. Or. Huggins Suggested the adding of Chemical Abstracts numbers to each of our abstracts. Dr. Hunter spoke in favor of the present method of preparing journals, an attempt to standardize nomenclature, a place to express o pinions and also space for detailed announcements on definite ways in which the Bulletin might be expanded. Dr. Fischer agreed to work along these lines with Dr. Bingham. Dr. Hunter

moved, and it was voted that all articles of rheological interest should at least be listed in the Bulletin as a matter of information for the membership

The Rheology Bulletin, Vol 14, No. 4 November 1943, pg. 44 (489)

RHEOLOGY PROGRESS ABSTRACTS

THEORY

VISCOSITY, MOLECULAR FORM, AND SOLVATION.--H. Fromm, Kolloid-Z., 102, 86-90 (1943)

This paper constitutes a proposal for a standardized and rationalized (German) nomenclature in the rheology of colloidal solutions. It is argued that the limiting value of (η_{sp}) is not a measure of viscosity, but rather of the rate of increase of viscosity with concentration. The word “Zahigung", or "viscosation", rather than the familiar "Zahigkeit", is suggested as the basic part of any expression relating to such a quantity.

The Rheology Bulletin, Vol 15, No. 1 February 1944, pg. 40 (543)

LETTERS TO THE EDITOR

Sir: I have today received a parcel of "Rheology Bulletins" which are not only full of interest to me but will also be most valuable to our Club.

I know that our Committee will be especially interested in the correspondence on Nomenclature but I thought that I should explain that these bulletins have only just reached us in case you might feel that our Committee's long delay in replying to the suggestions was due to some lack of interest. I can assure you that we are all profoundly

interested in your activities and we look forward to the end of the war when our co-operation need not be so long delayed. Again many thanks for sending the Bulletins.

Yours very truly,
GEORGE W. SCOTT BLAIR
Hon. Secretary
The British Rheologist's Club

The Rheology Bulletin, Vol 15, No. 3 August 1944, pg. 14 (625)

ANNUAL MEETING OF THE BRITISH RHEOLOGISTS' CLUB at Oxford, July 6-9, 1944

On Friday evening Dr. S. Whitehead lead an informal discussion on "Relaxation". On Saturday Morning, Dr. Hugh O'Neill of the L.M.S. Railway Co., introduced a discussion on "Rheological Nomenclature of Symbols, Metallurgical and Non-Metallurgical" in which a number of metallurgists join ed in debate with other rheologists on the vexed question of nomenclature.

The Rheology Bulletin, Vol 16, No. 3 August 1945, pg. 53 (732)

RHEOLOGICAL SYSTEMATICS

MARKUS REINER (JERUSALEM)

1.

For many years in America, Bingham (1929, 1930) and his school have made various proposals for a consistent nomenclature in rheology as a prerequisite for a rheological systematics. These proposals were partly accepted by the American Society of Testing Materials. Later, the Amsterdam Academy of Sciences seconded Bingham in its two Reports on Viscosity and Plasticity. In recent years these efforts were followed by British rheologists, notably Scott Blair and collaborators (1940, 1942). The present writer was active in this field some years ago (1931, 1932), basing the classification not so much on properties as on ideal materials having certain properties which can be treated mathematically and serve as ““pictures”’ of the real materials. It was proposed these ideal materials be named after the scientists who first postulated their existence. This method of designation indicates the character of the different “bodies” as mathematical idealizations. Accordingly, the term “Euclid solid” was used for the (nonexisting) “rigid’’ body and “Pascalian liquid’ for the (equally non existing) ““ideal’’ liquid. These two bodies, in which no rheological material constants appear, belong to mechanics and are outside the range of rheology. Rheology is entered with the Hooke solid from one side and the Newtonian liquid from the other. The former is a perfectly elastic solid, the latter a “simple” viscous liquid. If we add to these the St. Venant body,* a solid which has a yield point below which

its deformation is elastic, but above which it flows plastically at constant stress, it will be shown in the present paper that all other ‘rheological bodies’’ can be considered as built up through a combination of these three, a statement subject to qualification in regard to “dilatancy,” which is discussed in section 6.

2.

However, we must first take account of a basic phenomenon affecting any classification of rheological properties. It is a fact that, *under hydrostatic pressure, all materials—solids and liquids alike—behave in the same manner.* When uniformly compressed from all sides they behave in the manner of perfectly elastic solids, provided, of course, that the test piece is truly homogeneous. As Lord Kelvin pointed out, “‘by compression between the dies used in coining the density of gold may be raised from 19.258 to 19.367.” But such “permanent set’’ is due to the porosity of the coin and an indication of its structure; it is not inherent to the material forming the structure. If the pressure is truly hydro static, which requires that it was acting within the pores also, there is no pemanent set and no plastic deformation in a solid, no flow in a liquid, only an entirely recoverable elastic volumetric strain, the result of volume elasticity. In other words, materials do not differ rheologically in cubical dilatation (positive or negative), *all rheological differences are revealed under shear only* (or, more generally, distortion).

3.

Considering shear only, Scott Blair (1946) has compiled a list of eight “essential rheological properties.’ For reasons which will become clear later on, his list here is slightly modified and extended as follows:

I.	Viscosity (Fluidity)	VI.	Structural Viscosity
II.	Elasticity	VII.	Strain Hardening
III.	Firmoviscosity	VIII.	Strength
IV.	Elasticoviscosity	IX.	Thixotropy
V.	Plasticity	X.	Dilatancy

In a rheological table, Scott Blair listed a number of industrial materials with gradings of zero to five for each of the rheological properties in a semiquantitative way, as: (0) nonexistent, (1) very low, (2) low, (3) moderate, (4) fairly high, (5) very high. It then became evident that the materials could be grouped in classes in accordance with the more or less marked presence of the different rheological properties. For instance, milk, honey, and oil together form (with water and others) the class of true fluids in which the “‘species” milk, honey, etc., are distinguished by the different values of their viscosities. If this procedure is followed up and the rheological table extended to cover all materials for which rheological data are available and brought up to date as rheological research

advances, one fact should not be overlooked. It can be considered as a rheological axiom that every real material possesses all rheological properties in varying degrees in accordance with the conditions of the case. In thin layers, a liquid such as water can exhibit rigidity; in the shape of a thin rod, a solid like lead will show viscous flow. Only the ideal materials are limited in their proper ties—because this is how they are defined. They can therefore serve for the representation of real materials in different degrees of approximation. In a first ap proximation, steel can be considered as a Hooke body, in a second approximation as a St. Venant body, and in a third as a Bingham body (about which see later), etc.

4.

From Scott Blair's classification of rheological properties and his table of real materials we therefore proceed to the classification of the ideal materials. This can be done in two stages: (7) qualitatively, by a "structural formula"; (2) quantitatively, by a "rheological equation." The qualitative classification starts from the three simplest rheological bodies, the Hooke solid, the Newtonian liquid, and the St. Venant plastic, which we denote symbolically by H, N, and StV, respectively. To build up materials with more complicated properties, we employ a well-known device as apparently first used by Poynting and Thom son and later developed by Burgers and the writer (1943). We represent the Hooke solid by a perfectly elastic spring, the Newtonian liquid by a dashpot, and the St. Venant plastic by a weight resting on a table with "solid" friction between both (see Fig. 1). The only rheological property of the H spring is property IT, its elasticity measured by the modulus of rigidity, ; of the N dashpot, property I, its viscosity, measured by the coefficient of viscosity, 7; and of the StV weight, property V, its plastic resistance measured by the yield stress, 3 (significantly named by Bingham, the "'friction constant').

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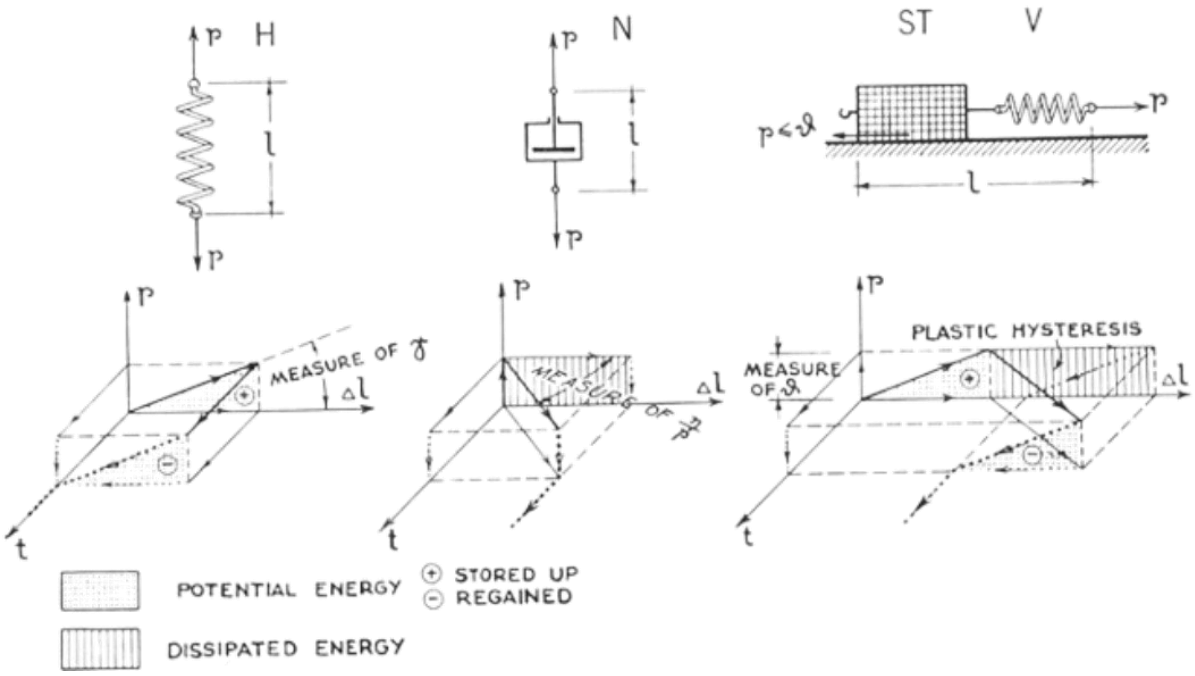


Fig. 1. – Structural elements

As in Broome's graphs, the dotted lines represent behavior on removal of load. In working out the diagrams, the inertia of the mechanism has been neglected and the elastic strain assumed as following instantaneously the applied force.

If we now combine the H and N elements, as shown in Figure 2, it will be seen that this represents a model for a solid which is elastic, but exhibits: (a) elastic fore- and aftereffects, i.e., a time lag in the appearance and disappearance of elastic strains; (b) elastic hysteresis and therefore also viscous damping of oscillations. These "imperfections of elasticity'' have been included by Jeffreys (1929) under the term "'firmoviscosity," which is listed above (as in Scott Blair's list, albeit under a different designation) as property III. The corresponding ideal material has been called Kelvin solid. Similarly, Figure 3 shows a model (a) for a liquid which exhibits "instantaneous" (as Bingham calls it) elasticity, or (b) for a ""solid'"' the elastic stresses of which relax and which exhibits creep. These properties have been included by Jeffreys under the term "elasticoviscosity," listed above as property [TV (not mentioned separately by Scott Blair in his list). The corresponding ideal material has been called a Maxwell liquid.

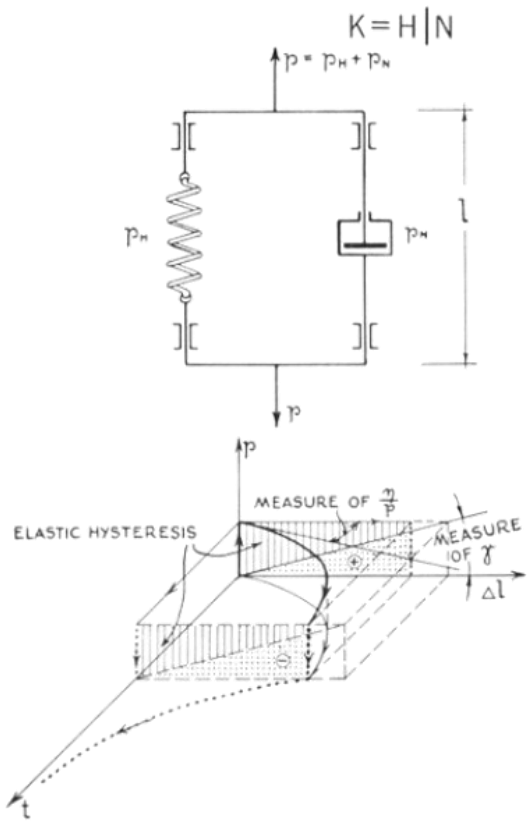


Fig. 2. – Model for a Kelvin solid.

The mechanisms in the figures show extensions and pulls, but one should not forget that these stand for shearing strains and shearing stresses, respectively. Also, the stress-strain relations have been pictured as linear; but the models can be applied in a general manner for nonlinear relations as well

In Figure 2 the spring and the dashpot are combined in parallel arrangement; in Figure 3 they are combined in series. Indicating “parallel coupling” by a vertical dash (|) and “coupling in series” by a horizontal dash (—), we can write the following structural formulas, using obvious abbreviations:

$$\mathrm{M} = \mathrm{H} - \mathrm{N}; \quad \mathrm{K} = \mathrm{H} \mid \mathrm{N} \tag{1}$$

In the working of these models no other rheological properties enter than the elasticity of the spring and the viscosity of the dashpot. Therefore, the proper ties ‘elastic foreeffect,’’ ‘elastic aftereffect,”’ “elastic hysteresis,” vie GOS “instantaneous elasticity,’ ‘relaxation of stresses,” and “creep” can all be expressed by the two fundamental properties of elasticity and viscosity. A material involving the StV body is the one with structural formula (2)

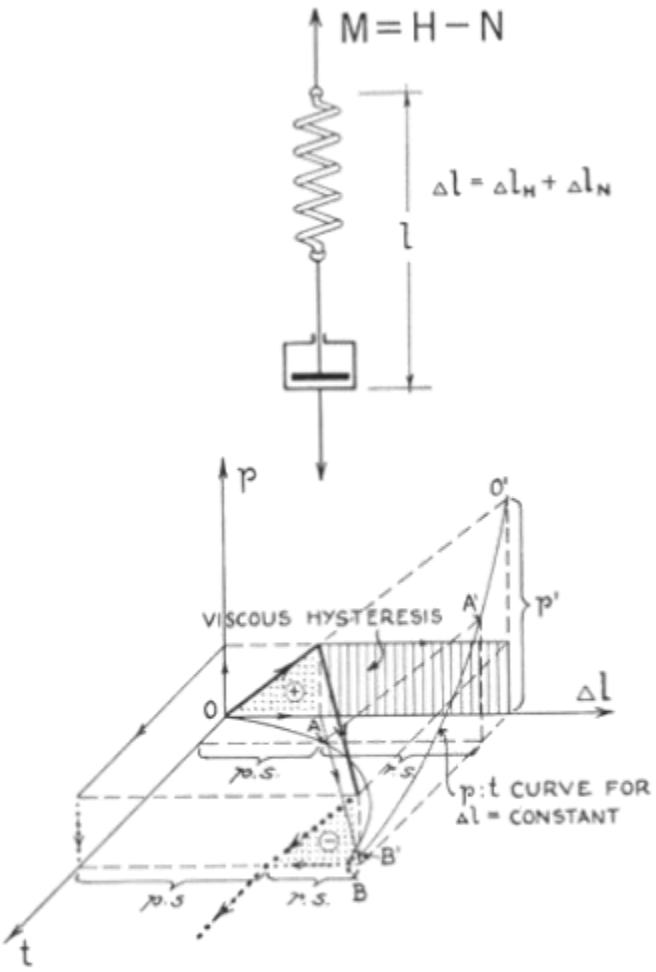


Fig. 3.—Model for a Maxwell liquid.

OAB = elastic recovery for elongation Δl
O′A′B′ = relaxation curve for stress p'
p.s. = permanent strain
r.s. = recovered strain

$$B = N - StV \tag{2}$$

Its model is shown in Figure 4. It will be recognized that this is a Bingham body

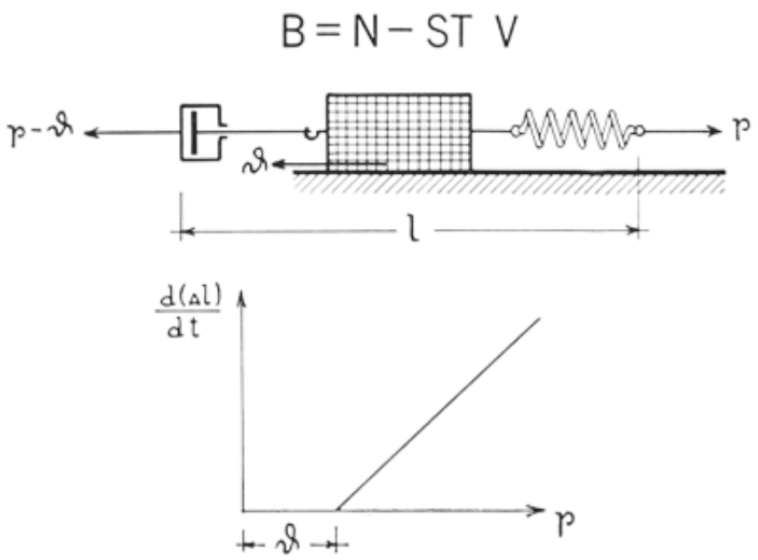


Fig. 4.—Model for a Bingham body.

It should be noted that, in contradistinction to the spring and the dashpot, the StV element is not symmetrical in respect to its ends. When coupling a StV element in series, it is therefore necessary to indicate whether the connection is to the "weight" or to the "spring."" When using the symbol StV we shall assume that "St" stands for the weight and "V"' for the spring.

A body in which three elements are combined is given by structural formula

$$\text{Schw} = \text{H} - \text{N} - \text{StV} = \text{M} - \text{StV} = \text{H} - \text{B} \qquad (3)$$

It represents the 0.5% gelatin sol with which Schwedoff experimented in 1890 in a coaxial cylinder instrument. He exerted a torque upon the internal cylinder which was rotated against the elastic resistance of the gelatin by a certain angle. To maintain a constant deflection, the torque had gradually to be reduced as in a Maxwell liquid; but Schwedoff found that the sol, in contrast to a Maxwell liquid, apparently maintained a small residual deformation permanently. This requires the connection, in series, of a StV element to the M body.

It is not necessary to work out models for all the different materials. *For the purpose of the qualitative classification of a material, it is sufficient to write the structural formula from the information provided by the tests to which the material has been subjected*.

5.

As an example of the classification of a given material qualitatively, let us treat the investigations of Schofield and Scott Blair (1932, 1933a, 1933b, 1937) on flour dough. The authors first stretched out long cylindrical pieces of dough for a measured time, at the end of which the cylinders were cut loose. It was found that part of the extension was

recovered, but part was permanent (Fig. 1 in Schofield and Scott Blair, part I). This showed that while flour dough (FD), because of the permanent extension of the cylinders, is not a Hooke body, it has a Hooke component; and in a first approximation we write FD = H — X, where X represents another element or elements to be deduced from further experiments. In the next series of experiments, it was found that the elastic recovery decreased with time under stress in accordance with an exponential law (Fig. 4 in the same paper). Comparison with our Figure 3 here shows that we must write N for X, or that in a second approximation FD = H —N = M. This was confirmed by further experiments, in which the decay of internal stresses in pieces of dough which had been stretched and held stretched, was followed (Fig. 6 in Schofield and Scott Blair, part 1). The shape of the curve con formed to our Figure 3 for Δl = const. The second report of these authors describes observations in which the rate of extension of cylinders of dough, hung vertically and allowed to elongate under the action of gravity, is related to the stress. It was found that, while the rate of extension generally decreased with decreasing stress, there was a finite stress at which the rate of extension vanished, in fact, a yield point. This shows that a St. Venant element must further be added; and in a third approximation FD = H — N — StV = M —StV = H —B = Schw. It had, however, been noticed that a "considerable time often elapses between the release of the stress and the cessation of contraction." This indicated an elastic aftereffect, to the investigation of which the third report is devoted. Foran elastic aftereffect, a K body must be coupled on. As the structural formula for FD contains a StV element, the question arises as to which end of the StV element the K body is to be connected. (Compare the paragraph in small type on page 58.) Experiments (Fig. 2 in report III) showed that the elastic aftereffect makes its appearance in strains *below the yield point*. 'This means that the K body must be connected at the spring end. It could be formed there by parallel coupling of the spring with an N element. The same figure, however, illustrates that, beside the "lagging" elastic recovery, there is also ""instantaneous" recovery, *i.e.*, the spring of the StV element is not impaired in its working and the K body is therefore connected to it "'in series." We accordingly obtain, in a fourth approximation, formula (4).

$$\mathrm{SchScB} = \mathrm{H} - \mathrm{N} - \mathrm{StV} - \mathrm{K} = \mathrm{M} - \mathrm{StV} - \mathrm{K} = \mathrm{H} - \mathrm{B} - \mathrm{K} - \mathrm{Schw} - \mathrm{K} \qquad (4)$$

Figure 4 in report III confirms the formula for strains above the yield point. The formula embodies all observations made by Schofield and Scott Blair, but in addition also those of Wolarowitsch and Samarina (1935). Scott Blair, on page 27 of his book (1938), remarks that to apply ""Bingham's principles to. . . flour doughs. . .is contrary to the author's [Scott Blair's] experience." But in view of the presence of a B element in SchScB = H — B — K, it can be understood that in *steady flow* flour dough must behave as a Bingham body.

6.

This example shows that a material with the most complicated rheological behavior can be adequately represented by a structural formula. The example also shows that, in the relations between shearing stress, shear, and their rates, which give rise to the *most complicated properties*, these *can be reduced qualitatively to the three fundamental rheological properties*, I, II, and V, listed in section 3. Of these, *elasticity* is that property by virtue of which the strain work per formed by the external forces in a deformation is stored up as potential energy and the deformation (which in this case is called "strain'') is therefore *recoverable*.[1]

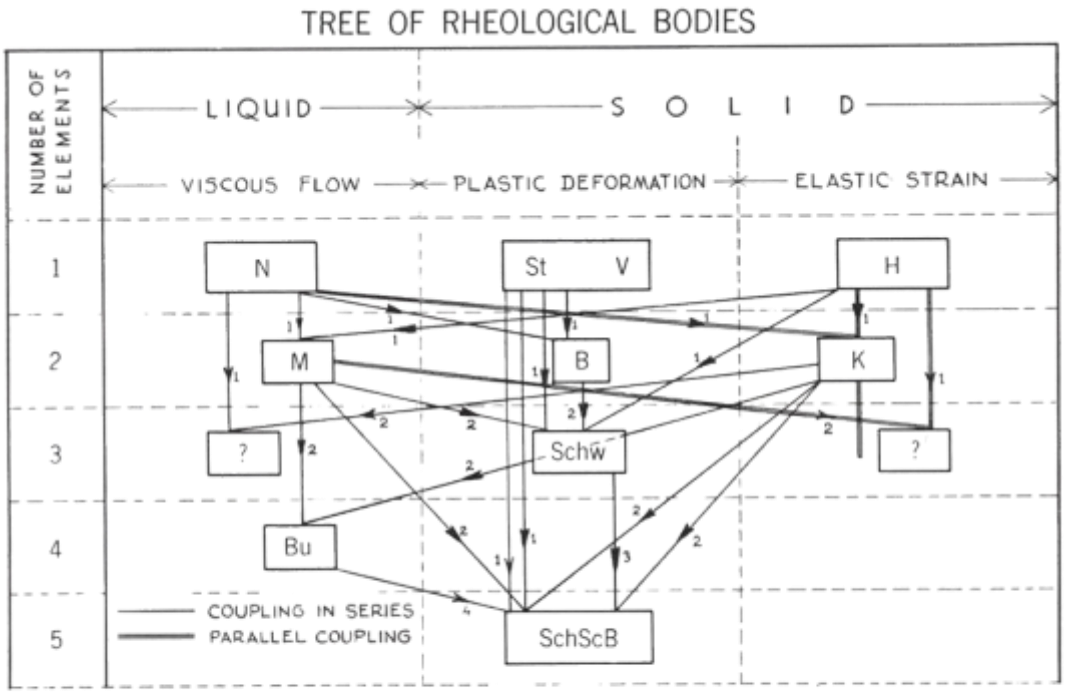


Fig. 5.—Tree of rheological bodies.

N = Newton	StV = St. Venant	H = Hooke
M = Maxwell	B = Bingham	K = Kelvin
Schw = Schwedoff	Bu = Burgers	SchScB = Schofield–Scott Blair

The figure beside the arrow indicates the number of elements.

When the strain work is dissipated (converted into heat) and the deformation (which in this case is called ''flow'') is nonrecoverable, we speak of *viscosity* if the resistance to flow depends upon the rate of deformation (viscous flow), and of *i* if it is independent of the rate of deformation (plastic f low). When the elastic energy is gradually dissipated through

[1] Compare *Nature,* **149**, 702 (1942)

lapse of time, we speak of *relaxation*, which, however, as we have shown, is not an independent property

There is, however, a fourth fundamental property which cannot be derived from the above three: property X, *dilatancy*. Dilatancy refers, not to a relation between shear and shearing stress, but to a change of volume caused by shear. Wherever dilatancy comes into play, another element must be added in the structural formula. Disregarding dilatancy for the present and consulting Figure 5,[2] we see how a system of rheological bodies can be built up in *accordance with the number of fundamental structural elements making up the body*. We see that, while three ideal bodies (M, B, K), each one made up of two elements are known, rheology has so far made use of one only, consisting of three, four, and five elements. Many others can be imagined, and they all can be arranged in a hierarchy of rheological bodies, starting with the three *groups* of solids, " "plastics," and "'liquids,"' each one divided into the *families* of Newtonian, Maxwellian, etc., liquids. St. Venant, Bingham, Schwedoff, etc., plastics, and Hooke, Kelvin, etc., solids, respectively

7.

We have so far dealt with rheological properties I, I, III, IV, V, and X, of which I, II, V, and X are fundamental, while III and IV are derived and can be expressed through I and II. We now turn to properties VI and VII. These express *quantitative* relations in respect to fundamental properties. For instance, VII implies an increase of yield stress with strain; but de Waele has drawn attention to the existence of strain *softening*. Property VI refers to a decrease of viscosity with increasing rate of strain, while theoretically, an increase of viscosity is also possible (the latter has been described, wrongly it seems to me, as a case of dilatancy). Schofield and Scott Blair have discovered in flour dough an increase of viscosity with strain (not *rate* of strain) similar to the ordinary strain hardening. Generally these properties show that η, γ, and ϑ must not necessarily be constants, but are to be considered as parameters or coefficients which themselves may depend upon stress and strain and their rates. If, for instance, η is not a constant, but decreases with increasing stress, such a material shows the property of what Ostwald called "structural viscosity"; it belongs to the class of what Green called 'pseudoplastics' or what Reiner (1929) called 'non-Newtonian liquids.' The last term has been widely accepted; but, being purely negative, it was originally meant to be provisional only. Considering that a Maxwell liquid also is *not* Newtonian, a better scheme would seem to be to call all liquids whose only fundamental property is viscosity *Newtonian*, whether η is a constant or not. If it is desired to distinguish the latter from the former, the latter can be called *generalized* Newtonian liquids and the former *simple* Newtonian liquids. This principle of designation can generally be applied, and we would have to divide each "family" into different ''classes'" of Newtonian liquids, Maxwell liquids, Hooke solids, etc., etc. However, here the structural formula breaks down, and we must proceed from qualitative to quantitative classification by means of rheological equations. A rheological equation is, according to Hersey (1932),

[2] Compare Treloar's suggestions in *Nature* (1942)

the relation connecting shearing stress with shear (and their derivatives in respect to time). Denoting shearing (tangential) stress by p_t, and shearing strain by e_t, and using for the time derivatives Newton's dot, the following are the fundamental rheological equations (in which t is dropped for simplifying the writing):

$$\begin{cases} \text{Newtonian (viscous) liquid,} & \mathrm{p} = \eta\dot{e} \\ \text{Hooke (elastic) solid} & \mathrm{p} = \gamma \mathrm{e} \\ \text{St. Venant (plastic) solid} & \mathrm{p} = \vartheta \end{cases} \qquad (5)$$

The rheological equations of materials built up of two elements are accordingly

$$\begin{cases} \text{Maxwell liquid,} & \dot{e} = p/\eta + \dot{p}/\gamma_l \\ \text{Kelvin Solid} & \mathrm{p} = \gamma \mathrm{e} + \eta_s \dot{e} \\ \text{Bingham body} & \mathrm{p} - \vartheta = \eta_{\mathrm{pl}} \dot{e} \end{cases} \qquad (6)$$

No essentially new constant appears in any of these equations, but indices must be used in order to avoid confusion. We accordingly write η_s; for what Lord Kelvin (1875) first called ‘“‘solid” viscosity, γ_l, for the elastic modulus of a liquid in accordance with Maxwell, following green, η_{pl} for the plastic viscosity, which Bingham calls “‘stiffness.”

The rheological equation of a material built up of three elements is the following one of the Schwedoff body:

$$\dot{e} = (p - \vartheta)/\eta_{pl} + \dot{p}/\gamma_l \qquad (7)^3$$

It should be noted that the first two equations of (6), being of the form:

$$dy/dx + My = N \qquad (8)$$

Permit of solutions of the form:

$$y = e^{-\int M dx}\left(\int N e^{\int M dx} + C\right) \qquad (9)$$

where e is the basis of natural logarithms. We accordingly have for the Maxwell liquid:

$$p = e^{-\gamma/\eta \cdot t}\left(p_0 + \gamma \int_0^t \dot{e}\, e^{\gamma/\eta \cdot t} dt\right) \qquad (10)$$

and for the Kelvin solid:

[3] Compare Reiner (1932), where the notation and the terminology only are different.

$$e = e^{-\gamma/\eta \cdot t}\left(e_0 + \gamma \int_0^t p\, e^{\gamma/\eta \cdot t} dt\right) \tag{11}$$

8.

We shall now show how, if the structural formula is known, the rheological equation can be derived, and shall take our example again from the flour dough experiments of the Schofield and Scott Blair. These workers express the rheological equation of their material, translated into our notation, as follows:

$$e = p/\eta + \dot{p}/\gamma - \dot{\alpha} \tag{12}$$

where $\dot{\alpha}$ represents the elastic aftereffect. We note that they do not take into account that flour dough possesses a yield point. Their rheological equation, therefore, represents only part of their experimental findings; as a matter of fact it corresponds to structural formula (13):

$$\mathrm{Bu} = \mathrm{M} - \mathrm{K} \tag{13}$$

which can be derived from the second form of formula (4) by omitting the StV element. Its model can easily be pictured. It appears in Burger's article, al ready quoted. It was used by Lee and Markwick to explain the rheological behavior of bituminous materials. To the rate of strain $\dot{e}$ of the complete body, the strains e_M, and e_K of the components M and K contribute, so that:

$$\dot{e} = \dot{e}_M + \dot{e}_K = \frac{p}{\eta} + \frac{\dot{p}}{\gamma_l} + \frac{d}{dt}\left[e^{-\gamma/\eta_s \cdot t}\left(e_0 + \gamma \int_0^t p\, e^{\gamma/\eta_s \cdot t} dt\right)\right] \tag{14}$$

After carrying out the differentiation this gives as the rheological equation of the to) body:

$$\dot{e} = p\frac{\eta + \eta_s}{\eta\eta_s} + \dot{p}/\gamma_l - \frac{\gamma}{\eta_s} e^{-\gamma/\eta_s \cdot t}\left(e_0 + \frac{1}{\eta_s}\int_0^t p\, e^{\gamma/\eta_s \cdot t} dt\right) \tag{15}$$

If we compare this expression with (12), we see that our method has carried us one step further, revealing at the same time that, in the summary expression $\dot{\alpha}$, two material constants are contained

9.

To give an example of the last division of the rheological families into rheological classes, it must be remembered that the postulation of nonlinear laws is a comparatively recent development. The existence of generalized bodies was realized by Maxwell when, in his quoted paper, he said "'it is possible that in some bodies τ (the time of relaxation) may

be (not a constant, but) a function of p."' It was, however, not until the existence of structural viscosity was discovered that nonlinear laws were proposed.[4] Considering these, the simple Newtonian liquid for which the viscosity η is a constant or $\varphi = 1/\eta = c$, forms a class within the family of Newtonian liquids together with the Weissenberg liquid for which

$$\varphi = \varphi_0 + cp^2 \tag{16}$$

the Kraemer-Williamson liquid for which

$$\varphi = \frac{\varphi_0 + \varphi_\infty p^2/c}{1 + p^2/c} \tag{17}$$

and a liquid proposed by Reiner and Rievlin (1927) for which:

$$\varphi = \varphi_\infty - (\varphi_\infty - \varphi_0)e^{-p^2/c} \tag{18}$$

In the "tree" of rheological bodies we shall ultimately have many such names under each family designating the classes. However, as has been stated, this division is not yet advanced very far; there are very few names which could be entered for the classes now; and this is where the classification of Scott Blair and Caffyn is very useful

10.

As Reiner (1933) has shown, every monotonously increasing function, $y = f(x)$, which can be extrapolated toward the origin ($x = 0, y = 0$) can be approximated by a power function, $y = \alpha x^\beta$ where α and β are two parameters. This analytical fact is at the root of many "'power laws" which make their appearance in different branches of physics and physical chemistry. Reiner has discussed the objections by which these "'laws'' must be regarded as convenient interpolation formulas only; but as such they are useful. In our case, until we have rheological equations for the many classes of nonlinear bodies at our disposal, we can, with Scott Blair and Caffyn, use power formulas to fill in the blanks in our rheological tree. Granted that all rheological behavior is dependent upon values of the four variables $p, \dot{p}, e$, and $\dot{e}$, we can define a coefficient (α_0) by:

$$\alpha_0 = p^{\alpha_1}\dot{p}^{\alpha_2}e^{\alpha_3}\dot{e}^{\alpha_4} \tag{19}$$

where $\alpha_0, \alpha_1, \alpha_2, \alpha_3$, and α_4 are characteristic for the material under consideration. For instance, for the simple Newtonian liquid $\alpha_0 = \eta,\ \alpha_1 = 1, \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = -1$, while for the generalized Newtonian liquid, $\alpha_1 > 1$; for the simple Hooke solid, $\alpha_0 = \gamma, \alpha_1 = 1, \alpha_2 = 0, \alpha_3 = -1, \alpha_4 = 0$, while for the generalized Hooke solid $\alpha_1 > 1$.

[4] Hodgkinson postulated for cast iron a generalized Hooke solid for which $p = \gamma e - ce^2$ but this law fell into oblivion.

If it is desired to include true plastics, p must be understood as the excess of the actual stress over the yield stress. Scott Blair and Caffyn write their equation in the form (using our notation):

$$\psi = p^{\beta} e^{-1} t^{k} \tag{20}$$

but this equation is not essentially different from (19). By differentiating both parts of Equation (20) in respect to time, we introduce $\dot{p}$ and $\dot{e}$, and if we eliminate time t from this equation and (20), we get an equation in which $p, \dot{p}, e$, and $\dot{e}$,and certain coefficients appear in which the t does not appear explicitly and which again can be approximated by (19).

Rheological Equations

Number of Elements	Liquid	Plastic Solid	Elastic Solid
1	**N** $\dot{e} = p/\eta$ ∞	**StV** $p - \vartheta = 0$ 0	**H** $p = \gamma e$ 0
2	**M** $\dot{e} = p/\eta + \dot{p}/\gamma_l$ γ_l	**B** $\dot{e} = (p - \vartheta)/\eta_{pl}$ η_{pl} ∞	**K** $p = \gamma e + \eta_s \dot{e}$ η_s
3	$\dot{e} = p\frac{\eta + \eta_s}{\eta\eta_s} - \frac{\gamma}{\eta_s} e^{-\gamma/\eta_s \cdot t}$ $\left(e_0 + \frac{1}{\eta_s}\int_0^t p\, e^{\gamma/\eta_s \cdot t} dt\right)$	**Schw** $\dot{e} = \frac{p - \vartheta}{\eta_{pl}} + \frac{\dot{p}}{\gamma_l}$ γ_l	$p = \gamma e + e^{-\gamma/\eta \cdot t}\left(p_0 + \gamma_l \int_0^t \dot{e}\, e^{\gamma/\eta \cdot t} dt\right)$
4	**Bu** $\dot{e} = p\frac{\eta + \eta_s}{\eta\eta_s} + \frac{\dot{p}}{\gamma_l} - \frac{\gamma}{\eta_s} e^{-/\eta_s \cdot t}$ $\left(e_0 + \frac{1}{\eta_s}\int_0^t p\, e^{\gamma/\eta_s \cdot t} dt\right)$		$p = \gamma e + \eta_s \dot{e} + e^{-\gamma/\eta \cdot t}$ $\left(e_0 + \frac{1}{\eta_s}\int_0^t p\, e^{\gamma/\eta \cdot t} dt\right)$
5		**SchScB** $\dot{e} = (p - \vartheta)\frac{\eta_{pl} + \eta_s}{\eta_{pl}/\eta_s} + \frac{\dot{p}}{\gamma_l} - \frac{\gamma}{\eta_s} e^{-\gamma/\eta_s \cdot t}$ $\left(e_0 + \frac{1}{\eta_s}\int_0^t (p - \vartheta)\, e^{\gamma/\eta_s \cdot t} dt\right)$	

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11.

For the rest, the reader is referred to the report of Scott Blair and Caffyn. As the whole subject is still in a very preliminary stage, names for different classes have not been entered in Figure 5. Nor have their rheological equations been listed in the accompanying set of rheological equations of the families only, where the rheological coefficients can either be constants for the "simple" bodies, or depend upon stress and strain and other coefficients. When the latter is the case, *i.e.*, when the fundamental properties of the material change their magnitude in the course of the rheological process through which the material is passing, this is generally an indication of a change in the internal structure of the material. For instance, the internal structure of a strain-hardened mild steel is different from one coming out of the rolling mill, the change consisting in a breaking up of the large into small crystals; the internal structure of a rubber solution in flow is different from its structure when at rest, the change consisting in a liberation of immobilized solvent. From the rheological behavior of a material, therefore, conclusions can be drawn about its internal structure. This was carried out by Reiner and Schonfeld-Reiner for rubber—toluene solutions and termed "'structural analysis." We can accordingly combine properties VI and VII with many similar properties under the term "rheological destruction," where "'destruction" should be understood to mean "reduction of structure."'

Property IX, thixotropy, is also a property resulting from a change of structure of the material. Its distinction from structural viscosity consists in that, in the case of thixotropy, *time* enters *explicitly* as a determinant. Another property of this kind is the "*aging*" of steel (compare Nadai, p. 25) by which its yield point increased during a period of rest. Property IX should include all such phenomena and may generally be termed *aging*.

12.

We still must deal with property VIII, "'strength."" Every body can sustain certain states of stress and strain between zero and a definite upper limit. When this limit is exceeded, the body decomposes into two or several parts: it *breaks*. This upper limit is the strength of the material. Reiner and Weissenberg have expressed the strengths of the different kinds of rheological materials in terms of *strain energy*. From an atomistic point of view, strength is not an independent property (*vide* Andrade), but such considerations belong to "'metarheology."'

Revising, in the light of the above, the list of essential rheological properties, we would have the five *fundamental* properties of viscosity, elasticity, plasticity, dilatancy, and strength, through a combination of which other properties such as firmoviscosity, elasticoviscosity, plasticoviscosity, etc., as well as such properties as toughness, brittleness, etc., can be expressed. In addition, the two phenomena of *rheological*

destruction and *aging* give rise to such properties as structural viscosity, strain hardening, strain softening, etc.

There are many technological properties which do not come under Scott Blair's essential properties and which defy classification. Sometimes they can be expressed in terms of essential properties. Green (1941), for instance, has reduced "tack" to a combination of viscosity and yield stress. Such properties, however, as the "covering power" of a pigment (Green, 1942) or "'seepage,"' etc., would simply have to be listed in a glossary

SUMMARY

A rheological systematics is built up through an *hierarchy of ideal materials* as shown in Figure 5. Of these, three are elemental; and it is shown how all others can be expressed qualitatively through these three by means of *structural formulas*. Rheological equations as listed in the table, page 65, describe the quantitative behavior of the materials. The parameters appearing in the rheological equations are characteristic coefficients or material "'constants,'' and the rheological properties are determined by the numerical values of these coefficients. Leaving cubical dilatation apart, we find five *fundamental properties,* viz., viscosity, elasticity, plasticity, dilatancy, and strength, from which other essential properties can be derived. The coefficients may be constants, but in many cases their values are affected by the phenomena of *rheological destruction* and *aging*.

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British Rheologists Club

NOMENCLATURE

Last spring the editor of the Rheology Bulletin received a copy of the following committee report from England. It is believed to be as complete a discussion of rheological nomenclature as appears anywhere in the literature. As such, it is of prime importance to all members of the Society, which is the reason for its publication here. You may f ind that some of the concepts differ from your own. Consequently, criticism and discussion are welcome and will be published in a later Rheology Bulletin as well as forwarded to our British colleagues. This report is reprinted through the courtesy of Professor G. W. Scott Blair, editor of the bulletin of the British Rheologists Club

The Meaning and Use of Certain Rheological Terms
Report by the Committee of the British Rheologists' Club

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Transcribed by James J Griebler, April 2026

1. General Information

In any discussion of nomenclature, certain difficulties are continually recurring. Of these the most serious are probably (1) the use of the same term in different senses in different sections of a science, and (2) the use of terms which have a general or qualitative everyday meaning, as well as a quantitative scientific significance. Nomenclature in rheology encounters not only these general difficulties, but also another kind of difficulty characteristic of a newly developed subject, namely, that many of the phenomena with which it is concerned are themselves frequently not sufficiently well understood to be unambiguously defined and classified.

This paper is an attempt to clarify certain terms and concepts, to point out the nature of any confusion which may have attended their usage, and when possible, to suggest means whereby such confusion may be avoided. At the present stage it would be too much to hope for finality or completeness, and the suggestions put forward are in tended mainly to provide a basis for further discussion and development.

In the selection of terms, the Committee recommend the adoption of the following three guiding principles: .

Principle 1.—Terms should be related as directly as possible to phenomena which are observable (or in principle observable), rather than to theoretical constructions or analytical interpretation of phenomena.

Principle 2.—Whenever possible, terms should be either self-explanatory, or designed to convey some indication of their meaning (directly or etymologically).

Principle 3—Words commonly employed in an every day or semi-scientific sense should not be given a quantitative connotation. They may, however, be retained as qualitative terms when they can be shown to fulfill a definite need.

It is emphasized that these are only guiding principles, not absolute rules, and there may well be exceptional cases in which the choice of suitable terms may be governed by other considerations. Thus, for example, the term “‘Hookean elasticity”? is justified by the almost universal familiarity with Hooke’s law, so that it is in effect self-explanatory, even though it does not conform literally with the second of our principles. Again, terms like force and energy, though used in a loose sense in everyday speech, have nevertheless a well-defined quantitative meaning in mechanics. This double usage does not appear to lead to confusion, because of the wide difference between the general and the scientific meanings. Other cases may be quoted, however, in which the position is not so happy. Two such cases are worthy of special consideration

(a) Consistency

With regard to the word *consistency* the (American) Society of Rheology have proposed two usages, (1) qualitatively, as "the resistance to deformation,' and (2) quantitatively, as "the ratio of shear stress to rate of shear,''' when this ratio is not constant.

The Committee feel that the use of the term in these two different senses is a source of confusion, and propose that it should be restricted to the first, or qualitative usage.

(b) Stiffness

The term *stiffness* has been used not only in a qualitative sense, meaning rigidity or resistance to deformation, but also in several different quantitative senses, viz. (1) as reciprocal of, "mobility'' in Bingham flow, (2) as the quantity constant in the Nutting equation, by Buist and Seymour, (3) as rigidity modulus, or closely-related quantity. Whilst drawing attention to the present position, the Committee do not feel able to recommend a solution to the difficulty without further discussion

2. Systems of Classification

Terms cannot be considered in isolation, on account of the relations which they bear one to another. Inevitably therefore, nomenclature implies some system of classification. In this section we consider some of the systems of classification which have been put forward.

(a) The British Rheologists' Club's "Table of Deformations"

A scheme of classification was put forward by the B.R.C. originally in 1942, and in a modified and more convenient form later in the same year. The later form is reproduced in Fig. 1. [This figure was not received—*Ed*.] Based generally on the principles enunciated in the preceding section, this scheme is largely self-explanatory. It is worth noting that only *deformations* are dealt with; no attempt was made to introduce real materials or types of matter. This limitation appeared to be advantageous, at least in a f irst attempt, in that it avoided the difficulty that a real material is generally capable of a diversity of types of behavior according to the conditions under which it is examined (e.g., pitch, glass, rubber). Bilmes' variation of the Table A circular layout of the B.R.C. Table has been proposed by Bilmes. His arrangement involves no intrinsic departure from the original classification, but may have advantages in a more effective pictorial presentation of the relationships between the types of deformation.

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a more effective pictorial presentation of the relationships between the types of deformation

(b) Reiner's System of Classification

A rather different approach has been suggested by Reiner. In his system a number of characteristic "'bodies"' are defined. These "bodies'' are defined in terms of their law of deformation, so that in effect the definitions may be regarded as definitions of types of deformation. The following five fundamental bodies are distinguished:

E. *Euclid solid*. A perfectly rigid (i.e., non-deformable) body.

P. *Pascalian liquid*. A perfectly non-viscous fluid.

H. *Hooke solid*. A body which gives a Hookean elastic deformation.

ST. V. *St. Venant body*. A body characterized by the existence of a yield stress (and which presumably behaves like E at lower stresses and like P at higher stresses).

N. *Newtonian liquid*. A body which is capable of a Newtonian viscous deformation.

Types E and P have no real counterparts in rheology, and need not be further considered. The remaining three may be combined in various ways to produce more complex types of deformation. The three principal derived types are:

M. *Maxwell liquid*. This is represented H and N in series. (Spring and dashpot in series.) This is a special case of the B.R.C. Viscoelastic deformation.

B. *Bingham body*. Defined as ST. V and N in series. This is equivalent to the B.R.C. Bingham plastic deformation.

K. *Kelvin solid*. Defined as H and N in parallel. (Spring and dashpot in parallel.) This is a special case of the B.R.C. Non-ideal completely recoverable deformation.

By further combinations between the fundamental and derived types still more complicated properties may be represented. The properties of all these bodies may be represented by means of rheological equations, the parameters in which correspond to the values of certain physical constants.

It is not proposed here to make any judgment on the relative value of Reiner's and the B.R.C. schemes, but it may not be out of place to make some reference to the underlying divergences in the respective methods of ap proach. We observe, firstly, that Reiner's system is based on the assumption that the behavior of a given system can be represented in terms of a number of elements, each obeying a simple law. This theoretical analytical approach is in contrast to the empirical approach of the B.R.C., represented Principle (1). Secondly, Reiner's terms are not self-explanatory, but are all associated with the names of original investigators.

(c) The Nomenclature of the Dutch School

Burgers, Saal, and Bienzeno have published (in Dutch) a rather comprehensive report entitled "Basis for a Nomenclature of Deformations,"! which attempts to cover the whole range of types of deformation. It is hardly possible to convey an adequate idea of the scope of this work in summary form, but it may be said that the method of approach is identical in principle with that adopted by the B.R.C., though there is naturally some difference in emphasis and in detailed terminology. The following table contains a list of the principal types of deformation considered, together with the most nearly comparable B.R.C. types.

	Dutch System	Nearest B.R.C Equivalent
A0	Completely elastic deformation	T Ideal elastic deformation T Non-ideal completely recoverable
Al*a*T	Thresholdless limited deformation	Non-ideal completely recoverable
Al*b*T Bl*a*T Bl*b*T	Plastic limited deformation	Plastico-elastic deformation (?)
A2	Thresholdless creeping deformation ($d\gamma/dt$ continually decreasing)	Non-Newtonian viscous flow (special case)
B2	Plastic creeping deformation ($d\gamma/dt$ continually decreasing)	Non-Bingham plastic flow (special case)
A3	Thresholdless constant speed deformation ($d\gamma/dt$ const.)	Newtonian flow
B3	Plastic constant speed deformation ($d\gamma/dt$ const.)	Bingham flow
A4	Thresholdless accelerating deformation ($d\gamma/dt$ continually increasing)	Non-Newtonian viscous flow (special case)
B4	Plastic accelerating deformation ($d\gamma/dt$ continually increasing)	Non-Bingham plastic flow (special case)

The general similarity between the two systems is ap parent from this table. Apart from the actual terms used, the chief difference is that the Dutch system contains rather more subdivisions of the flow group, and rather fewer of the elastic group than the B.R.C. classification. We turn now to the consideration of certain terms and phenomena in greater detail

3. Stress, Strain, and Related Quantities

Since stress and strain are fundamental quantities in all rheological measurements, it may be well to begin the discussion by a consideration of their definition. On this subject a fairly complete system of definitions is provided by the classical theory of elasticity. In applying the results of this theory in the field of rheology, however, it is important to

distinguish between propositions which are true for deformations of any magnitude and those which are valid only in the limiting case of small deformations. In this paragraph the point of view adopted is that which has been expounded in Love's Mathematical Theory of Elasticity. Some of the more detailed mathematical expositions are given in the Appendix. Before defining stress, it is necessary to consider traction.

Traction

Traction is defined as the force per unit area acting across a specified plane within a body or at the surface of a body. The components of this force respectively perpendicular and parallel to the plane considered are termed the normal and tangential tractions.

Stress

Stress. The complete specification of the stress at a point O within a body involves a knowledge of the traction across all planes passing through *QO*. In particular, if the tractions across a set of three mutually perpendicular planes through *O* are given, then the tractions across any other plane may be calculated. The tractions on the three mutually perpendicular planes, resolved into normal and tangential components, yield the six components of stress with respect to a given coordinate system.

Notes

(1) Traction is a *vector* quantity. Stress, being related to six vector components as defined above is a *tensor* quantity
(2) Some authors (e.g., Morley, Strength of Materials, and Southwell, Theory of Elasticity) use the term stress in the sense of both traction and stress as defined by Love. Thus, if a bar is loaded in simple tension, the load per unit area of cross section is referred to as the "stress." This ambiguous usage is a source of confusion, and should be avoided.
(3) The stress at all points of a body at rest or in un accelerated motion is determined solely by the applied forces. It is not, therefore, a function of the material.

Shear stress is of particular importance in rheology. It is defined as a system of two equal tangential tractions acting on planes at right angles to one another. In these planes there is no normal traction

Note that the importance of distinguishing between stress and traction is here apparent. In planes other than those on which the normal traction vanishes there will be both normal and tangential tractions, varying in magnitude with the direction of the plane; and determined in magnitude by the direction and magnitude of the shear stress

Strain

Strain. The concept of strain involves a comparison between two states, (a) the undeformed, and (6) the deformed. In a homogeneous strain (i.e., one which is the same at all points of a body) a sphere is transformed in general to an ellipsoid with three unequal axes. This is known as the *strain ellipsoi*d. Conversely, it is possible to find a particular ellipsoid in the unstrained body which is transformed on deformation into a sphere. The original ellipsoid is called the *reciprocal strain ellipsoid*.

The general homogeneous strain may be defined in terms of the principal axes of either the strain ellipsoid or. the reciprocal strain ellipsoid. Strain, like stress, is a tensor quantity, and may be represented in terms of six components. (See appendix.)

Simple shear

Simple shear is a type of strain in which every point of a body is displaced parallel to a given line by an amount proportional to its distance from a fixed plane containing that line. The magnitude of the shear is the ratio of the displacement of a point to its distance from the fixed plane.

Notes

(1) In a *small* shear strain the principal axes of the strain ellipsoid make an angle of 45° with the direction of sliding. In a large shear this is no longer true.
(2) *The unstrained state*. Whilst mathematically any state has an equal claim to be regarded as the unstrained state, there may be physical reasons for preferring a particular state to represent the state of zero strain. An ideally elastic rubber, for example, is isotropic in one and only one state, and it is naturally simplest to consider this to be the unstrained state. A Newtonian fluid, on the other hand, is isotropic after any deformation, and there is no reason for choosing any particular state in preference to another as unstrained state.
(3) The *magnitude of 'strain.'* The general strain requires six numbers for its specification. In special cases, however, the strain may be represented by fewer than six numbers. In these simpler cases the classical components of strain may not be the most appropriate for the specification of the amount of strain. Shear strain, for example, may be represented by a single parameter, the amount of shear.

The type of strain undergone by a bar in simple tension cannot, in general, be represented by a single number, but requires the specification of at least two parameters (length and thickness). If, however, we are concerned only with the change in length, we may call this, for our immediate purpose, the "strain,'' which may be measured or defined in a number of ways, viz.

(i) *Engineering strain*. Defined as $(l - l_0)/l_0$. This is the simplest measure to adopt when the strain is small. It is also frequently used in an empirical manner when the strain is large.

(ii) *Natural strain* (Hencky). Defined as $\ln(l/l_0)$ or dl/l_0. This is likely to be useful in the case of a material which remains unchanged in its properties during deformation

(iii) *Extension ratio,* or l/l_0. This is particularly useful in large elastic deformation, on account of its direct connection with the strain ellipsoid.

(4) Elastic Moduli and their Reciprocals

The three principal moduli are:

(i) For simple tension (length l increased to $l + \delta l$)

$$\text{Young's Modulus} = \frac{\text{Normal Traction}}{\pm l/l_0}$$

(ii) For simple shear (line normal to plane of slide rotates through angle ϕ).

$$\text{Rigidity Modulus} = \frac{\text{Tangential Traction}}{\tan\phi}$$

Reciprocal, Shear coefficient

(iii) For hydrostatic compression (volume V reduced by $\pm V$).

$$\text{Modulus of Compression (or Bulk Modulus)} = \frac{\text{Hydrostatic Pressure}}{\pm V/V}$$

Reciprocal, Coefficient of Compressibility

(iv) In addition, we have in the case of simple tension

$$\text{Poisson's Ratio} = \frac{\text{Relative lateral contraction}}{\text{Relative longitudinal extension}}$$

Notes

(1) The definition of Rigidity Modulus is valid for larger strains. The remaining three quantities above are not defined for large strains.

(2) The use of the terms 'tangent' and "'secant'' moduli, ($dS/d\sigma$ and S/σ) by Lazan and Yorgiadis may be convenient in some cases. "Elastic compliance" (Alfrey) for shear coefficient, has the disadvantage that the type of strains is not indicated in the term. There is nothing to be said, however, for the common usage in rubber technology of the expression "modulus at (say) 300 percent extension, when what is meant is "force per unit area of the unstrained material at 300 percent extension."'

4. Viscosity

Viscosities range from about 10 for liquid helium to about 10 for glass at 500°C, with many common liquids such as water (10) and golden syrup (10^2) occupying positions near

the middle of the range. In hydrodynamics the viscosity of a liquid, even though numerically small, may be of great significance in determining the type of flow, namely, whether it is streamlined or turbulent. This influence is recognized in the so-called Reynolds Number, involving viscosity, which in a certain sense measures the hydrodynamical scale of any given field of motion

(a) Newtonian Liquids

Newton, who attributed viscosity in fluids to "defectus lubricitantis" or internal friction, "attritus," clearly defined viscosity in terms of the tangential traction S and the velocity gradient du/dx in a simple shear. His equation is

$$S = \eta du/dx$$

The unit of viscosity, η, is then that viscosity which corresponds to unit stress when the velocity gradient is unity, and is called the *Poise*. Its reciprocal, the unit of fluidity, is known as the *Rhe* (Bingham).

From Newton's equation, viscosity (η) has dimensions $ML^{1}T^{1}$.

Relaxation and orientation times

Poisson and Maxwell compared a liquid with an elastic solid in which the elastic reaction is continuously dissipated. If λ is the time taken for an instantaneously applied stress to fall to $1/e$ of its initial value (under constant strain), and n is the rigidity modulus, then the following relation may be shown to hold:

$$\eta = n \times \lambda$$

or viscosity – rigidity ×time of relaxation.

The Maxwell model of a liquid, and the time of relaxation defined thereby, provides a natural interpretation of the viscosity of liquids when these are approached from the solid state. If, on the other hand, we approach the liquid state from the basis of the gaseous state, we are led to regard viscosity as a diffusion process arising from the transfer of momentum between molecules in different parts of the flowing liquid.

Maxwell's model is mathematically equivalent to a spring and dashpot in series. In an alternative system, comprising a spring and dashpot in parallel, the application of a stress causes an asymptotic approach to a final limiting deformation. The deformation then follows the law

$$\sigma = A(1 - e^{t/\mu})$$

This model is identical with Reiner's Kelvin solid (Section 2) and is appropriate as a first approximation to a certain class of non-ideal elastic deformations, as exemplified by high-molecular systems. The constant μ in the above equation has been called the "'orientation time.

(b) Non-Newtonian Liquids

Many liquids do not follow Newton's simple law of flow. In such cases it is necessary to consider whether the rate of shear is determined solely by the shear stress, or whether time effects are present.

If the rate of shear is determined solely by the shear stress, a unique relation exists between stress and rate of shear. This relation completely defines the properties of the material. Suggested equations for such materials contain two or more constants, to which special names have been given. We do not propose to follow up these suggestions in detail, but will restrict ourselves to the general comment that where such non-linear equations are employed the symbol η and the term viscosity should not be used to indentify the constants. (It is considered permissible, however, to make use of the term "'apparent viscosity" when referring to the reading of a particular instrument, or in a similar empirical sense, provided that it is made clear that the term is not used in the absolute sense of Newton's equation.

(c) Structural Viscosity

Structural viscosity (or structure viscosity) is a special case of non-Newtonian viscosity in which the rate of flow is determined only by the shear stress, but in which the apparent viscosity $S/(d\gamma/dt)$ falls with increasing stress. The term is not a good one, since it implies a theoretical breakdown of structure, but it is probably too well established to be discontinued

(d) Kinematic Viscosity

In cases of varying motion we are often concerned not so much with viscosity itself as with the ratio of viscosity to the density or inertia of the fluid. The quantity η/d or η/i is known as the kinematic viscosity, and is measured in Stokes. Kinematic viscosity has the dimensions of a dif fusion coefficient, i.e., $ML^{-1}T^{-1}/ML^{-3}=L^{2}T^{-1}$.

In industrial processes the rate of flow of liquids such as petroleum may be sufficiently high to be in the region where turbulence occurs, and for purposes of calculation the kinematic viscosity becomes important.

The Committee feel that the term "kinematic viscosity" should be retained, without, however, eliminating the term "'viscosity'' (as had been suggested by some authors)

(e) Specific and Intrinsic Viscosity

The term "specific viscosity'' has frequently been used in connection with solutions. It is defined by the equation]

$$\eta_{sp} = (\eta - \eta_0)/\eta_0$$

where η and η_0 are the viscosities of the solution and solvent respectively. Specific viscosity is a pure number.

"Intrinsic viscosity" is defined as

$$\eta = \lim_{c \to 0}(\eta_{sp}/c)$$

where c is the concentration. An equivalent definition is

$$[\eta] = \left(\frac{d\eta_{sp}}{dc}\right)_{c \to 0}$$

The concentration c is variously expressed as (a) g per liter, (b) gram per 100 cc, or (c) g per g.

Note

There is some opposition to the use of terms involving the word viscosity for quantities whose dimensions are not those of viscosity. If "intrinsic viscosity"' is to be used, it is strongly recommended that c be measured in g per g

5. *Reversible and Irreversible Changes in Consistency*

Under this heading will be considered a number of more complex rheological phenomena in which the deformation is accompanied by a change in the state or properties of a material.

(a) Thixotropy

The clearest example of thixotropic behavior is that of a substance which changes from a state in which it is able to resist shearing stresses (below a yield value) to one in which it is unable to withstand shearing stresses as a result of the application of a shear, but which on resting gradually and completely returns to its original state. The term *thixotropic* is also applied by some authors to describe any material which suffers a loss of consistency on shearing, and which gradually reverts wholly or partially to its initial state on removal of the shear stress. Other workers have applied the term "false body" to this second type of behavior. Unfortunately *false body* is also used in quite a different sense in the paint industry, and the distinction between the two terms (thixotropy and false body) is by no means always clear.

Since it seems to us that the vanishing of a yield value is only a special case of reduction of consistency, we suggest that the term *thixotropy* be used to cover all such types of behavior, and that the term *false body* be avoided. The proposed definition of thixotropy is given below:

Thixotropy is a reversible time-dependent loss of consistency accompanying the application of a shear, or the increase in the amount of the applied shear

Notes

(i) The change of consistency may be related to the magnitude of the stress, or to the magnitude of the strain. These classes may be distinguished by the terms *stress-thixotropy* and *strain-thixotropy*.

(ii) The difference between structural viscosity and thixotropy lies in the absence of an observable time-lag in the former case.

(b) Work-Hardening and Work-Softening

Work-hardening is a commonly used term implying a nonreversible increase of consistency on shearing. Strain hardening is sometimes used synonymously with work hardening, but it seems desirable to distinguish the possibility of either the amount of strain or the amount of stress being the determining factor. We suggest, therefore, the use of *strain-hardening* and *stress-hardening*, with work hardening as a general term including both types. *Stress softening, strain-softening,* and *work-softening* may be used in corresponding senses when the consistency is irreversibly reduced on shearing.

The term "rheodestruction'' has been suggested to indicate a non-reversible loss of consistency on shearing. This term is etymologically unfortunate, and is not favored by the Committee. "Shear breakdown"' has been used in industry, e.g., to describe the loss in viscosity of polymer thickened fluids with use in aircraft.

(c) Rheopexy

The term *rheopexy* has been applied to the phenomenon exhibited by certain thixotropic systems (e.g., V_2O_5 sols) in which the rate of resetting is accelerated by slowly rotating the containing vessel.

(d) Dilatancy

The term *dilatancy* originally implied the expansion into an open packing which a system of closely packed spheres must necessarily undergo when stressed, before flow can take place. This rearrangement usually involves an increasing resistance to shearing, and in the simplest case the amount of "hardening'' would be a function of the amount of strain. Following Williamson and Heckert's finding that for cornflour at least, the

concentrations at which the phenomena occur do not correspond to close-packing, a problem in nomenclature arose. Should these phenomena be included under the term "dilatancy"" irrespective of mechanism, or should some other term (e.g., "inverted plasticity'' or "pseudo-dilatancy"') be introduced?

Since it seems dangerous to speak of dilatancy in systems which do not dilate, and since in some cases, the extent of hardening may be a function of stress rather than of strain, it would seem preferable to speak of strain-hardening or stress-hardening in such cases, without reference to mechanism. Phenomenologically, dilatancy is a special case of strain-hardening, and the former term should be reserved for cases where a change of packing is proved

6. *"SPINNBARKEIT"*

Many materials, such as white of egg and adhesive like seccotine, can be readily drawn into threads. In some cases these threads can be hardened to give fibers which can be spun, but in many instances this is not so. Erbring, who has worked extensively on this property uses the term "Spinnbarkeit"' for the phenomenon, and his term is now in fairly common use among British Rheologists (e.g., Pryce-Jones, Clift). The term is clearly not sufficiently wide even in the original German, and is quite unsuitable in English. Attempts at literal translation (spinnability, fibrosity, etc.) are equally unsatisfactory.

There would seem to be a good case for an international word of classical origin. The first proposal (Pirie) of "'leptogenesis"' appears to mean "producing thinness," and the later alternative (McClean) "nematagenesis'' would be preferable but for the circumstance that Bernal has adopted this term in the sense of "producing the nematic condition."' It appears that the term "nematapoeic'' has been introduced by McClean, and other possibilities which have been discussed are "'mitogenic'' and "closmagenic."' These last three all seem to mean "thread producing,'' in the sense of a thread of cotton, which is not perfect but seems to be the best we can do at present. Investigations on this question are continuing.

APPENDIX – THE COMPONENTS OF STRAIN

The state of strain is completely determined when we know the lengths in the strained and unstrained states of corresponding lines. If ds_1 and ds are the lengths of corresponding linear elements in the two states, the relation between ds_1 and ds is a function of the direction cosines (l, m, n) of ds which may be expressed in the form

$$\left(\frac{ds_1}{ds}\right)^2 = (1 + 2\epsilon_{xx})l^2 + \left(1 + 2\epsilon_{yy}\right)m^2 + (1 + 2\epsilon_{zz})n^2 + 2\epsilon_{yz}mn + 2\epsilon_{zx}nl + 2\epsilon_{xy}ln$$

where l, m, n are the direction cosines of ds. The components of strain ϵ_{xx}, ... , ϵ_{yz}, ... are functions of the partial differentials of the components of displacement u, v, w of a point, i.e.

$$\epsilon_{xx} = \frac{\partial u}{\partial x} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial x}\right)^2\right]$$

$$\epsilon_{yy} = \frac{\partial v}{\partial y} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right]$$

$$\epsilon_{zz} = \frac{\partial w}{\partial z} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2\right]$$

$$\epsilon_{yz} = \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} + \frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} + \frac{\partial v}{\partial y} \cdot \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \cdot \frac{\partial w}{\partial z}$$

$$\epsilon_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} + \frac{\partial v}{\partial z} = \frac{\partial v}{\partial x} + \frac{\partial w}{\partial z} \cdot \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial u}{\partial x}$$

$$\epsilon_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial w}{\partial x} = \frac{\partial w}{\partial y} + \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \cdot \frac{\partial v}{\partial y}$$

(1)

strain, like stress, is a tensor quantity.

For any homogeneous strain it is possible to find an ellipsoid which becomes a sphere on deformation. This ellipsoid is called the *reciprocal strain ellipsoid*, and its axes the *principal axes of strain*. The quantity ds/ds-1 is called the extension of the linear element, and the extensions of the axes of the reciprocal strain ellipsoid are called the principal extensions

SIMPLE SHEAR

Defined by

$$x^1 = x + \sigma y$$

$$y^1 = y$$

$$z^1 = z$$

The proportionality constant σ is the amount of the shear, and may be shown to be equal to difference of the two principal extensions in the plane (x, y). Moreover, the area of a figure in the (x, y) plane is unaltered by the shear strain

SMALL DEFORMATIONS

When the strain is small the squares and products of $\partial u/\partial x$ etc in (1) vanish, and the components of the strain take on a particularly simple form, i.e.

$$\epsilon_{xx} = \frac{\partial u}{\partial x}, \qquad \epsilon_{yy} = \frac{\partial v}{\partial y}, \qquad \epsilon_{zz} = \frac{\partial w}{\partial z}$$

$$\epsilon_{yz} = \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}, \qquad \epsilon_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}, \qquad \epsilon_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

The Rheology Bulletin, Vol 18, No. 3 and 4 December 1947, pg. 12 (891)

International Rheological Congress in Holland, September, 1948

Tuesday, September 21st: Morning: General lecture on Recent Developments in the Theory of Viscosity. Afternoon: Communications from the Joint Committee. General lectures on: (a) Nomenclature; (b) Fundamental Experimental Methods

The Rheology Bulletin, Vol 18, No. 3 and 4 December 1947, pg. 14 (893)

British Rheological Nomenclature, A Letter to the Editor

Dear Sir:

In connection with Section 3 and the Appendix of the British report on rheological nomenclature published in the June issue of the Rheology Bulletin, we should like to present the following comments.

1. *Traction and stress*. The adoption of .a terminology which allows clear distinction between the force per unit area transmitted across a specified surface and the state of stress at a point is highly desirable. The terms “traction” and “stress’’ serve this purpose, but the fact that “traction” then covers tensile, compressive and shearing forces may create certain difficulties. An alternative procedure would be to reserve the term “stress’’ for the force per unit area transmitted across a specified surface and use the term “state of stress’ or “‘stress tensor’’ when referring to the complete specification of the stresses trans mitted across various surface elements through a specified point.
2. *Stress not a function of the material*. The statement in the first paragraph of Section 3 under the heading *Stress* might be misinterpreted. Even for an elastic body in equilibrium, the stresses at a generic point are not solely determined by the forces applied to its surface, but may depend also on Poisson’s ratio.

3. *Sehar stress*. In the second paragraph of Section 3 under the heading *Stress*, the term shear stress is used in a rather restricted sense to denote a state of simple shear (stress). This may cause confusion in so far as the term *shear stress* is customarily used to denote what, according to the present proposal, would be called tangential traction.
4. *Moduli*. In the definitions of Young's modulus and bulk modulus, the denominators should read $\delta l/l$ and $\delta V/V$, respectively
5. *Appendix*. The strain components given in the Appendix are not tensor components. It seems highly desirable to define the shear components ϵ_{xy}, etc., with the factor ½ so that the quantities $\epsilon_{xx}, \ldots, \epsilon_{xy}, \ldots$ constitute the components of a tensor (strain tensor). If this seems desirable, the conventional strain components given in the Appendix might be denoted by $\epsilon_{xx}, \ldots, \gamma_{xy}, \ldots$

Sincerely yours,

D. C. Drucker, Associate Professor of Engineering Brown University

W. Prager, Professor of Applied Mechanics, Providence, R. I.

The Rheology Bulletin, Vol 21, No. 2 Fall, 1952, pg. 1 (28)

The Annual Meeting, 1952

Lecture Program, October 30-31

H. LEADERMAN, NATIONAL BUREAU OF STAN DARDS, WASHINGTON, D.C.

PROPOSED NOMENCLATURE AND SYMBOLISM FOR LINEAR VISCOELASTIC BEHAVIOR OF HIGH POLYMERS."

There is at present a lack of uniformity in the use of terms and symbols describing the observable and derived quantities associated with the linear viscoelastic behavior of high polymers. A logical system of nomenclature which is likely to be adopted must be modified to take account of convenience and of current usage.

The Rheology Bulletin, Vol 22, No. 1 Spring, 1953, pg. 2 (36)

MINUTES : Business meeting of the Society of Rheology Philadelphia, October 30, 1952

President Spencer announced the appointment of: a Committee on Nomenclature with H. Leaderman as its chairman for 1953; a Committee on a second Viscosity Standard with W. F. Fair, Jr., as chairman; and a Committee on Publication with the Editor in the Chair

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 22, No. 2 Fall, 1953, pg. 6 (45)

The Second International Congress on Rheology

The Second International Congress on Rheology, the Program of which was circulated to our members in our Bulletin Supplement in the Spring of 1953, was an even greater success than the first Congress held in Holland five years ago. There were about 250 visitors from twelve countries in attendance. A total of 56 papers were presented. The quality of most papers was high, and there was a great deal of stimulating discussion. Only 37 of the papers presented were available in preprint form before the meeting, but all of those presented, with the discussions, will be included in the final Proceedings of the Congress, and in this country will be available through the Academic Press.

Two general meetings were held on Tuesday and Thurs day evenings during which discussions were held on the organization of an International Union of Rheology and on Nomenclature of Rheology. A number of factors bearing on the advisability and usefulness of an International Union, as well as questions of financing and affiliating with either the International Union of Chemistry, that of Physics, or that of Applied Mechanics were discussed. The general feeling was that the principal function of an International Union of Rheology should be to insure continuity of an organization which would organize periodic Congresses. The possibilities of such a body considering officially questions of nomenclature and of publishing an abstract journal were also mentioned. The attitude of our Society will be discussed at the next Executive Meeting, after which our membership will be consulted.

In the discussion on nomenclature, Dr. Leaderman presented results to date of the Nomenclature Committee of the Society of Rheology of the U.S.A. in its endeavors to agree on nomenclature and symbols for linear viscoelastic behavior

The Rheology Bulletin, Vol 23, No. 1 Spring, 1954, pg. 2 (47)

MINUTES : Business meeting of the Society of Rheology, October 29 1953

Dr. H. Leaderman reported on the activities of the Committee on Nomenclature. A full report on nomenclature for linear viscosity is expected to be available at the next Annual Meeting

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 23, No. 3 December 1954, pg. 4 (59)

INTERNATIONAL COMMITTEE ON RHEOLOGY

As Members will remember, such a Committee was formed at the occasion of the Second International Congress on Rheology in Oxford, in 1953. All the delegates from the various countries have since been nominated to this Committee. Their names are given below, since they should be useful for contacts in the respective countries:

B. Gross (Brazil), L. R. G. Treloar (England), H. Weiss (France), F. H. Muller (Germany), M. Reiner (Israel), R. N. J. Saal (Netherlands), E. Forslind (Sweden), R. S. Spencer (U.S. A.), A. Peterlin (Yugoslavia).

The International Committee is setting up a Subcommittee on Nomenclature which will prepare a summary of present usage (terms, definitions, symbols) to be referred to the various rheological societies for discussion. Eventually, a more definite report will be prepared for consideration at the Third International Congress on Rheology.

The Rheology Bulletin, Vol 24, No. 2 Fall, 1955, pg. 1 (66)

The Bingham Medal, 1955

True to its tradition, the Committee for the Bingham Medal has arrived at another fortunate selection for the 1955 medalist, in choosing Dr. Herbert Leaderman.

Known to every physicist in the field of mechanical behavior of natural and synthetic high polymers by his classical monograph on fiber deformation, by his fundamental contributions to the behavior of dielectrics in high frequency fields and especially to the construction of radar domes, and last not least for his tenacious uphill struggle for rhyme and reason in the jungles of viscoelastic nomenclature, Dr. Leaderman combines best old-world traditions with individualism, unconventionality and originality of approach in an outstanding manner.

Dr. Leaderman has been for many years at the National Bureau of Standards where he directs research on viscoelasticity. It is a happy thought that in his case the award of the Bingham Medal does not come late in a career but represents a timely recognition of past achievements and an encouragement for others to come.

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 25, No. 1 January 1956, pg. 3 (72)

MINUTES : Business meeting of the Society of Rheology, November 3, 1955

Dr. Herbert Leaderman, Chairman, Committee on Nomenclature, discussed the work of his Committee and noted that a new report on nomenclature for viscoelastic behavior was prepared and circulated in August 1955 to members of his Committee and to about 40 other rheologists both here and abroad. The Committee was requested to complete the questionnaire submitted and comments from the others were also solicited. The replies to date have been encouraging. A committee of the Dutch Rheological Society is working on a similar problem and plans to issue a report which will be similar to that submitted by our Committee. A final report will be available in a few months. Anyone interested should communicate with Dr. Leaderman.

The Rheology Bulletin, Vol 26, No. 1 January 1957, pg. 3 (86)

MINUTES : Business meeting of the Society of Rheology, November 8, 1956

7.

The Report of the Committee on Nomenclature submitted by Dr. Leaderman August 1956, was highly commended and notice given that it will be published and distributed to the membership in the near future. Thanks and appreciation were expressed to Dr. Leader man and his committee

8.

The meeting was advised that 50 copies of the Nederlandse Reologische Vereniging “Proposals on Rheo logical Nomenclature and Definition” have been received and some distributed to individuals particularly interested, with a small number still available.

Nomenclature—This committee is now inactive. Its report of recommended nomenclature has been gratefully received by the Executive Committee and will be published in the forthcoming Transactions

Transcribed by James J Griebler, April 2026

TRANSACTIONS OF THE SOCIETY OF RHEOLOGY, VOL I, 213-222 (1957)

Proposed Nomenclature for Linear Viscoelastic Behavior[5]

Herbert Leaderman, *National Bureau of Standard, Washington 25, D.C.*

I. Introduction

A systematic standardized nomenclature is very desirable to facilitate communication among physicists working in the same field or in associated fields; such a nomenclature gives names and symbols to mensurable related quantities. After the fundamentals of a branch of physics are fairly well established, it is then possible to draw up a logical complete nomenclature system. Unfortunately, by that time the field is usually burdened with several systems which are not generally accepted and which are not completely logical; these systems may or may not be related to nomenclature systems in associated branches of physics.

Ideal nomenclature systems which are prepared subsequent to the understanding of the relationships between measurable quantities in a branch of physics are often proposed, They may represent radical departures from more or less established usage; under these circumstances the proposals do not gain general acceptance. A practical improvement on a haphazard nonstandardized nomenclature can only be gained by adhering as closely as possible to existing usage. In this report an attempt has been made to devise a nomenclature following this principle.

When certain materials are deformed such that the stored elastic energy and the rate of dissipation of mechanical energy are sufficiently small, then the mechanical behavior of these materials approximates to the idealized behavior known as *linear viscoelastic behavior*, which may be defined in general by equations such as (1) and (2) given below. This report deals with a proposed nomenclature for primary measurable quantities and for related quantities applicable to isotropic materials manifesting (approximate) linear viscoelastic behavior. While this nomenclature is planned primarily in order to systematize the somewhat extensive literature on the behavior of amorphous polymeric systems, it is also adaptable, perhaps with some modifications, to other materials such as inorganic glasses and metals. The *proposed nomenclature* is given in Table II, which contains *primary measurable quantities*. In Table III are given certain *auxiliary quantities*, namely, limiting values and spectra. No at tempt has been made to make this nomenclature consistent with the nomenclature for dielectric behavior, since in practice only one of the four primary

[5] This is the final report of the Committee on Nomenclature of the Society of Rheology.

Transcribed by James J Griebler, April 2026

methods of presenting data considered in this report is used in the specification of dielectric behavior. Further more, no attempt has been made to prepare a *complete* system of nomenclature. For materials manifesting linear viscoelastic behavior, many different methods of specifying behavior are possible; in the present proposed nomenclature a minimum number of measurable and related quantities are defined. The text of this report gives the basis for the nomenclature, and some of the relations between the quantities specified in Tables II and III. For the reasons given above, generally accepted usage has been followed as much as possible,

II. Step-Function and Dynamic Moduli and Compliances

The linear viscoelastic behavior of a material (at a given temperature) is generally specified by the response of the material in certain types of experiment. While any reasonable type of experiment is theoretically acceptable, in practice four types of experiment are customarily considered for the specification of viscoelastic behavior. Linear viscoelastic behavior may be specified in general by either of the equivalent relations:

$$\epsilon(t) = \int_{-\infty}^{t} \frac{d\sigma(u)}{du} k(t-u)du \tag{1}$$

$$\sigma(t) = \int_{-\infty}^{t} \frac{d\epsilon(u)}{du} m(t-u)du \tag{2}$$

where $\epsilon(t)$ is the strain and $\sigma(t)$ is the stress at time t, and $k(t)$ and $m(t)$ are time-dependent function of the dimensions, respectively, of compliance and modulus; these functions represent the viscoelastic properties of the material. It is seen that, if a *unit stress* is applied instantaneously, that is, *stepwise (*and maintained constant thereafter), then the strain at time t is given by $k(t)$. Similarly, if a unit strain is applied stepwise at zero time, then the stress at time t is $m(t)$. The functions $k(t)$ and $m(t)$ define, respectively, the *creep compliance* and *relaxation modulus* functions.

From equation (1) it can be shown that if the stress is varying sinusoidally with time with circular frequency ω, then in the steady state the strain varies sinusoidally with time, and the amplitude of the strain is proportional to the amplitude of the stress. In general, the strain differs in phase from the stress, so under these conditions the ratio of strain to stress can be represented by a frequency-dependent *complex* compliance $k^*(\omega)$. Thus the complex compliance represents in phase and amplitude the strain response, as a function of fre quency, to a sinusoidal stress of unit amplitude. Similarly, if the stress is varying sinusoidally with time, then from equation (2) it can be shown that the ratio of stress to strain can be represented by a frequency-dependent complex modulus $m^*(\omega)$.

Table I

Symbols and Terminology for Modulus and Compliance

Type of Deformation	Remarks	Modulus, m	Compliance, k
Shear		G	J
Bulk (volume)		K	B
Longitudinal (a)	No transverse stress. Strain measured longitudinally	E	D
Longitudinal (b)	No transverse strain. Stress measured longitudinally	M	(N)
Longitudinal (c)	No transverse stress. Strain measured laterally	-	-
Longitudinal (d)	No transverse strain. Stress measured laterally	(λ)	-

The above time- or frequency-dependent moduli and compliances are related to the moduli and compliances of classical elasticity theory. These are given in Table 1. It is seen from the table that there are four cross-measures, involving a transverse strain (or stress) and a longitudinal stress (or strain); one of these is the Lamé constant λ. These four measures seem to be of limited use in representation of viscoelastic behavior, and will not be considered further. The compliance N has not yet been used in viscoelasticity; it is introduced here for completeness.

III. Relations between step-function moduli and compliances

In this report definitions are proposed for three creep compliances, namely, the *shear creep compliance* $J(t)$ the *bulk* (or *volume*) *creep compliance* $B(t)$, and the longitudinal creep compliancc $D(t)$. The corresponding relaxation moduli are denoted by $G(t), K(t),$ and $E(t)$. The relations between the moduli and compliances of classical elasticity theory are analogous to the relations between the Carson transforms of the time-dependent step-function moduli and com pliances given above, where the Carson transform of a creep compliance $k(t)$ is given by:

$$p\int_0^\infty e^{-pt}k(t)dt = p\mathcal{L}\{k(t)\}$$

and similarly for the Carson transform of a relaxation modulus. Thus, for example, the relation between a creep compliance $k(t)$ and its associated relaxation modulus $m(t)$ is given by:

$$p\mathcal{L}\{k(t)\} = \frac{1}{p\mathcal{L}\{m(t)\}}$$

IV. Methods of representation of dynamic behavior

Dynamic behavior refers to the response of a material manifesting linear viscoelasticity when the stress and strain vary sinusoidally with time. The strain can be considered as consisting of two components, one in phase with the stress and the other lagging 90°. The ratio of the amplitude of the former component to the amplitude of the stress is the *storage compliance,* $k'(\omega)$. The ratio of the amplitude of the latter component to the amplitude of the stress is the *loss compliance,* $k''(\omega)$. Alternatively, the stress can be considered as consisting of two components, one in phase with the strain, and the other leading by 90°. The ratio of the amplitude of the former component to the amplitude of the strain is the *storage modulus,* $m'(\omega)$; The ratio of the amplitude of the latter component to the amplitude of the strain is the *loss modulus,* $m''(\omega)$. Thus in general:

$$k^*(\omega) = k'(\omega) - jk''(\omega) \tag{3}$$

and:

$$m^*(\omega) = m'(\omega) + jm''(\omega) \tag{4}$$

The *absolute modulus*, $|m(\omega)|$, is given by

$$|m(\omega)| = [m'^2(\omega) + m''^2(\omega)]^{\frac{1}{2}} \tag{5}$$

and the *absolute compliance*, $|k(\omega)|$, is given by

$$|k(\omega)| = [k'^2(\omega) + k''^2(\omega)]^{\frac{1}{2}} \tag{6}$$

The *loss tangent* or *damping,* $\tan\delta$, is given by

$$\tan\delta = m''(\omega)/m'(\omega) = k''(\omega)/k'(\omega) \tag{6}$$

The quantity $m''(\omega)/\omega$ has the dimensions of viscosity and is generally called the *dynamic viscosity,* $\eta'(\omega)$

Table II

Primary Measurable Quantities

	Type of deformation			
Quantity	Shear	Bulk	Longitudinal (a) (no lateral stress)	Longitudinal (b) (no lateral strain)
Creep compliance	$J(t)$	$B(t)$	$D(t)$	-
Relaxation modulus	$G(t)$	$K(t)$	$E(t)$	-

Storage compliance	$J'(\omega)$	$B'(\omega)$	$D'(\omega)$	-
Loss compliance	$J''(\omega)$	$B''(\omega)$	$D''(\omega)$	-
Storage modulus	$G'(\omega)$	$K'(\omega)$	$E'(\omega)$	$M'(\omega)$
Loss modulus	$G''(\omega)$	$K''(\omega)$	$E''(\omega)$	$M''(\omega)$
Complex compliance	$J^*(\omega)$	$B^*(\omega)$	$D^*(\omega)$	
Complex modulus	$G^*(\omega)$	$K^*(\omega)$	$E^*(\omega)$	
Absolute compliance	$\lvert J(\omega) \rvert$	$\lvert B(\omega) \rvert$	$\lvert D(\omega) \rvert$	
Absolute modulus	$\lvert G(\omega) \rvert$	$\lvert K(\omega) \rvert$	$\lvert E(\omega) \rvert$	$\lvert M(\omega) \rvert$
Loss tangent (damping)	$\tan\ \delta_s$	$\tan\ \delta_v$	$\tan\ \delta_t$	$\tan\ \delta_p$
Dynamic viscosity	$\eta'_s(\omega)$	$\eta'_v(\omega)$	$\eta'_l(\omega)$	$\eta'_p(\omega)$

In principle, six moduli and compliances can be defined (compare Table I). In practice, three compliances and four moduli are sufficient. The three complex compliances thus the *complex shear compliance*, $J^*(\omega)$, the *complex bulk compliance*, $B^*(\omega)$, and the *complex longitudinal compliance* $D^*(\omega)$. The four moduli, are, respectively, $G^*(\omega)$, $K^*(\omega)$, $E^*(\omega)$, and $M^*(\omega)$. The relations between the complex moduli and compliances are the same as those existing between the corresponding moduli and compliances of classical elasticity theory. When necessary, values of $\tan\delta$ and $\eta'(\omega)$ corresponding to different moduli may be distinguished by subscripts, thus $\tan\delta_s$, $\tan\delta_v$, $\tan\delta_t$ and $\tan\delta_p$ represent the loss tangents associated, respectively, with $G^*(\omega)$, $K^*(\omega)$, $E^*(\omega)$, and $M^*(\omega)$. Similarly, the corresponding dynamic viscosities may be denotes as $\eta'_s(\omega)$, $\eta'_v(\omega)$, $\eta'_l(\omega)$, and $\eta'_p(\omega)$.

The behavior of a material in step-function or dynamic tests can thus be expressed in terms of the primary measurable quantities given in Table II

V. Limiting Values

In principle, certain limiting values exist, corresponding to zero and infinite time in step-function experiments, and zero and infinite frequency in dynamic experiments. These limiting values are defined as follows:

$$\left\{\begin{array}{c} k_g = \lim_{t\to 0} k(t) = \lim_{\omega\to\infty} k'(\omega) \\ m_g = \lim_{t\to 0} m(t) = \lim_{\omega\to\infty} m'(\omega) \\ k_g = 1/m_g \end{array}\right\} \tag{8}$$

The quantities k_g and m_g, are called, respectively, the *glass compliance* and *glass modulus*. The quantity defined usually by

$$\left\{\begin{array}{c} 1/\eta = \lim_{t\to\infty}\{dk(t)dt\} \\ \eta = \lim_{\omega\to 0} \eta'(\omega) \end{array}\right\} \tag{9}$$

is called the *viscosity*. Similarly, the quantities k_e and m_e are defined by the relations

$$\left\{\begin{array}{c} k_e = \lim_{t\to\infty}\{k(t) - t/\eta\} \lim_{\omega\to 0} k'(\omega) \\ m_e = \lim_{t\to\infty} m(t) = \lim_{\omega\to 0} m'(\omega) \\ k_e = 1/m_e \quad \text{if} \quad \eta = \infty \end{array}\right\} \tag{10}$$

The quantity k_e may be called the *equilibrium compliance* for a crosslinked material, for which the viscosity is infinite; otherwise, it may be called the *steady-state elastic compliance*. The quantity m_e differs from zero only when the viscosity is infinite, in which case it may be ealled the *equilibrium modulus*. In practice these limiting values may be experimentally inaccessible: or they may not be determinable experimentally.

VI. Equivalent Continuous Spectra

The creep compliances are related to the complex compliances, and the relaxation moduli are related to the complex moduli through retardation and relaxation spectra, respectively. These spectra may be in principle discrete, continuous, or mixed. Molecular theories of viscoelasticity of amorphous polymers lead to discrete spectra. Since it is not possible to obtain these from experimental data, it is customary to obtain from experimental data the equivalent continuous spectra defined below.

The *retardation spectrum* $L(t)$ is defined by the relationships

$$\left\{\begin{array}{c} k(t) - k_g - t/\eta = \int_{-\infty}^{\infty} \left(1 - e^{-t/\tau}\right) L(\tau) d\ln\tau \\ k'(\omega) - k_g = \int_{-\infty}^{\infty} \dfrac{L(\tau)}{1+\omega^2\tau^2} d\ln\tau \\ k''(\omega) - 1/\omega\eta = \int_{-\infty}^{\infty} \dfrac{\omega\tau L(\tau)}{1+\omega^2\tau^2} d\ln\tau \end{array}\right\} \tag{11}$$

and the *relaxation spectrum* $H(t)$ is defined by the relationships

$$\left\{\begin{array}{c} m(t) - m_e = \int_{-\infty}^{\infty} e^{-t/\tau} H(\tau) d\ln\tau \\ m'(\omega) - m_e = \int_{-\infty}^{\infty} \dfrac{\omega^2\tau^2 H(\tau)}{1+\omega^2\tau^2} d\ln\tau \\ m''(\omega) = \int_{-\infty}^{\infty} \dfrac{\omega\tau H(\tau)}{1+\omega^2\tau^2} d\ln\tau \end{array}\right\} \tag{12}$$

When *necessary*, the retardation and relaxation spectra for shear and for bulk deformation can he distinguished by subscripts, thus we may write $H_s(t)$ and $L_s(t)$ for shear, and $H_v(t)$ and $L_v(t)$ for bulk (volume) deformation. The terms *relaxation spectrum* and *retardation spectrum* refer to the function over the range of values of t from zero to

infinity. The value of $H(t)$ at a specified value of t may be termed the *spectral strength* at time t. The auxiliary quantities, namely, the limiting values and spectra, are listed in Table III. It will be noticed that in this table the viscosity is characterized by a suffix when it is required to distinguish between the longitudinal and shear viscosities; the bulk viscosity of course does not exist.

Table III
Auxiliary Quantities

	Type of deformation		
Quantity	Shear	Bulk	Longitudinal (no lateral stress)
Glass compliance	J_g	B_g	D_g
Equilibrium compliance	J_e	B_e	D_e
Steady-state elastic compliance			
Glass modulus	G_g	K_g	E_g
Equilibrium modulus	G_e	K_e	E_e
Viscosity	η_s	-	η_l
Retardation spectrum	$L_s(t)$	$L_v(t)$	$L_l(t)$
Relaxation spectrum	$H_s(t)$	$H_v(t)$	$H_l(t)$

VII. Discussion

In this report a minimum of quantities has been defined. It is observed that time- or frequency-dependent quantities are systematically labeled, while limiting quantities are denoted by suffixes. In the system used in this report, the connection between linear viscoelasticity and classical elasticity theory is apparent. For this reason, the more commonly used symbols for elastic moduli (G, K, E, and M) are adopted here. With regard to compliances, J is a generally used symbol, as is also B. The symbol D has been adopted for longitudinal compliance, replacing the previously used symbol F, to which objection was raised. Similarly, the symbol for the "dummy" compliance k replaces the earlier symbol c.

There are some difficulties with regard to nomenclature for the limiting values k_g, m_g, k_e, and m_e. The suffixes zero and infinity do not give rise to confusion in the analogous but not so complete system of nomenclature for dielectric behavior; with linear

viscoelasticity, however, their use is unjustified. The terms "initial" and "instantaneous" modulus and compliance seem uncertain. For amorphous polymers the appropriate modulus and compliance are often obtained as limiting values when the timescale is short enough at any given temperature, the material behaving under these conditions as an organic glass. On this account the terms glass modulus and glass compliance have been chosen, and these quantities have been denoted by m_g and k_g. For other materials, for example, metals, another term would be preferable. It does not seem likely that a term for these limiting values suitable for all eategories of materials can be proposed. For the other limiting values, namely, k_e, and m_e, the terms "static" and "ultimate" have been suggested. Here again these do not seem to be an improvement over the terms proposed in this report, The letter suffix for the symbol should of course be related to the name adopted for the limiting value. For papers published in languages other than English, it would be desirable that the letter suffixes adopted be associated with the nomenelature for the limiting values in other languages. This desirable state of affairs would seem to be diffieult to accomplish.

In the case of nonlinear behavior, the term "viscosity" is sometimes used to denote the ratio of shear stress to rate of shear in general, while the limiting value at very small values of the parameters is denoted by a term such as "Newtonian" viscosity. Since in this report we are dealing only with linear behavior, it seems that such a modifying term is unnecessary here, Objections have been raised to the term "dynamic viscosity" for $m''(\omega)/\omega$. Since this name is well established for a useful quantity, it has heen adopted in this report.

Much of the material of previous preliminary versions of this report has been eliminated, in line with the objective of defining a minimum number of quantities in order to promote uniform methods of reporting data. In theoretical treatments it may be convenient to define further functions, for example:

$$\{k(t) - k_g - t/\eta\}/(k_e - k_g)$$

It does not seem either necessary r desirable to standardize a nomenclature for such quantities at this stage.

The Rheology Bulletin, Vol 27, No. 1 Spring, 1958, pg. 1 (98)

International Rheology Congress

The program will include a number of lectures by invited guests, and technical sessions devoted to: (1) Fundamental basis of rheology, (2) Measuring techniques, (3) Experimental results, and (4) Practical applications. The scope of the program is considered to include the entire field of deformation and flow of matter, including

molecular interpretation and engineering applications, but not the classical theories of elasticity, aerodynamics, atomic processes (dislocations), etc., which are considered to be branches of applied mechanics. A discussion on nomenclature is planned, an exhibit of measuring equipment and apparatus is to be held, and a ladies' program will be arranged. Papers may be given in English, French, German or Russian, and manuscripts likewise submitted in these languages. It is planned to have the lectures which are delivered in Russian translated simultaneously into English, French and German

TRANSACTIONS OF THE SOCIETY OF RHEOLOGY, VOL III, 205-206 (1959)

Communication Problems in Industrial Rheology

Committee Recommendations for Describing Flow Phenomena

Increased study of the flow characteristics of polymers and dispersions has revealed a lack of authoritative definitions and standardized terminology for stress-dependent flow phenomena. Confusion resulting from these deficiencies has become so acute that a Committee on Communication Problems in Rheology has been organized to establish areas of agreement in describing the flow characteristics of materials.

The Committee is sponsored by several American Chemical Society divisions (Agricultural and Food, Colloid, Industrial and Engineering, Paint, Petroleum, and Polymer), the Society of Rheology, and the American Society for Testing Materials.

Before delving into more controversial details, including the definition of viscosity, the Committee considered that it must first establish and define those broad aspects of flow concerning which no serious disagreement exists among practicing rheologists. It was decided that the following realities must be recognized as the framework upon which definitions and terms can be built:

1. The basic measured quantities are shear stress (S), shear rate (D), and time.
2. Shear-dependent properties are necessarily time-dependent, and confusion results from attempts to measure both phenomena simultaneously.
3. The terms used to describe flow behavior are the main source of confusion; the concepts are fairly well established.

The recommendation is made that flow phenomena be described using a broad classification of behavior based on the dependence of applied stress on shear rate. Three broad areas of flow behavior are recognized. They are based on the ratio, S/D. A constant

S/D ratio is characteristic of a Newtonian system; a decreasing ratio with increased D is shear thinning, and the material it characterizes is a shear-thinning system; an increasing ratio with increased D is shear thickening, and the material it characterizes is a shear-thickening system.

The Committee recommends further that flow phenomena be expressed in terms of the directly measured variables S and D and that reference to terms such as viscosity and fluidity either be avoided or qualified whenever the possibility exists for their misinterpretation in a quantitative sense. It is essential that the dependence of S on D be determined under steady-state conditions.

The usage of new words was considered by the Committee which sampled the opinions of academic and industrial personnel engaged in making and describing measurements of flow behavior. Sentiment was strongly in favor of reducing the number of terms used to describe the three main types of flow behavior, rather than adding two additional terms.

The scientific community is strongly urged to adopt the terms shear thinning and shear thickening in future communications regarding shear-sensitive flow behavior.

The Rheology Bulletin, Vol 38, No. 3 December 1969, pg. 1 (215)

Rheology Nomenclature

The Executive Committee is interested in receiving expressions of interest from members concerning nomenclature. The time appears ripe for a second ad hoc committee on communication problems in rheology with emphasis this time on symbols and with viscoelasticity in general. [See Transactions SOR, Volume III, 205-206 (1959)] Inform your Editor if you are interested in chairing or serving on such a committee. Even if your answer to that question is no, we would be interested in your advice.

The Rheology Bulletin, Vol 39, No. 2 August 1970, pg. 3 (10)

Society of Rheology Executive Committee Meeting

18) A committee on rheological nomenclature will be appointed by the President.

The Rheology Bulletin, Vol 39, No. 3 December 1970, pg. 2 (12)

The Nomenclature Committee will make recommendations for increasing the uniformity of usage of rheological terms and symbols.

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 40, No. 1 April 1971, pg. 2 (17)

Nomenclature

After one meeting and an exchange of correspondence the recently appointed Nomenclature Committee has approved some notations which are offered as an interim report by chairman Eirich as follows:

Tentative agreement has been arrived at between most Committee members on the following items: 'Steady Shearing Flow', π, Normal Stress Coefficients', θ and β; 'Stress Ratio', $(p_{11} - p_{22})/p_{21}$; 'Velocity Gradient', γ; endorsement for the term 'Intrinsic Viscosity and rejection of the term, 'Apparent Viscosity'; replacement of 'Real' and 'Imaginary' by 'In-phase' and 'Out-of Phase', Work on other definitions relating to Time Dependent, Dynamic, and Second Order quantities is in progress. Anyone who feels very strongly on these points is invited to communicate with the Committee Chairman, F. R. Eirich

The Rheology Bulletin, Vol 42, No. 2 September 1973, pg. 2 (54)

1954 Winter Meeting The Society of Rheology

President's Message

The reconstituted Nomenclature Committee, with C. Sieglaff as chairman, has made substantial progress on agreeing on a nomenclature in dilute and moderately dilute systems. The proposals will be brought up for final discussion and approval before the Executive Committee meeting in Montreal. Anyone who feels strongly on certain points of nomenclature and wishes to communicate with this Committee is encouraged to do so. The importance of arriving at a consistent, acceptable and physically meaningful rheological nomenclature has increased greatly in recent years due to the need for broader interdisciplinary communication. The Committee on Nomenclature of the Colloid Division of IUPAC will consider nomenclature proposals at the IUPAC meeting in Munich in August. Your president will contribute to that meeting in view of the many rheological terms that are employed by colloid and surface chemists, and vice versa.

The Rheology Bulletin, Vol 43, No. 2 August 1974, pg. 2 (66)

Rheology Nomenclature

The nomenclature committee (C. Sieglaff, Chairman, R. Bird, R. Dickie, F. Eirich, H. Markovitz, R. Myers) has circulated to the Executive Committee of the Society of Rheology the attached nomenclature list. The Executive Committee has accepted this list for use by the Society. It is recommended that the members of the Society adhere to this nomenclature for their communications to the Society.

Transcribed by James J Griebler, April 2026

Reference should be made to earlier communications by H. Leaderman and by a committee established by the Society and chaired by R. R. Myers. These recommendations were published in the Transactions 1, 213 (1957) and 3, 205 (1959).

These recommendations will be published in the Transactions after a reasonable time for communications by the membership to Dr. Sieglaff, Diamond Shamrock Co., Painesville, Ohio

The Rheology Bulletin, Vol 43, No. 2 August 1974, pg. 6 (70)

Nomenclature Committee Recommendations

Nomenclature for Simple Steady Shear Flow

$[v_x = f(v_y)$ or $v_1 = f(x_2)$

Symbol	Name	Definition	Units
X,1	Geometric direction of flow		
Y,2	Direction of velocity change		
Z,3	Neutral Direction		
$\sigma \equiv \sigma_{21}$	Shear stress in simple steady shear flow		dynes cm^{-2} (newtons m^{-2})
$\dot{\gamma}$	Shear rate	dv_x/dy	sec^{-1}
η	Viscosity	$\sigma/\dot{\gamma}$	poise (dyne sec cm^{-2})
N_1	First normal stress function	$\sigma_{11} - \sigma_{22}$	dynes cm^{-2} (newtons m^{-2})
N_2	Second normal stress function	$\sigma_{22} - \sigma_{33}$	dynes cm^{-2} (newtons m^{-2})
Ψ_1	First normal stress coefficient	$N_1/\dot{\gamma}^2$	dynes sec^2 cm^{-2}
Ψ_2	Second normal stress coefficient	$N_2/\dot{\gamma}^2$	dynes sec^2 cm^{-2}
η_0	Limiting viscosity at zero shear rate	$\lim_{\dot{\gamma}\to 0} \sigma/\dot{\gamma}$	poise (dyne sec cm^{-2})
η_∞	Limiting viscosity at infinite shear rate	$\lim_{\dot{\gamma}\to \infty} \sigma/\dot{\gamma}$	poise (dyne sec cm^{-2})
η_s	Viscosity of solvent or continuous medium	-	poise (dyne sec cm^{-2})
η_r	Realtive viscosity	η/η_s	Dimensionless
η_{sp}	Specific viscosity	η_r^{-1}	Dimensionless

Transcribed by James J Griebler, April 2026

$[\eta]$	Intrinsic viscosity	$\lim_{C \to 0.} \frac{\eta_r - 1}{C}$	cm^{-3} or units of C

Nomenclature for Small Amplitude Oscillatory Motion

Symbol	Name	Units
ω	Angular frequency	sec^{-1}
t	Direction of velocity change	sec
τ	Relaxation time or retardation time	sec
η^*	Complex dynamic shear viscosity	dyne sec cm^{-2}
η'	Dynamic viscosity	
η''	Out-of-phase component of η^*	
B^*	Bulk dynamic complex compliance	dyne cm^{-2}
B'	Bulk storage compliance	
B''	Bulk loss compliance	
$B(t)$	Bulk creep compliance	
D^*	Tensile dynamic complex compliance	dyne cm^{-2}
D'	Tensile storage compliance	
D''	Tensile loss compliance	
$D(t)$	Tensile creep compliance	
E^*	Tensile dynamic complex modulus	dyne cm^{-2}
E'	Tensile storage modulus	
E''	Tensile loss modulus	
$E(t)$	Tensile creep modulus	
G^*	Shear dynamic complex modulus	dyne cm^{-2}
G'	Shear storage modulus	
G''	Shear loss modulus	
$G(t)$	Shear creep modulus	
J^*	Shear dynamic complex compliance	cm^2 $dyne^{-1}$
J'	Shear storage compliance	
J''	Shear loss compliance	
$J(t)$	Shear creep compliance	
K^*	Bulk dynamic complex modulus	dyne cm^{-2}
K'	Bulk storage modulus	
K''	Bulk loss modulus	
$K(t)$	Bulk creep modulus	
μ^*	Complex dynamic Poisson's ratio	dimensionless
μ'	In-phase component of μ^*	
μ''	Out-of-phase component of μ^*	

Nomenclature for Small Deformation Linear Elasticity

Symbol	Name	Units
Simple Shear		

G	Shear modulus – modulus of rigidity	dyne cm^{-2}
J	Shear compliance	cm^2 $dyne^{-1}$
Bulk (Isotropic) Compression		
K	Bulk modulus	dyne cm^{-2}
B	Bulk compliance	cm^2 $dyne^{-1}$
Tensile Extension		
E	Young's modulus (tensile modulus)	dyne cm^{-2}
D	Tensile compliance	cm^2 $dyne^{-1}$
μ	Poisson's ratio	dimensionless

The Rheology Bulletin, Vol 44, No. 2 August 1975, pg. 2 (79)

VIIth International Congress on Rheology

Nomenclature

The final version of the report on the Proposed Nomenclature for Steady Shear Flow and Linear Viscoelastic Behavior has been prepared by the Nomenclature Committee and will be published in a forthcoming issue of the Transactions.

The Rheology Bulletin, Vol 46, No. 1 August 1977, pg. 3 (99)

Volume 20 appeared on its regular quarterly schedule. It contained 654 pages, comprising 34 papers, three notes and a report from the Nomenclature Committee. This volume honored Past President Eirich in Issue 1 and Bingham Medalist Gent in Issue 2.

Transactions of The Society of Rheology 20, 311–317 (1976)

Proposed Nomenclature for Steady Shear Flow and Linear Viscoelastic Behavior

Eighteen years ago Leaderman [Trans. Soc. Rheol., 1,213 (1957)] published a nomenclature for linear viscoelasticity. Since that time the science of rheology has matured and its use in other disciplines has grown. In order to facilitate communication between these diverse disciplines, a systematic organization of names and symbols is very desirable.

In 1970, F. R. Eirich, as vice-president of the Society of Rheology, formed an ad hoc committee to review the status of rheological nomenclature. This committee first convened at the Princeton Annual Society meeting. The committee then agreed that a uniform system of nomenclature for material properties, such as moduli, viscosities, etc., was needed. It was also agreed that the existing prevalent usages should serve as guidelines for proposed nomenclature.

Subsequent to the original meeting, several approaches to a uniform system of nomenclature were explored. The results were distilled down to the nomenclature list proposed in the Rheology Bulletin, Vol. 43. Several corrections and comments have been included in the final nomenclature published in this report. A major addition is the use of SI units, as well as the cgs system. The committee recommends that the Society of Rheology members adopt this nomenclature in their communications to the Society. The committee expressly recommends standardization of those symbols and notations representing material constants, material functions, and notations for the directions of shearing flow. The present committee did not consider it necessary or advisable to develop a dictionary of (phenomenological) behavior. A dictionary of this type by M. Reiner and G. W. Scott Blair is currently available (Rheology, Vol. 4, Academic Press, New York, 1967). Also, the systemization of non-linear nomenclature was considered to be premature. A future committee should be organized to consider this vital area of rheology.

The chairman wished to thank the committee members: R. B. Bird R. A. Dickie }'. R. Eirich H. Markovitz R. R. Myers for their diligent efforts in compiling the nomenclature list. Also, the committee wishes to express its gratitude to the Society members who made many thoughtful inputs to this endeavor.

Transcribed by James J Griebler, April 2026

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			Units				
			SI System			cgs System	
Symbol	Name	Definition	Name	Symbol	Dimensions	Name	Units
$x, 1$	Geometric direction of flow	–	–	–	–	–	–
$y, 2$	direction of velocity change	–	–	–	–	–	–
$z, 3$	neutral direction	–	–	–	–	–	–
$\sigma \equiv \sigma_{21}$	shear stress in simple steady shear flow	–	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-2}$ or (N/m^2)	–	dyne cm^{-2}
$\dot{\gamma}$	shear rate	dv_x/dy	s^{-1}	s^{-1}	s^{-1}	–	s^{-1}
η	viscosity	$\sigma/\dot{\gamma}$	Pascal second	Pa·s	$m^{-1}\cdot kg\cdot s^{-1}$ or $(N\cdot s/m^2)$	Poise	dyne·s·cm^{-2}
N_1	first normal stress function	$\sigma_{11} - \sigma_{22}$	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-2}$ or (N/m^2)	–	dyne cm^{-2}
N_2	second normal stress function	$\sigma_{22} - \sigma_{33}$	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-2}$ or (N/m^2)	–	dyne cm^{-2}

Transcribed by James J Griebler, April 2026

Ψ_1	first normal stress coefficient	$N_1/\dot{\gamma}^2$	–	Pa·s^2	m^{-1}·kg	–	dyne·s^2·cm^{-2}
Ψ_2	second normal stress coefficient	$N_2/\dot{\gamma}$	–	Pa·s^2	m^{-1}·k	–	dyne·s^2·cm^{-2}
η_0	limiting viscosity at zero shear rate	$\lim_{\dot{\gamma}\to 0} \sigma/\dot{\gamma}$	Pascal second	Pa·s	m^{-1}·kg·s^{-1}	Poise	dyne·s·cm^{-2}
η_∞	limiting viscosity at infinite shear rate	$\lim_{\dot{\gamma}\to \infty} \sigma/\dot{\gamma}$	Pascal second	Pa·s	m^{-1}·kg·s^{-1}	Poise	dyne·s·cm^{-2}
η_s	viscosity of solvent or continuous medium	–	Pascal second	Pa·s	m^{-1}·kg·s^{-1}	Poise	dyne·s·cm^{-2}
η_r	relative viscosity	η/η_s	dimensionless			dimensionless	
η_{sp}	specific viscosity	$\eta_r - 1$	dimensionless			dimensionless	
$[\eta]$	intrinsic viscosity	$\lim_{C\to 0} \frac{\eta_r - 1}{C}$	–	–	m^3·kg^{-1}	–	cm^3·g^{-1} or specified concentration units

Transcribed by James J Griebler, April 2026

Nomenclature for Small Amplitude Oscillatory Motion

		Units				
		SI System			cgs System	
Symbol	Name	Name	Symbol	Dimensions	Name	Units
ω	angular frequency	hertz or radians per second	Hz rad/s	s^{-1}	—	s^{-1}
t	time	second	s	—	second	s
τ	relaxation time or retardation time	second	s	—	second	s
η^*	complex dynamic shear viscosity	Pascal second	Pa·s	$m^{-1}\cdot kg\cdot s^{-1}$ or $(N\cdot s/m^2)$	Poise	$dyne\cdot s\cdot cm^{-2}$
η'	dynamic viscosity					
η''	out-of-phase component of η^*					
B^*	bulk dynamic complex compliance	—	Pa^{-1}	$m\cdot kg^{-1}\cdot s^2$ or (m^2/N)	—	$dyne^{-1}\ cm^2$
B'	bulk storage compliance					
B''	bulk loss compliance					
$B(t)$	bulk creep compliance					
D^*	tensile dynamic complex compliance	—	Pa^{-1}	$m\cdot kg^{-1}\cdot s^2$ or (m^2/N)	—	$dyne^{-1}\ cm^2$
D'	tensile storage compliance					
D''	tensile loss compliance					
$D(t)$	tensile creep compliance					

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E^*	tensile dynamic complex modulus	Pascal	Pa	$m^{-1} \cdot kg \cdot s^{-2}$	—	$dyne \cdot cm^{-2}$
E'	tensile storage modulus					
E''	tensile loss modulus					
$E(t)$	tensile creep modulus					
G^*	shear dynamic complex modulus	Pascal	Pa	$m^{-1} \cdot kg \cdot s^{-2}$	—	$dyne \cdot cm^{-2}$
G'	shear storage modulus					
G''	shear loss modulus					
$G(t)$	shear creep modulus					
J^*	shear dynamic complex compliance	—	Pa^{-1}	$m \cdot kg^{-1} \cdot s$	—	$dyne^{-1}\ cm^2$
J'	shear storage compliance					
J''	shear loss compliance					
$J(t)$	shear creep compliance					
K^*	bulk dynamic complex modulus	Pascal	Pa	$m^{-1} \cdot kg \cdot s^{-2}$	—	$dyne \cdot cm^{-2}$
K'	bulk storage modulus					
K''	bulk loss modulus					
$K(t)$	bulk creep modulus					
μ^*	complex dynamic Poisson's ratio	dimensionless				–
μ'	in-phase component of μ^*					
μ''	out-pf-phase component of μ^*					

Nomenclature for Small Deformation Linear Elasticity

Symbol	Name	Units				
		SI System			cgs System	
		Name	Symbol	Dimensions	Name	Units
		Simple Shear				
G	shear modulus; modulus of rigidity	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-2}$	—	$dyne\cdot cm^{-2}$
J	shear compliance	—	Pa^{-1}	$m\cdot kg^{-1}\cdot s$	—	$dyne^{-1}\ cm^{2}$
		Bulk (Isotropic) Compression				
K	bulk modulus	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-1}$	—	$dyne\cdot cm^{-2}$
B	bulk compliance	—	Pa^{-1}	$m\cdot kg^{-1}\cdot s$	—	$dyne^{-1}\ cm^{2}$
		Tensile Extension				
E	Young's modulus (tensile modulus)	Pascal	Pa	$m^{-1}\cdot kg\cdot s^{-2}$	—	$dyne\cdot cm^{-2}$
D	tensile compliance	—	Pa^{-1}	$m\cdot kg^{-1}\cdot s$	—	$dyne^{-1}\ cm^{2}$
μ	Poisson's ratio	dimensionless			—	

Transcribed by James J Griebler, April 2026

The Rheology Bulletin, Vol 50, No. 1 May 1981, pg. 2 (12)

Minutes of Executive Committee Meeting

It was also noted that J. Dealy has expressed concern over the need for nomenclature clarification in extensional flow. It was agreed that Dealy should be appointed chairman of a nomenclature committee and should present a proposal to the Executive Committee on the objectives to be achieved, with proper regard for the prior work done under the Chairmanship of C. Sieglaff.

The Rheology Bulletin, Vol 50, No. 2 July 1981, pg. 2 (18)

Minutes of Executive Committee Meeting

President Bagley announced the creation of a new Nomenclature Committee (J. Dealy, Chairman; R. Bird; W. Graessley; and K. Wissbrun). The committee will be asked to focus on nomenclature for extensional and transient flows, to coordinate their efforts with the ICR, and to submit the results of Editor Myers for publication in the Journal of Rheology

The Rheology Bulletin, Vol 51, No. 1 January 1982, pg. 2 (20)

Minutes of Executive Committee Meeting

The report of the Nomenclature Committee (J. Dealy, chairman) was read by President Bagley, who also related that the question of the Canadian Division has been satisfactorily resolved without an amendment. The Society's representative to the International Congress on Rheology, J. R. Knox, reported on progress toward a Congress in Mexico in 1984. Bagley has volunteered to the ICR the services of The Society to help with the planning, but advised President-elect A. Gent to contact Baltasar Mena of the Mexican Society of Rheology directly

The Rheology Bulletin, Vol 52, No. 2 August 1983, pg. 2 (35)

The January 1983 Bulletin contained six pages, plus an abstract form and the report of the nomenclature committee. This bulletin was shipped to AIP in New York on January 18; our copy arrived on February 21.

Transcribed by James J Griebler, April 2026

Journal of Rheology, 28, 181-195 (1984)

Official Nomenclature for Material Functions Describing the Response of a Viscoelastic Fluid to Various Shearing and Extensional Deformations

Introduction

In the spring of 1981, the Executive Committee of the Society appointed an ad hoc nomenclature committee to recommend standard names and symbols for material functions arising in the study of nonlinear viscoelasticity and of extensional flows. Two previous nomenclature committees':" had prepared nomenclature lists for linear viscoelastic functions and for viscometric flows, and the recommendations of those committees are summarized in Tables I and II

Table I Society of Rheology Nomenclature for Steady Simple Shear

Quantity	Symbol	S.I. units	CGS units
Direction of flow	x_1 or x	m	cm
Direction of velocity gradient	x_2 or y	m	cm
Neutral Direction	x_3 or z	m	cm
Shear stress	σ	Pa	dynes cm^{-2}
Shear strain	γ	—	
Shear rate	$\dot{\gamma}$	s^{-1}	s^{-1}
Viscosity	η	Pa s	P (poise)
First normal stress function	N_1	Pa	dynes cm^{-2}
Second normal stress function	N_2	Pa	dynes cm^{-2}
First normal stress coefficient	Ψ_1	Pa s^2	dynes sec^2 cm^{-2}
Second normal stress coefficient	Ψ_2	Pa s^2	dynes sec^2 cm^{-2}
Limiting viscosity at zero shear rate	η_0	Pa s	P
Limiting viscosity at infinite shear rate	η_∞	Pa s	P
Viscosity of solvent or continuous medium	η_s	Pa s	P
Realtive viscosity	η_r	—	—
Specific viscosity	η_{sp}	—	—
Intrinsic viscosity	$[\eta]$	m^3kg^{-1}	cm^3g^{-1}

Transcribed by James J Griebler, April 2026

Table II Society of Rheology Nomenclature for Linear Viscoelasticity

Quantity	Symbol	S.I. units
Simple shear		
Shear strain	γ	—
Shear modulus (modulus of rigidity)	G	Pa
Shear relaxation modulus	$G(t)$	Pa
Shear compliance	J	Pa^{-1}
Shear creep compliance	$J(t)$	Pa^{-1}
Equilibrium shear compliance	J_e	Pa^{-1}
Steady-state shear compliance	J_s^0	Pa^{-1}
Complex viscosity	$\eta^*(\omega)$	Pa s
Dynamic viscosity	$\eta'(\omega)$	Pa s
Out-of-phase component of η^*	$\eta''(\omega)$	Pa s
Complex shear modulus	$G^*(\omega)$	Pa
Shear storage modulus	$G'(\omega)$	Pa
Shear loss modulus	$G''(\omega)$	Pa
Complex shear compliance	$J^*(\omega)$	Pa^{-1}
Shear storage compliance	$J'(\omega)$	Pa^{-1}
Shear loss compliance	$J''(\omega)$	Pa^{-1}
Tensile Extension		
Hencky strain	ϵ	—
Young's modulus	E	Pa
Tensile relaxation modulus	$E(t)$	Pa
Tensile compliance	D	Pa^{-1}
Tensile creep compliance	$D(t)$	Pa^{-1}

However, the increased use of large transient shear flows and extensional flows to study viscoelastic materials made it desirable to augment these lists. A tentative proposal was prepared by the Committee and sent to members of the Society at the end of 1982 along with a request for corrections and comments. The Committee studied the responses to this request and prepared a final report that was presented to the Executive Committee at the Knoxville meeting in October 1983. Presented here is the nomenclature approved at that time. The Committee wishes to acknowledge the helpful suggestions received from a number of members. Particularly useful were the thoughtful and extensive comments received from C.J.S. Petrie, J. Meissner, and W. Philippoff. This presentation commences with descriptions of a number of flows that have been introduced in the last 15years for the study of viscoelastic materials. A number of material functions are defined, and symbols are specified for these. The names of the newly defined material functions and the corresponding symbols are listed in Tables III and IV

Table III Society of Rheology Nomenclature for Nonlinear Viscoelasticity in Shear

Quantity	Symbol	S.I. units
Start-up flow		
Shear stress growth function	$\sigma^+(t,\dot{\gamma})$	Pa
Shear stress growth coefficient	$\eta^+(t,\dot{\gamma})$	Pa s
First normal stress growth function	$N_1^+(t,\dot{\gamma})$	Pa
First normal stress growth coefficient	$\Psi_1^+(t,\dot{\gamma})$	Pa s^2
Second normal stress growth function	$N_2^+(t,\dot{\gamma})$	Pa
Second normal stress growth coefficient	$\Psi_2^+(t,\dot{\gamma})$	Pa s^2
Cessation of steady shear flow		
Shear stress decay function	$\sigma^-(t,\dot{\gamma})$	Pa
Shear stress decay coefficient	$\eta^-(t,\dot{\gamma})$	Pa s
First normal stress decay function	$N_1^-(t,\dot{\gamma})$	Pa
First normal stress decay coefficient	$\Psi_1^-(t,\dot{\gamma})$	Pa s^2
Second normal stress decay function	$N_2^-(t,\dot{\gamma})$	Pa
Second normal stress decay coefficient	$\Psi_2^-(t,\dot{\gamma})$	Pa s^2
Step strain		
Shear stress relaxation function	$\sigma(t,\gamma)$	Pa
Shear stress relaxation modulus	$G(t,\gamma)$	Pa
First normal stress relaxation function	$N_1(t,\gamma)$	Pa
Second normal stress relaxation function	$N_2(t,\gamma)$	Pa
Creep and recoil		
Shear creep compliance	$J(t,\sigma)$	Pa^{-1}
Steady-state compliance	$J_s(\sigma)$	Pa^{-1}
Recoil strain	$\gamma_r(t,\sigma)$	–
Recoil function	$R(t,\sigma)$	Pa^{-1}
Ultimate recoil	$\gamma_\infty(\sigma)$	–
Ultimate recoil function	$R_\infty(\sigma)$	Pa^{-1}
Superimposed steer and oscillatory shear		
Parallel complex viscosity	$\eta_\parallel^*(\omega,\dot{\gamma}_m)$	Pa s
Orthogonal complex viscosity	$\eta_\perp^*(\omega,\dot{\gamma}_m)$	Pa s

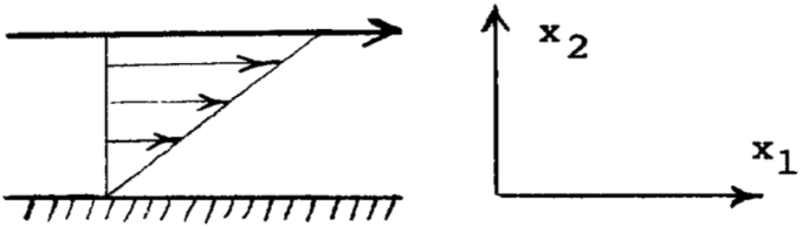
Table IV Society of Rheology Nomenclature for Nonlinear Viscoelasticity in Extension

Quantity	Symbol	S.I. units
Tensile (simple extension)		
Tensile strain	ϵ	—
Strain rate (≥ 0)	$\dot{\epsilon}$	—
Net tensile stress	σ_E	Pa
Tensile stress growth function	$\eta_E^+(t,\dot{\epsilon})$	Pa s
Tensile viscosity	η_E	Pa s
Tensile stress decay coefficient	$\eta_E^-(t,\dot{\epsilon})$	Pa s
Tensile creep compliance	$D(t,\sigma_E)$	Pa^{-1}
Steady-state tensile compliance	$D(\sigma_E)$	Pa^{-1}
Tensile recoil function	$\epsilon_r(t,\sigma_E)$	—
Tensile recoil coefficient	$S(t,\sigma_E)$	Pa^{-1}
Ultimate tensile recoil	$\epsilon_\infty(t,\sigma_E)$	—
Ultimate tensile recoil coefficient	$S(t,\sigma_E)$	Pa^{-1}
Tensile stress relaxation modulus	$E(t,\epsilon)$	Pa
Biaxial extension (symmetric)		
Biaxial strain/strain rate (≥ 0)	$\epsilon_B/\dot{\epsilon}_B$	$-/s^{-1}$
Net stretching stress	σ_B	Pa
Biaxial stress growth coefficient	$\eta_B^+(t,\dot{\epsilon}_B)$	Pa s
Biaxial stress decay coefficient	$\eta_B^-(t,\dot{\epsilon}_B)$	Pa s
Biaxial extension viscosity	η_B	Pa s
Asymmetric extension		
Largest principal strain rate	$\dot{\epsilon}$	s^{-1}
Strain ratio	m	—
First net stretching stress	$\sigma_1^{(m)}$	Pa
Second net stretching stress	$\sigma_2^{(m)}$	Pa
First extensional viscosity	$\eta_1^{(m)}$	Pa s
Second extensional viscosity	$\eta_2^{(m)}$	Pa s

Transient shear flows

All the tests and material functions defined below are based on a homogeneous simple shear flow, described as follows:

$$v_1 = \dot{\gamma} x_2 ,$$
$$v_2 = v_3 = 0$$



1. Start-up flow

A sample, initially in its rest state, is subjected to a constant shear rate $\dot{\gamma}$ at time $t = 0$. The quantities measured are the shear stress σ and the normal stress differences N_1 and N_2, all as functions of t and $\dot{\gamma}$. The measurable material functions are as follows:

Shear stress growth function:

$$\sigma^+(t,\dot{\gamma}) \equiv \sigma$$

Shear stress growth coefficient:

$$\eta^+(t,\dot{\gamma}) \equiv \sigma^+/\dot{\gamma}$$

First normal stress growth function:

$$N_1^+(t,\dot{\gamma}) \equiv \sigma_{11} - \sigma_{22}$$

First normal stress growth coefficient:

$$\Psi_1^+(t,\dot{\gamma}) \equiv N_1^+/\dot{\gamma}^2$$

Second normal stress growth function:

$$N_2^+(t,\dot{\gamma}) \equiv \sigma_{22} - \sigma_{33}$$

Second normal stress growth coefficient:

$$\Psi_2^+(t,\dot{\gamma}) \equiv N_2^+/\dot{\gamma}^2$$

Note 1: If the stresses approach steady-state values, we can make the following correspondences with the viscometric functions:

$$\lim_{t\to\infty}[\eta^+(t,\dot{\gamma})] = \eta(\dot{\gamma})$$

$$\lim_{t\to\infty}[N_1^+(t,\dot{\gamma})] = N_1(\dot{\gamma})$$

$$\lim_{t\to\infty}[N_2^+(t,\dot{\gamma})] = N_2(\dot{\gamma})$$

Note 2: If linear viscoelastic behavior is exhibited by the fluid of interest at very low :y, the following relationships are valid:

$$\lim_{\dot{\gamma}\to 0}[\eta^+(t,\dot{\gamma})] = \eta^+(t)$$

$$\lim_{\dot{\gamma}\to 0}[\Psi_1^+(t,\dot{\gamma})] = \Psi_1^+(t)$$

$$\lim_{\dot{\gamma}\to 0}[\Psi_2^+(t,\dot{\gamma})] = \Psi_2^+(t)$$

2. Cessation of Steady Shear Flow

A fluid that has been subjected to steady simple shear flow at a rate $\dot{\gamma}$, until its stresses are steady, is brought suddenly to rest at time $t = 0$. The stresses are monitored as functions of time, and the measurable material functions are as follows:

Shear stress decay function:

$$\sigma^-(t, \dot{\gamma}) \equiv \sigma$$

Shear stress decay coefficient:

$$\eta^-(t, \dot{\gamma}) \equiv \sigma^-/\dot{\gamma}$$

First normal stress decay function:

$$N_1^-(t, \dot{\gamma}) \equiv \sigma_{11} - \sigma_{22}$$

First normal stress decay coefficient:

$$\Psi_1^-(t, \dot{\gamma}) \equiv N_1^-/\dot{\gamma}^2$$

Second normal stress decay function:

$$N_2^-(t, \dot{\gamma}) \equiv \sigma_{22} - \sigma_{33}$$

Second normal stress decay coefficient:

$$\Psi_2^-(t, \dot{\gamma}) \equiv N_2^-/\dot{\gamma}^2$$

Note: If the fluid of interest exhibits linear viscoelasticity in the limit of small shear rates, the following relationships are valid:

$$\lim_{\dot{\gamma}\to 0}[\eta^-(t, \dot{\gamma})] = \eta^-(t)$$

$$\lim_{\dot{\gamma}\to 0}[\Psi_1^-(t, \dot{\gamma})] = \Psi_1^-(t)$$

$$\lim_{\dot{\gamma}\to 0}[\Psi_2^-(t, \dot{\gamma})] = \Psi_2^-(t)$$

3. Step strain

A material initially in its rest state is subjected to a sudden shear strain of magnitude γ at time $t = 0$, and the stresses are observed as functions of time.

Shear stress relaxation function:

$$\sigma(t, \gamma) \equiv \sigma$$

Shear stress relaxation modulus:

$$G(t,\gamma) \equiv \sigma/\gamma$$

Note: In the case of linear viscoelastic behavior

$$\lim_{\gamma \to 0}[G(t,\gamma)] = G(t)$$

First normal stress relaxation function:

$$N_1(t,\gamma) \equiv \sigma_{11} - \sigma_{22}$$

Second normal stress relaxation function:

$$N_1(t,\gamma) \equiv \sigma_{22} - \sigma_{33}$$

4. Creep

A material initially in its rest state is subjected to a constant shear stress σ at time $t = 0$. The shear strain γ is monitored as a function of time.

Shear creep compliance:

$$J(t,\sigma) \equiv \gamma/\sigma$$

If the material of interest is a fluid, so that the strain ultimately becomes linear with time, the compliance curve can be extrapolated to $t = 0$ to determine the steady-state compliance J_s. Thus, in the linear portion of the curve, the compliance is given by

$$J(t,\sigma) = J_s(\sigma) + t/\eta$$

where η is evaluated at the shear stress σ.

Note: When the material of interest exhibits linear viscoelastic behavior at small shear stresses, we have:

$$\lim_{\sigma \to 0}[J(t,\sigma)] = J(t)$$

and after the strain becomes a linear function of time

$$J(t) = J_s^0 + t/\eta_0,$$

where J_s^0 is the steady-state compliance in the limiting case of small stress.

5. Recoil

A fluid in which the shear stress σ and the shear rate $\dot{\gamma}$ are constant in time has its shear stress reduced to zero at time $t = 0$ while being constrained in the x_2 direction. The recoil strain is monitored as a function of time, and is considered positive when it is in a direction opposite to that of the original shearing motion:

Recoil strain:

$$\gamma_r \equiv \gamma(0) - \gamma(t)$$

Recoil function:

$$R(t,\sigma) \equiv \gamma_r/\sigma$$

Ultimate recoil:

$$\gamma_\infty(\sigma) = \lim_{t\to\infty}[\gamma_r(t,\sigma)]$$

Ultimate recoil function:

$$R_\infty(\sigma) = \lim_{t\to\infty}[R(t,\sigma)]$$

Note 1: Since $\sigma = \eta\dot{\gamma}$ at $t = 0$, the strain rate can be used in place of the stress as an independent variable.

Note 2: If linear viscoelastic behavior is exhibited in the limit of small initial stress, we have

$$\lim_{\sigma\to 0}[R(t,\sigma)] = R(t) = J(t) - t/\eta_0$$

$$\lim_{\sigma\to 0}[R_\infty(t)] = R_\infty^0 = J_s^0.$$

Note 3: Recoil experiments are sometimes performed by releasing the stress in a creep experiment at some time t_0 before the strain has become linear in time. In this case the functions defined above can still be used, but the new independent variable must be noted. For example,

$$\gamma_r(t - t_0, t_0, \sigma) \equiv \gamma(t_0) - \gamma(t)$$

6. Superposed Parallel Steady and Oscillatory Shear Flow

The shear rate is the sum of a constant (mean) value ($\dot{\gamma}_m$) and an oscillatory component

$$\dot{\gamma}(t) = \dot{\gamma}_m + \gamma_0\omega\cos\omega t$$

If the amplitude of oscillation, γ_0, is sufficiently small that the resulting shear stress is a sum of a mean value σ_m and a sinusoidal component, we have

$$\sigma_m = \eta(\dot{\gamma}_m)\dot{\gamma}_m$$

$$\sigma = \sigma_m + \sigma_0\sin(\omega t + \delta)$$

where σ_0 is the amplitude of the sinusoidal component and δ is the mechanical loss angle.

The parallel complex viscosity has the following real and imaginary components:

$$\eta'_{\parallel}(\omega, \dot{\gamma}_m) \equiv (\sigma_0/\omega\gamma_0)\sin\delta$$

$$\eta''_{\parallel}(\omega, \dot{\gamma}_m) \equiv (\sigma_0/\omega\gamma_0)\cos\delta$$

7. Superposed Orthogonal Steady and Oscillatory Shear Flow

In this case the steady shear is in the 1-2 plane, while the oscillatory component is in the 2-3 plane:

$$v_1 = \dot{\gamma}_m x_2$$

$$v_2 = 0$$

$$v_3 = (\gamma_0\omega\cos\omega t)x_2$$

In other words, the rate of deformation tensor[6] has the following form

$$\begin{bmatrix} 0 & \dot{\gamma}_m & 0 \\ \dot{\gamma}_m & 0 & (\gamma_0\omega\cos\omega t) \\ 0 & (\gamma_0\omega\cos\omega t) & 0 \end{bmatrix},$$

If the amplitude of oscillation, γ_0, is sufficiently small that the shear stress component σ_{23} is sinusoidal, we have:

$$\sigma_{23} = \sigma_0\sin(\omega t + \delta)$$

and the orthogonal complex viscosity has the following real and imaginary components:

$$\eta'_{\perp}(\omega, \dot{\gamma}_m) \equiv (\sigma_0/\omega\gamma_0)\sin\delta$$

$$\eta''_{\perp}(\omega, \dot{\gamma}_m) \equiv (\sigma_0/\omega\gamma_0)\cos\delta$$

[6] The rate of deformation tensor is defined here as $(\nabla\mathbf{v} + \nabla\mathbf{v}^T)$

8. Extensional Flows

A. Tensile (Simple) Extension

The material functions defined below are based on the homogeneous simple extension of a fluid with constant density.

$$v_1 = \dot{\epsilon} x_1$$

$$v_2 = -\frac{1}{2}\dot{\epsilon} x_2$$

$$v_3 = -\frac{1}{2}\dot{\epsilon} x_3$$

where $\dot{\epsilon} \geq 0$

Note 1: This is an axisymmetric flow and can be described alternatively in terms of cylindrical coordinates

$$v_z = \dot{\epsilon} z$$

$$v_r = -\frac{1}{2}\dot{\epsilon} r$$

Note 2: The parameter $\dot{\epsilon}$ appearing in the velocity distribution is the Hencky strain rate. Thus ϵ is the Hencky strain, defined as

$$\epsilon \equiv \ln(L/L_0)$$

1. Tensile Start-Up Flow

A sample initially in its rest state is subjected to a constant extensional (Hencky) strain rate $\dot{\epsilon}$ at time $t = 0$. The "net tensile stress" σ_0 is monitored as a function of time

$$\sigma_E \equiv \sigma_{11} - \sigma_{22} = \sigma_{11} - \sigma_{33} = \sigma_{zz} - \sigma_{rr}$$

Tensile stress growth coefficient:

$$\eta_E^+(t, \dot{\epsilon}) = \sigma_E/\dot{\epsilon}$$

Tensile viscosity:

$$\eta_E(\dot{\epsilon}) = \lim_{t\to\infty}[\eta_E^+(t, \dot{\epsilon})]$$

Note1: If the material of interest tends toward linear viscoelastic behavior at small strain rates, the following relationships are valid:

$$\lim_{\dot{\epsilon}\to 0}[\eta_E^+(t, \dot{\epsilon})] = \eta_E^+(t) = 3\eta^+(t)$$

$$\lim_{\dot{\epsilon}\to 0}[\eta_E(\dot{\epsilon})] = 3\eta_0$$

Note 2: Some researchers prefer to use the strain ϵ as an independent variable in place of time when presenting tensile stress growth results.

2. Cessation of Steady Tensile Extension

A material that has been subjected to steady tensile extension at a strain rate $\dot{\epsilon}$ until the net tensile stress is constant in time is brought suddenly to rest at time $t = 0$, and the net tensile stress is monitored as a function of time.

Tensile stress decay coefficient:

$$\eta_E^-(t, \dot{\epsilon}) = \sigma_E/\dot{\epsilon}$$

Note: If the material of interest tends toward linear viscoelastic behavior at small strain rates:

$$\lim_{\dot{\epsilon}\to 0}[\eta_E^-(t, \dot{\epsilon})] = \eta_E^-(t) = 3\eta^-(t)$$

3. Tensile creep

A sample initially in its rest state is subjected to a constant net tensile stress σ_E, at time $t = 0$, and the Hencky strain ϵ is monitored as a function of time

Tensile creep compliance:

$$D(t, \sigma_E) \equiv \epsilon/\sigma_E$$

After the strain becomes linear with time, the tensile creep compliance is given by

$$D(t, \sigma_E) = D_s(\sigma_E) + t/\eta_E,$$

where D_s is the steady-state tensile compliance, and η_E is evaluated at the steady strain rate corresponding to σ_E

Note: If the behavior of the material of interest tends toward linear viscoelasticity at small values of stress:

$$\lim_{\sigma_E\to 0}[D(t, \sigma_E)] = D(t) = \frac{1}{3}J(t)$$

4. Tensile Recoil

A material in which the stresses and the strain rate $\dot{\epsilon}$ are steady in time has its tensile stress σ_E reduced suddenly to zero at time $t = 0$, and the recoil strain ϵ_r is monitored as a function of time

Tensile recoil strain (a positive quantity):

$$\epsilon_r \equiv \epsilon(0) - \epsilon(t)$$

Tensile recoil function:

$$S(t, \sigma_E) \equiv \epsilon_r / \sigma_E$$

Ultimate tensile recoil:

$$\epsilon_\infty(\sigma) = \lim_{t \to \infty} [\gamma_\epsilon(t, \sigma_E)]$$

Ultimate tensile recoil function:

$$S_\infty(\sigma) = \epsilon_\infty(\sigma_E) / \sigma_E$$

Note 1: Since $\sigma_E = \eta_E \dot{\epsilon}$ at $t = 0$, the strain rate can be used in place of the stress as an independent variable

Note 2: If the material of interest tends toward linear viscoelastic behavior at small stresses:

$$\lim_{\sigma_E \to 0} [S(t, \sigma_E)] = S(t) = \frac{1}{3} R(t)$$

Note 3: Recoil experiments are sometimes performed by releasing the stress in a creep experiment at some time t_0 before the strain has become linear in time. In this case the functions defined above can still be used, but the new independent variable must be noted. For example,

$$\epsilon_r(t - t_0, t_0, \sigma_E) \equiv \epsilon(t_0) - \epsilon(t)$$

5. Tensile Step Strain

A material initially in its rest state is subjected to a sudden strain of magnitude ϵ at time $t = 0$, at the net tensile stress is observed as a function of time.

Tensile stress relaxation modulus:

$$E(t, \epsilon) \equiv \sigma_E / \epsilon$$

Note: If the material of interest tends toward linear viscoelastic behavior at small values of strain:

$$\lim_{\epsilon \to 0} [E(t, \epsilon)] = E(t) = 3G(t)$$

B. Biaxial Extensional Flow (Axisymmetric)

The material functions defined below are based on the homogeneous simple extension of a fluid with constant density.

$$v_1 = \dot{\epsilon}_B x_1$$

$$v_2 = \dot{\epsilon}_B x_2$$

$$v_3 = -2\dot{\epsilon}_B x_3$$

where $\dot{\epsilon} \geq 0$

Note: This flow is axisymmetric and can be described in a cylindrical coordinate system as follows:

$$v_r = \dot{\epsilon}_B r$$

$$v_z = -2\dot{\epsilon}_B z$$

1. Biaxial Start-Up Flow

A sample initially in its rest state is subjected to a constant biaxial strain rate $\dot{\epsilon}_B$ at $t = 0$, and the net stretching stress σ_B is monitored as a function of time.

$$\sigma_B \equiv \sigma_{11} - \sigma_{33} = \sigma_{22} - \sigma_{33} = \sigma_{rr} - \sigma_{zz}$$

Biaxial stress growth coefficient

$$\eta_B^+(t, \dot{\epsilon}_B) \equiv \sigma_B / \dot{\epsilon}_B$$

Biaxial extensional viscosity

$$\eta_B(\dot{\epsilon}_B) = \lim_{t \to \infty} [\eta_B^+(t, \dot{\epsilon}_B)]$$

Note 1: If the material of interest tends toward linear viscoelastic behavior at small strain rate:

$$\lim_{\dot{\epsilon}_B \to 0} [\eta_B^+(t, \dot{\epsilon}_B)] = \eta_B^+(t) = 6\eta^+(t)$$

$$\lim_{\dot{\epsilon}_B \to 0} [\eta_B(\dot{\epsilon}_B)] = 6\eta_0.$$

Note 2: Those calculating the predictions of constitutive equations may prefer to show results for both steady tensile flow and steady biaxial extension on a single plot. This can be accomplished by noting that axisymmetric biaxial extensional flow can be considered to be tensile flow at a negative strain rate. For example, let $\bar{\eta}(\dot{\epsilon}_B)$ be an extensional viscosity defined only for axisymmetric flows, where $\dot{\epsilon}$ can take on both positive and negative values. This material function is then defined as follows:

$$\bar{\eta} \equiv (\sigma_{zz} - \sigma_{rr})/\dot{\epsilon}$$

The following correspondences can then be made with the functions described above:

$$\bar{\eta}(\dot{\epsilon}) = \eta_E(\dot{\epsilon}), for\ \dot{\epsilon} \geq 0$$

$$\bar{\eta} = \frac{1}{2}\eta_B \text{ with } \dot{\epsilon} = 2\,\dot{\epsilon}_B, \text{for } \dot{\epsilon} \leq 0$$

C. Extensional Flow (General)

Tensile extension and biaxial extension are two special cases of a class of flows for which the components of the rate of deformation tensor have been shown by Stevenson, Chung, and Jenkins to have the following form:

$$\dot{\epsilon}\begin{bmatrix} 1 & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & -(1+m) \end{bmatrix}$$

Meissner et al, have pointed out that if $\dot{\epsilon}$ is interpreted as the largest principal strain rate, and m does not vary with time, then every possible extensional flow corresponds to some value of m between -0.5 and +1.0.

Unless the flow is axisymmetric, two independent normal stress differences can, in principle, be measured:

$$\sigma_1 \equiv \sigma_{11} - \sigma_{22},$$

$$\sigma_2 \equiv \sigma_{22} - \sigma_{33}.$$

Thus, two independent material functions can be defined for a given value of m and a given strain or stress history. For example, if $\dot{\epsilon}$ is constant, two extensional viscosities can be defined:

$$\eta_1^{(m)} \equiv \sigma_1/\dot{\epsilon}$$

$$\eta_2^{(m)} \equiv \sigma_2/\dot{\epsilon}$$

We note that $m = -1/2$ corresponds to simple extension, while $m = 1$ corresponds to biaxial extension. However, in these cases there is only one independent normal stress difference, and the special symbols proposed for these flows should be used rather than the ones defined above. Another special case of interest is "planar" extension, for which $m = 0$. The most easily measures stress difference is σ_1, and a planar extensional viscosity can be defined as follows:

$$\eta_p(\dot{\epsilon}) \equiv \eta_1^{(0)}(\dot{\epsilon})$$

The Rheology Bulletin, Vol 62, No. 2 July 1993, pg. 2 (86)

JOHN M. DEALY RECEIVES DISTINGUISHED SERVICE AWARD

The Society's Distinguished Service Award was presented to Dr. John M. Dealy at the 64th Annual Meeting held in Santa Barbara. This is the third time the Award has been made

since its establishment in 1984. The criterion for the Award is "...members whose service in responsible positions has extended over long periods and has been of great value to the Society". A glance at the history of John's service shows that it meets this criterion fully and admirably. He has served on many Society Committees, including the Executive Committee, on which his attention to issues of Society governance has been of inestimable value. He served as Chairman and principal mover of the Nomenclature Committee, whose report he wrote; it has been reprinted most recently in Volume 37(1) of the Journal of Rheology.

The Rheology Bulletin, Vol 81, No. 2 July 2012, pg. 21 (153)

Official Nomenclature Being Revised: Input Solicited

A committee has been appointed to revise and update the Official Nomenclature of the Society of Rheology. The members are John Dealy (Chair), Jeff Morris, Faith Morrison, and Dimitris Vlassopoulos. Members are invited to submit their suggestions for amendments and additions to the nomenclature. You can see the present nomenclature from the Society's web page. Please send your input to john.dealy@mcegill.ca or to another member of the committee. Your help is vitally neede

Official Nomenclature of US and European Societies of Rheology, 2018 Edition

First edition: vol. 39, p. 253 (1995). Second edition: vol. 57, p. 1047 (2013). Update assembled by J. M. Dealy and D. Vlassopoulos, with advice from G. G. Fuller, R. G. Larson, J. Mewis, D. J. Read, J. Vermant, N. J. Wagner, and G. McKinley

Table I Society of Rheology Nomenclature for Steady Simple Shear

Quantity	Definition	Symbol	S.I. units[7]
Direction of flow (simple shear)		x_1	m
Direction of velocity gradient (simple shear)		x_2	m
Neutral Direction (simple shear)		x_3	m
Shear or Normal Force		F	N
Velocity	dx/dt	v	ms^{-1}
Acceleration	dv/dt	a	ms^{-2}
Shear stress[8]	F/A	σ	Pa
Shear strain	dx_1/dx_2	γ	–
Shear rate	dv_1/dx_2	$\dot{\gamma}$	s^{-1}
Vorticity[9]	$\nabla v - (\nabla v)^T$	ω	s^{-1}

[7] SI allows either a dot between units or a space, as used here
[8] A is the area in m^2
[9] In the fluid dynamics community a prefactor of ½ is used in the definition of the vorticity

Viscosity	$\sigma/\dot{\gamma}$	$\eta(\dot{\gamma})$	Pa s
Yield stress		σ_y	Pa
Yield strain		γ_y	–
First normal stress function	$\sigma_{11}-\sigma_{22}$	N_1	Pa
Second normal stress function	$\sigma_{22}-\sigma_{33}$	N_2	Pa
First normal stress coefficient	$N_1/\dot{\gamma}^2$	Ψ_1	Pa s^2
Second normal stress coefficient	$N_2/\dot{\gamma}^2$	Ψ_2	Pa s^2
Normal stress ratio	$-N_1/N_2$	$\Psi(\dot{\gamma})$	–
Zero-shear viscosity (limiting low shear rate viscosity)	$\eta(\dot{\gamma}\to 0)$	η_0	Pa s
Zero-shear first normal stress coefficient	$\Psi_1(\dot{\gamma}\to 0)$	$\Psi_{1,0}$	Pa s^2
Consistency index in power law for viscosity	$\eta = K\lvert\dot{\gamma}\rvert^{n-1}$	n	–
Constant in power law for viscosity	See above eqn.	K	Pa s^n

Table II Linear Viscoelasticity

Quantity	Definition	Symbol	S.I. units
Simple shear			
Shear modulus of a solid	σ/γ	G	Pa
Relaxation modulus	$\sigma(t)/\gamma$	$G(t)$	Pa
Relaxation strength in discrete spectrum	–	g_i	Pa
Relaxation time in discrete spectrum	–	τ_i	s
Continuous relaxation spectrum	–[10]	$H(\tau)$	Pa
Orthogonal superposition complex modulus[11,12]	–	$G_\perp^*(\omega,\dot{\gamma})$	Pa
Parallel superposition complex modulus[11,12]	–	$G_\parallel^*(\omega,\dot{\gamma})$	Pa
Memory function	$-dG(s)/ds$	$m(s)$	Pa s^{-1}
Creep compliance (shear)	$\gamma(t)/\sigma$	$J(t)$	Pa^{-1}
Equilibrium compliance of solid	$J(t)(t\to\infty)$	J_e	Pa^{-1}
Recoverable compliance	$J(t)-t/\eta_0$	$J_r(t)$	Pa^{-1}

[10] $G(t)=\int_{-\infty}^{\infty}H(\tau)[\exp(-t/\tau)]d(\ln\tau)$

[11] The same subscripts apply to definitions of storage and loss moduli and viscosities

[12] Also applies to nonlinear phenomena in uniaxial extension

Steady-state compliance of fluid	$J(t) - t/\eta_0$ $(t \to \infty)$	J_s^0	Pa^{-1}
Continuous retardation spectrum[13]	$-$[7]	$L(\tau)$	Pa^{-1}
Small-amplitude oscillatory shear			
Strain amplitude	$\gamma(t) = \gamma_0 \sin(\omega t)$	γ_0	$-$
Loss angle (Phase angle)	$\sigma(t) = \sigma_0 \sin(\omega t + \delta)$	δ	rad
Stress amplitude	$\sigma(t) = \sigma_0 \sin(\omega t + \delta)$	σ_0	Pa
Complex modulus	$G' + iG''$	G^*	Pa
Absolute magnitude of G^*	σ_0/γ_0	$\lvert G^* \rvert$ or G_d	Pa
Storage modulus	$G_d \cos\delta$	G'	Pa
Loss modulus	$G_d \sin\delta$	G''	Pa
Complex viscosity	$\eta' - i\eta''$	η^*	Pa s
Absolute magnitude of η^*	$\sigma_0/\omega\gamma_0$	$\lvert \eta^* \rvert$	Pa s
Dynamic viscosity (in phase with strain rate)	G''/ω	η'	Pa s
Out-of-phase (with strain rate) component of η^*	G'/ω	η''	Pa s
Complex compliance	$J' - iJ''$	J^*	Pa^{-1}
Absolute magnitude of J^*	$\gamma_0/\sigma_0 = 1/G_d$	$\lvert J^* \rvert$	Pa^{-1}
Storage compliance	$\gamma_0/\sigma_0 \cos\delta$	J'	Pa^{-1}
Loss compliance	$\gamma_0/\sigma_0 \sin\delta$	J''	Pa^{-1}
Plateau modulus[14]	$-$	G_N^0	
Tensile (Uniaxial) Extension			
Net tensile stress	$\sigma_{zz} - \sigma_{rr}$	σ_E	Pa
Hencky strain	$\ln(L/L_0)$	ε	$-$
Hencky strain rate	$d(\ln L)/dt$	$\dot{\epsilon}$	s^{-1}
Tensile Relaxation modulus	$\sigma_E(t)/\varepsilon_0$	$E(t)$	Pa
Tensile Creep compliance	$\varepsilon_0(t)/\sigma_E$	$D(t)$	Pa^{-1}

[13] $J(t) = \int_{-\infty}^{\infty} L(\tau)[1 - \exp(-t/\tau)]d(\ln\tau)$

[14] Because there is not a true plateau in $G(t)$ or $G'(\omega)$, G_N^0 is inferred from $G'(\omega)$ and $G''(\omega)$ using methods reviewed by Lie *et al.* [Polymer, **47**, 4461-4479 (2006)]

Table III Shift factors for time-temperature superposition

Name	Definition	Symbol	S.I. units
Vertical shift factor[15]	$b_T G'(T, \omega a_T) = G'(T_0, \omega)$	b_T	–
Horizontal shift factor	$b_T G'(T, \omega a_T) = G'(T_0, \omega)$	a_T	–
First WLF coefficient	$\log_{10} a_T = \frac{-c_1(T - T_0)}{[c_2 + (T - T_0)]}$	c_1	–
Second WLF coefficient	$\log_{10} a_T = \frac{-c_1(T - T_0)}{[c_2 + (T - T_0)]}$	c_2	K
Activation energy for flow	$a_T = \exp\left[\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$	E_a	kJ/mol

Table IV Nonlinear viscoelasticity in shear

Name	Definition	Symbol	S.I. units
Stress relaxation (step strain)			
Strain amplitude		γ_0	–
Relaxation modulus (nonlinear)	$\sigma(t)/\gamma_0$	$G(t, \gamma_0)$	Pa
Damping function in shear	$G(t, \gamma_0)/G(t)$	$h(\gamma_0)$	–
First normal stress relaxation function	–	$N_1(t, \gamma_0)$	Pa
Second normal stress relaxation function	–	$N_2(t, \gamma_0)$	Pa
First normal stress relaxation coefficient	$N_1(t, \gamma_0)/\gamma_o^2$	$\Psi_1(t, \gamma_0)$	Pa
Second normal stress relaxation coefficient	$N_2(t, \gamma_0)/\gamma_o^2$	$\Psi_2(t, \gamma_0)$	Pa
Start-up of steady shear (at fixed shear rate)			
Shear stress growth function		$\sigma^+(t, \dot{\gamma})$	Pa
Shear stress growth coefficient	$\sigma^+(t, \dot{\gamma})/\dot{\gamma}$	$\eta^+(t, \dot{\gamma})$	Pa s
First normal stress growth function	$\sigma_{11} - \sigma_{22}$	$N_1^+(t, \dot{\gamma})$	Pa
First normal stress growth coefficient	$N_1^+(t, \dot{\gamma})/\dot{\gamma}^2$	$\Psi_1^+(t, \dot{\gamma})$	Pa s^2
Second normal stress growth function	$\sigma_{22} - \sigma_{33}$	$N_2^+(t, \dot{\gamma})$	Pa
Second normal stress growth coefficient	$N_2^+(t, \dot{\gamma})/\dot{\gamma}^2$	$\Psi_2^+(t, \dot{\gamma})$	Pa s^2
Stress Ratio	$N_1(\dot{\gamma})/\sigma(\dot{\gamma})$	SR	–
Cessation of steady shear	$\dot{\gamma} = 0$ from $t = 0$		
Shear stress decay function		$\sigma^-(t, \dot{\gamma})$	Pa
Shear stress decay coefficient	$\sigma^-(t, \dot{\gamma})/\dot{\gamma}$	$\eta^-(t, \dot{\gamma})$	Pa s
First normal stress decay function	$\sigma_{11} - \sigma_{22}$	$N_1^-(t, \dot{\gamma})$	Pa

[15] For rubbers, b_T is given by $T_0\rho_0/T\rho$, where ρ is mass density, with $\rho_0 = \rho(T_0)$ [J. D. Ferry, Viscoelastic properties of polymers, 3rd Ed., Wiley, NY, 1980]. It has also been found to be appropriate for polymers in general.

First normal stress decay coefficient	$N_1^-(t,\dot{\gamma})/\dot{\gamma}^2$	$\Psi_1^-(t,\dot{\gamma})$	Pa s^2
Second normal stress decay function	$\sigma_{22}-\sigma_{33}$	$N_2^-(t,\dot{\gamma})$	Pa
Second normal stress decay coefficient	$N_2^-(t,\dot{\gamma})/\dot{\gamma}^2$	$\Psi_2^-(t,\dot{\gamma})$	Pa s^2
Creep and creep recovery (recoil)			
Creep compliance	$\gamma(t,\sigma)/\sigma$	$J(t,\sigma)$	Pa^{-1}
Steady-state compliance[16]	$J(t\to\infty,\sigma)$	$J_s(\sigma)$	Pa^{-1}
Recoverable strain (after t_0 when $\sigma\to 0$)	$\gamma[t_0,\sigma]-\gamma[t,\sigma]$ $t>t_0$	$\gamma_r(t',\sigma)$ $t'=t-t_0$	–
Ultimate recoil	$\gamma_r(t'\to\infty,\sigma)$	$\gamma_\infty(\sigma)$	–
Steady-state recoverable compliance[10]	$\gamma_\infty(\sigma)/\sigma$	$J_r(\sigma)$	Pa^{-1}

Table V Nonlinear viscoelasticity in extension

Name	Definition	Symbol	S.I. units
Tensile (Uniaxial) extension			
Engineering strain[17]	$(L-L_0)/L_0$	ε	–
Engineering stress[1]	F/A_0	σ	Pa
Young's modulus of a solid	σ/ε	E	Pa
Net tensile stress (true)	$\sigma_{zz}-\sigma_{rr}$	σ_ε	Pa
Hencky strain	$\ln(L/L_0)$	ε or ϵ	–
Hencky strain rate	$d\varepsilon/dt$	$\dot{\varepsilon}$	s^{-1}
Tensile stress growth function		$\sigma_E^+(t,\dot{\varepsilon})$	Pa
Tensile stress growth coefficient	$\sigma_E^+(t,\dot{\varepsilon})/\dot{\varepsilon}$	$\eta_E^+(t,\dot{\varepsilon})$	Pa s
Extensional viscosity	$\eta_E^+(t,\dot{\varepsilon})\ t\to\infty$	$\eta_E(\dot{\varepsilon})$	Pa s
Tensile creep compliance	$\varepsilon(t)/\sigma_E$	$D(t,\sigma_E)$	Pa^{-1}
Recoverable strain (after t_0 where $\sigma_E=0$)	$\varepsilon[t_0,\sigma_E]-\varepsilon[t,\sigma_E]\ t>t_0$	$\varepsilon_r(t',\sigma_E), t\equiv t'-t_0$	–
Biaxial extension (symmetrical)			
Biaxial strain	$\ln(R/R_0)$	ε_B	–
Biaxial strain rate	$d(\ln R)/dt$	$\dot{\varepsilon}_B$	s^{-1}
Net stretching stress	$\sigma_{rr}-\sigma_{zz}$	σ_B	Pa
Biaxial stress growth function		$\sigma_B^+(t,\dot{\varepsilon}_B)$	Pa
Biaxial stress growth coefficient	$\sigma_B^+(t,\dot{\varepsilon}_B)/\dot{\varepsilon}_B$	$\eta_B^+(t,\dot{\varepsilon}_B)$	Pa s
Biaxial stress decay coefficient	$\sigma_B^-(t,\dot{\varepsilon}_B)/\dot{\varepsilon}_B$	$\eta_B^-(t,\dot{\varepsilon}_B)$	Pa s
Biaxial extensional viscosity	$\sigma_B^+(t\to\infty,\dot{\varepsilon}_B)/\dot{\varepsilon}_B$	$\eta_B(\dot{\varepsilon}_B)$	Pa s
Biaxial creep compliance	$\varepsilon_B(t)/\sigma_B$	$D(t,\sigma_B)$	Pa^{-1}

[16] Although measured in different ways, the steady-state compliance and the steady-state recoverable compliance should be equal to each other according to the Boltzmann principle.
[17] In the mechanics literature, the same symbols are often used for both engineering and true stress and strain, but they are only equivalent in the limit of very small deformations.

Table VI Rheometry

Name	Definition	Symbol	S.I. units
Capillary rheometers			
Apparent wall shear stress[18]	$P_d R/2L$	σ_A	Pa
Apparent wall shear rate	$4Q/\pi R^3$	$\dot{\gamma}_A$	s^{-1}
Wall shear stress	$-\sigma_{rz}(r=R)$	σ_W	Pa
Wall shear rate	$-dv/dr\ (r=R)$	$\dot{\gamma}_W$	s^{-1}
Cone-plate rheometers			
Cone angle	Figure 1	β	rad
Angular rotation	Figure 1	φ	rad
Angular velocity	$d\varphi/dt$	Ω	rad/s
Torque[19]	$2\pi R^3 \sigma_{\varphi\theta}/3$	M	N m
Normal thrust		F_Z	N

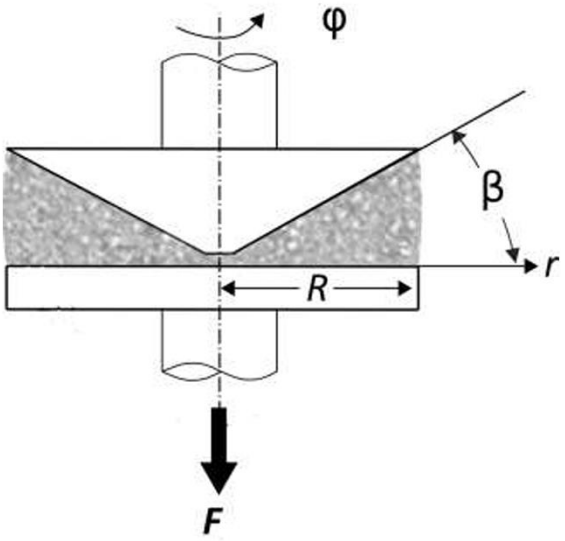


Figure 1. Symbols describing cone-plate geometry

[18] P_d =driving (reservoir) pressure (exit pressure neglected)
[19] Approximation valid for $\beta < 0.1$ rad

Table VII Solutions

Name	Definition	Symbol	S.I. units
Concentration		c	kg m^{-3} [20]
Overlap concentration		c^*	kg m^{-3} [20]
Solvent viscosity		η_s	Pa s
Relative viscosity	(η/η_s)	η_r	–
Specific viscosity	$(\eta_r - 1)$	η_{sp}	–
Reduced viscosity	η_{sp}/c	η_{red}	m^3 kg^{-1} [21]
Intrinsic viscosity	$\lim(\eta_{red}, \dot{\gamma} \to 0, c \to 0)$	$[\eta]$	m^3 kg^{-1} [21]
Viscosity of matrix[22]		η_m	Pa s
Solvent contribution to the stress tensor		$\boldsymbol{\sigma^s}$	Pa
Polymeric contribution to the stress tensor		$\boldsymbol{\sigma^p}$	Pa

Table VIII Suspensions

Name	Definition	Symbol	S.I. units
Volume fraction solid	$V_{solid}/V_{suspension}$	ϕ	–
Maximum packing fraction		ϕ_{max}	–
Suspending medium viscosity		η_m [23]	Pa s
Effective viscosity of suspension	$\Sigma_{12}/\dot{\gamma}$	η	Pa s
Relative viscosity of suspension	η/η_f	η_r	–
Particle contribution to η		η_P	Pa s
Local stress tensor		$\boldsymbol{\sigma}(\boldsymbol{x}, t)$	Pa
Total bulk stress	$\mathbf{\Sigma}^f + \mathbf{\Sigma}^\mathbf{p}$	$\mathbf{\Sigma}$	Pa
Fluid (suspending medium) contribution		$\mathbf{\Sigma}^f$	Pa
Particle contribution		$\mathbf{\Sigma}^\mathbf{p}$	Pa
First normal stress difference	$\Sigma_{11} - \Sigma_{22}$	N_1	Pa
Second normal stress difference	$\Sigma_{22} - \Sigma_{33}$	N_2	Pa
Dimensionless first normal stress difference	$-N_1/\eta_f\dot{\gamma}$	Υ_1	–
Dimensionless second normal stress difference	$-N_2/\eta_f\dot{\gamma}$	Υ_2	–
Particle pressure	$-\frac{1}{3}\left(\Sigma_{11}^P + \Sigma_{22}^P + \Sigma_{33}^p\right)$	Π	Pa

[20] Units of g/mL are often used

[21] Units of mL/g are often used

[22] Often used for nanocomposites, i.e., particles in a viscoelastic matrix

[23] Sometimes the subscript f is used to indicate the fluid viscosity, but it seems more adequate to use m which stands for "medium" if the suspending fluid is itself rheologically complex or "matrix" in nanocomposites. Also, μ is often used to denote fluid viscosity

Hydrodynamic particle stress	Σ^H	Pa
Interparticle stress	Σ^{IP}	Pa
Brownian stress	Σ^B	Pa

Table IX. Interfacial and surface rheology

Name	Definition	Symbol	S.I. units
Interfacial or surface tension[24]		σ_α, β	Pa m
Surface pressure	$\sigma^0_{\alpha,\beta} - \sigma_{\alpha,\beta}$	Π	Pa m
Interfacial shear stress	F^s/L	σ^s	Pa m
Interfacial shear strain	dX_1/dx_2	γ^s	–
Interfacial shear rate	dv_1/dx_2	$\dot{\gamma}^s$	s^{-1}
Interfacial dilatational strain	$\ln(A/A_0)$	α^s	–
Interfacial dilatational strain rate	$\mathrm{d}(\ln A)/dt$	$\dot{\alpha}^s$	s^{-1}
Interfacial concentration	–	Γ	kg m^{-2}
Steady shear and dilatation			
Interfacial shear viscosity	$\sigma^s/\dot{\gamma}^s$	η^s	Pa s m
Interfacial dilatational viscosity	$\sigma^s/\dot{\alpha}^s$	κ^s	Pa s m
Simple shear			
Interfacial shear modulus	σ^s/γ^s	G^s	Pa m
Relaxation modulus (shear)	$\sigma^s(t)/\gamma^s$	$G^s(t)$	Pa m
Pure dilatation			
Interfacial dilatational modulus	σ^s/α^s	K^s	Pa m
Dilatational storage modulus	$\sigma^s(t)/\alpha^s$	$K^s(t)$	Pa m
Gibbs elasticity (surfactants) [25]	$d\sigma_{\alpha,\beta}/d(\ln A)$	K_Π	Pa m
Small-amplitude oscillatory shear			
Strain amplitude	$\gamma^s(t) = \gamma_0^s \sin(\omega t)$	γ_0^s	–
Phase angle (loss angle)	$\sigma^s(t) = \sigma_0^s \sin(\omega t + \delta)$	δ	rad
Stress amplitude	$\sigma^s(t) = \sigma_0^s \sin(\omega t + \delta)$	σ_0^s	Pa m
Complex interfacial shear modulus	$G^{s\prime} + iG^{s\prime\prime}$	G^{s*}	Pa m
Absolute magnitude of G^{s*}	σ_0^s/γ_0^s	$\lvert G^{s*}\rvert$	Pa m
Storage modulus	$\lvert G^{s*}\rvert \cos\delta$	G'	Pa m
Loss modulus	$\lvert G^{s*}\rvert \sin\delta$	G''	Pa m

[24] In some cases the symbols α or Γ are used to denote surface tension (not to be confused with interfacial dilatational strain or surface concentration, respectively)

[25] Sometimes the symbol E is used instead of K, but this should not be confused with Young's modulus. The use of K for a compression modulus is recommended.

Complex shear viscosity	$\eta^{s\prime} - i\eta^{s\prime\prime}$	η^{s*}	Pa m s
Absolute magnitude of η^{s*}	$\sigma_0^s/\omega\gamma_0^s$	$\lvert\eta^{s*}\rvert$	Pa m s
Dynamic viscosity (in phase with strain rate)	$G^{s\prime\prime}/\omega$	$\eta^{s\prime}$	Pa m s
Out-of-phase (with strain rate) component of η^{s*}	$G^{s\prime}/\omega$	$\eta^{s\prime\prime}$	Pa m s
Small-amplitude oscillatory dilatation			
Dilatational strain amplitude	$\alpha^s(t) = \alpha_0^s \sin(\omega t)$	α_0^s	–
Complex dilatational modulus	$K^{s\prime} + iK^{s\prime\prime}$	K^{s*}	Pa m
Absolute magnitude of G^{s*}	σ_0^s/α_0^s	$\lvert K^{s*}\rvert$	Pa m
Storage modulus	$\lvert K^{s*}\rvert \cos\delta$	$K^{s\prime}$	Pa m
Loss modulus	$\lvert K^{s*}\rvert \sin\delta$	$K^{s\prime\prime}$	Pa m
Complex shear viscosity	$\kappa^{s\prime} - i\kappa^{s\prime\prime}$	κ^{s*}	Pa s m
Absolute magnitude of η^{s*}	$\sigma_0^s/\omega\alpha_0^s$	$\lvert\kappa^{s*}\rvert$	Pa s m
Dynamic viscosity (in phase with strain rate)	$K^{s\prime\prime}/\omega$	$\kappa^{s\prime}$	Pa s m
Out-of-phase (with strain rate) component of η^{s*}	$K^{s\prime}/\omega$	$\kappa^{s\prime\prime}$	Pa s m
Other properties			
Creep compliance (shear)	$\gamma^s(t)/\sigma^s$	$J^s(t)$	$Pa^{-1}\,m^{-1}$
Equilibrium compliance of solid	$J^s(t)(t \to \infty)$	J_0^e	$Pa^{-1}\,m^{-1}$
Recoverable compliance	$J^s(t) - t/\eta_0^s$	$J_r^s(t)$	$Pa^{-1}\,m^{-1}$
Steady-state compliance of fluid	$J^s(t) - t/\eta_0^s\ (t \to \infty)$	J_0^s	$Pa^{-1}\,m^{-1}$
Extensional viscosity	$\eta_E^s(t, \dot{\varepsilon}^s)(t \to \infty)$	$\eta_E^s(\dot{\varepsilon}^s)$	Pa s m

Table X. Molecular description of entangled polymers

a	tube diameter (typically in nm); average entanglement spacing / mesh size $\sqrt{< R^2 >_0 M_e/M}$
b_K	Kuhn segment length[26]
f	tension in a chain segment (typically in N)
f_{max}	maximum tension in a chain segment
k_B	Boltzmann's constant, 1.38×10^{-23} J/K
L	mean tube contour length
M	Molecular weight[27], kg mol^{-1}

[26] b_K and L_K are defined by the following relationships: $< R^2 \geq b_K^2 N_K : R_{max} = b_K N_K$

[27] IUPAC recommends molar mass (MM), which has SI units of g/mol. But molecular weight (MW) is widely used, and ACS accepts both terms. However, MW is formally a dimensionless ratio that is numerically very close to MM (g/mol), and one cannot "change its units". The number often called "molecular weight (kg/mol)"

M_C	Critical molecular weight for effect of entanglements on η_0 kg mol^{-1}
M_C'	Molecular weight for effect of entanglements on J_S^0, kg mol^{-1}
M_e	molecular weight between entanglements[28] ($\rho RT/G_N^0$)
N_K	number of Kuhn segments in equivalent freely jointed chain
p	packing length[29] ($M/\lfloor< R^2 >_0 \rho N_A\rfloor$)
p^2	parameter to denote degree of branch point hopping within tube (fraction of tube diameter)
R	end-to-end distance of polymer molecule
R_{max}	Fully extended chain length
s	tube contour variable (curvilinear coordinate along tube)
$\boldsymbol{S}$	Tube orientation tensor
Z	Number of entanglement segments per molecule [30] (M/M_e)

Greek Letters

α	Dilution exponent forM_e
ζ	Friction coefficient
ζ_0	Monomer friction coefficient
λ	Chain stretch; stretch ratio
ξ	Correlation length; characteristic size scale (blob)
τ_d	reptation (tube disengagement) time
τ_e	Rouse time of an entanglement strand ($\tau_R/3\pi^2Z^2$)
τ_p	Relaxation time of the pth mode (p is the mode index)
τ_r	Rouse reorientation relaxation time
τ_R	Rouse stress relaxation time [$(\varsigma N^2b^2)/(6\pi^2k_BT)$]
τ_s	Stretch relaxation time [31]

Table XI. Stress and strain tensors

Total stress tensor	$\boldsymbol{\sigma}$
Extra stress tensor	$\boldsymbol{\tau}$
Strain tensor for linear viscoelasticity	$\boldsymbol{\gamma}$
Cauchy tensor	$\boldsymbol{C}$
Finger tensor	$\boldsymbol{B}$ **or** $\boldsymbol{C^{-1}}$
Doi-Edwards strain tensor	$\boldsymbol{Q}$

is formally MW/1000 (no units). This quantity should more properly be called molar mass with units of kg/mol. But it is so widely called molecular weight that it seems hopeless to try to change it now.

[28] This is the definition originally proposed by John Ferry. The following alternative definition was introduced much later in the context of Doi-Edwards model. It should be used in the context of the development of, or comparisons with tube-model theories$M_e = \frac{4}{5}\rho RT/G_N^0$

[29] For a discussion of p see Fetters et al. [Macromolecules 27, 4639 (1994)]

[30] In tube-model theory, Z is called "number of tube segments" and is defined based on the M_e of footnote (c), i.e., using the Doi-Edwards 4/5 factor. Hence, when comparing with or using tube models, the 4/5 factor should be used".

[31] The Rouse reorientation relaxation time was called the rotational relaxation time by Doi and Edwards and is conjectured to be equal to $2\tau_R$

Rate-of-strain tensor[32]	$\dot{\boldsymbol{\gamma}} = \boldsymbol{\nabla v} + (\boldsymbol{\nabla v})^{\boldsymbol{T}}$
Vorticity tensor	$\boldsymbol{\Omega} = \boldsymbol{\nabla v} - (\boldsymbol{\nabla v})^{\boldsymbol{T}}$

Table XII. Dimensionless groups used to describe experimental regimes

Bingham number[33]	Bn	(yield stress)/(shear stress)
Boussinesq number	Bq	(surface shear stress)/[(bulk subphase shear stress)× (perimeter length along which the surface shear stress acts)
Capillary number	$Ca = \eta v/\sigma_{\alpha,\beta}$ Or $Ca = \eta\dot{\gamma}R/\sigma_{\alpha,\beta}$	(viscous forces)/(surface forces)
Deborah number[34,35]	De	(characteristic time of fluid)/(duration of deformation)
Peclet Number	$Pe \equiv \dot{\gamma}a/D_0$	*(rate of advection)/(rate of diffusion) (a =particle radius D_0 =particle diffusion coefficient)*
Poisson ratio[36]	$\nu = -d\varepsilon_{trans}/d\varepsilon_{axial}$	(negative ratio of transverse to axial strain)
Reynolds Number	$Re = \rho v d/\eta$	(inertial forces)/(viscous forces)
Weissenberg Number[34,37,38]	Wi	(characteristic time of fluid)×(rate of deformation)

Table XIII. Frequently used constants

N_A	Avogadro's number 6.023x10^{23} molecules/mole
R	Ideal gas constant, 8.314 Jmol^{-1}K^{-1}
k_B	Boltzmann's constant, 1.38 × 10^{-23} JK^{-1}

[32] An alternative definition, equal to ½ $\dot{\gamma}$, is widely used in fluid mechanics and is acceptable, (but the symbol $\boldsymbol{D}$ should be used for this tensor to avoid confusion: $D = \frac{1}{2}(\nabla v + \nabla v^T)$) , and in the fluid dynamics community a factor of a ½ is also used as a prefactor in the definition of the vorticity tensor

[33] Sometimes the wall shear stress is specifically used in the denominator of the definition, or the ratio (viscosity x velocity)/length.

[34]The definitions and uses of these groups are explained in detail in "Weissenberg and Deborah Numbers – Their definition and use", Rheol. Bulletin (The Society of Rheology), 79(2) p.14 (2010).

[35] Specific Deborah numbers are often used, depending on the choice of characteristic time. Popular examples are: Deborah number based on Rouse relaxation time (De_R) and Deborah number based on longest relaxation time (De_d).

[36] The Poisson ratio is also defined as function of Young's and shear modulus ($\nu = E/2G - 1$) in three dimensions (and differently in two dimensions).

[37] The Weissenberg number has sometimes been considered to be (N_1/σ). However, this is a ratio of dependent rather than independent variables and thus describes data rather than experimental conditions. The quantity (N_1/σ) is often called the stress ratio.

[38] Specific Weissenberg numbers are often used, depending on the characteristic time. Popular examples are: Weissenberg number based on Rouse relaxation time (Wi_R) and Weissenberg number based on longest relaxation time (Wi_D).

Transcribed by James J Griebler, April 2026